

FINITE-USE GEOMETRY AND LEARNING OF ORDERED BINARY QUANTUM MEASUREMENTS

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ABSTRACT. We study memoryless ordered binary quantum measurements that consume their input and return a classical label. A regularized midpoint Sylvester form is comparable to the finite-use adaptive distance on the complete effect body, with explicit dimension-dependent constants. At fixed dimension d , the small-error covering number has order $N^{d^2/2}[\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}$; all spectral boundaries are included. A common learner attains $Cd^4 N \delta^{-2} \log(d/\eta)$ calls, and a finite exact rational code attains the optimal reusable description order. We determine how a cap $b \leq N$ on independent quantum training blocks changes this cost. For fixed $d \geq 2$ the minimax order is $\Theta_d((N^2/b)\delta^{-2} \log(1/\eta))$, allowing classical feedback and collective processing of completed outputs. A conditional fidelity argument gives the finite-confidence converse. Separately, an information bound for weak binary measurements proves $\Omega(d^2 N \delta^{-2})$ calls even with coherent adaptive training. On the fixed interior $I/4 \preceq E \preceq 3I/4$, a dimension-free metric comparison and a reanalysis of an existing Choi estimator give the joint order $\Theta(d^2 N \delta^{-2})$ at fixed confidence. On the same interior we prove a dimension-uniform covering law, a finite exact index length $(d^2/2) \log_2 N + d^2 \log_2(1/\delta) + O(d^2)$, and simultaneous query-optimal learning and transmission. Allowing shared randomness and failure probability η changes the leading optimal expected prefix length by the factor $1 - \eta$. Thus independent one-call acquisition has linear future-horizon cost in this interior and quadratic cost uniformly on the full body. Our common learners also admit finite trusted preparation specifications with unchanged loss, confidence, and call orders. The full-body learning laws hold for sufficiently small absolute δ and $0 < \eta \leq 1/8$; the scalar case is stated separately.

1. INTRODUCTION

Repeated observations impose different resolutions on different perturbations of a quantum measurement. An interior change of an eigenvalue is resolved on the scale $N^{-1/2}$, whereas a rotation of a projective measurement can be resolved on the scale N^{-1} . The two scales meet at singular effects and repeated eigenvalues. We determine the resulting geometry on a compact matrix body, calculate its description entropy, and construct a common learning procedure whose loss is the distinguishability of all future experiments with at most N uses.

Fix a finite input dimension d and consider

$$\mathfrak{E}_d = \{E = E^* \in M_d(\mathbb{C}) : 0 \preceq E \preceq I_d\}.$$

The effect E specifies the memoryless ordered binary measurement

$$\mathcal{M}_E(\rho) = \text{tr}(E\rho)|1\rangle\langle 1| + \text{tr}((I_d - E)\rho)|0\rangle\langle 0|.$$

The input is consumed and the output is classical. The distance d_N is the supremum of the unhalved trace norm between the final states of a common experiment with at most N calls. The experiment may entangle its inputs, retain an arbitrary reference and quantum memory, use feedback, and stop publicly. Thus $0 \leq d_N \leq 2$; classical total variation is one half of the corresponding classical trace norm. The nonadaptive restriction d_N^{na} retains entangled input blocks and references and omits feedback between calls.

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1.1. An intrinsic comparison and its entropy. For $E, F \in \mathfrak{E}_d$ put

$$M = \frac{1}{2}(E + F), \quad H = F - E, \quad V = M(I_d - M),$$

and define the midpoint modulus

$$Q_N(E, F)^2 = N \langle H, (L_V + R_V + N^{-1} \text{Id})^{-1} H \rangle_{\text{HS}}.$$

The inverse acts on the real Hilbert space of Hermitian matrices. Theorem 2.1 proves, for every $d, N \geq 1$,

$$(1.1) \quad \frac{\min\{1, Q_N(E, F)\}}{8192d} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

No spectral margin or multiplicity assumption is imposed. In particular $d_N \leq 65536d d_N^{\text{na}}$. This is a fixed-dimensional comparison of optimal values. The modulus Q_N is neither identified with an optimal value nor assumed to satisfy a triangle inequality.

A horizontal measurement dilation yields the adaptive tangent bound. A regularized Sylvester equation separates an arbitrary tangent from a remainder controlled by the hybrid inequality. Operator concavity of $G(I_d - G)$ then integrates the tangent bound along a finite segment, including boundary endpoints. The converse uses Bernoulli inputs and exact two-dimensional compressions. All required qubit statements and their proofs are included below.

Let $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ denote the least number of legal memoryless binary centres needed to cover the body in d_N . For every fixed d , Theorem 3.1 gives

$$(1.2) \quad \mathcal{C}_N(\mathfrak{E}_d, \delta) \asymp_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}, \quad N \geq 1, \quad 0 < \delta \leq \delta_d.$$

The positive cap δ_d and the comparison constants may depend on d . The proof first bounds the weighted volume of actual operational balls uniformly over every boundary stratum. The total volume is then a spectral integral whose ordered endpoint sectors have cumulative exponents $S_j^2/2$, where S_j is a signed prefix sum. Balanced prefixes and matching nested endpoint boxes produce the logarithmic multiplicity $\lfloor d/2 \rfloor$.

1.2. Learning with a future operational loss. The statistical problem supplies calls to an unknown effect, without its matrix entries. For every fixed d , Theorem 5.1 constructs one learner, common to all $E \in \mathfrak{E}_d$, satisfying

$$(1.3) \quad \sup_E \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta, \quad M \leq C_d N \delta^{-2} \log(1/\eta).$$

Here $N \geq 1$, $0 < \delta \leq \delta_0$ for an absolute $\delta_0 > 0$, and $0 < \eta \leq 1/8$. The bound on the training calls M holds on every record. Each entangled block uses at most N calls. A scalar Bernoulli subfamily gives the matching lower order even for arbitrary reference-assisted adaptive training. More precisely, the upper bound is $Cd^4 N \delta^{-2} \log(d/\eta)$ with an absolute C . The dependence on d is not asserted to be optimal.

The construction begins with a qubit learner for unknown bias, contrast, and direction. Independent fair signs remove the bias from GHZ parity observations. A dyadic amplitude search stops at its first rejected scale, preserving a small-cone invariant; fresh observations complete the direction and endpoint estimates. Its confidence ledger controls both exceptional records and the maximum training cost.

For a general matrix effect, an additional calibration stage is essential. It estimates the two support boundaries in relative Loewner order at scale N^{-1} and controls the absolute matrix error at scale $N^{-1/2}$. In the calibrated eigenbasis, these two guarantees control the extra variance introduced by every two-dimensional compression. Qubit learning on those compressions then gives compatible matrix coordinates in the regularized quadratic form. Projection onto the legal effect body completes the estimate. The calibration and compression budgets are summed conditionally on the observed record, with no additional $\log N$ factor.

1.3. Independent training blocks and dimension. The maximal size of an independent quantum experiment is a second training resource. At the boundary of a block, a learner may jointly process all completed outputs and choose its next preparation from the classical record. The next block, including its reference and unused inputs, is isolated from earlier quantum registers until its last call. This permits collective final estimation and classical feedback. It excludes coherent transfer from older quantum memory into a block whose queries are not yet complete.

Let $1 \leq b \leq N$ be the maximal declared block length. For every fixed $d \geq 2$, Theorem 7.2 proves

$$(1.4) \quad M_b^*(d, N, \delta, \eta) = \Theta_d \left(\frac{N^2}{b\delta^2} \log \frac{1}{\eta} \right),$$

for an absolute small-error range and $0 < \eta \leq 1/8$. The explicit upper bound is

$$Cd^4 \frac{N^2}{b} \delta^{-2} \log(d/\eta).$$

A horizon-blocking inequality gives the common learner, while a history-weighted root-fidelity bound gives the converse even with arbitrary quantum storage and collective processing at block boundaries. In dimension one the device is a coin and the cost remains $\Theta(N\delta^{-2} \log(1/\eta))$, independently of b .

There is also a joint dimension obstruction. Theorem 8.2 proves $M \geq cd^2 N \delta^{-2}$, already on $I_d/4 \preceq E \preceq 3I_d/4$, under arbitrary coherent adaptive training. If every candidate effect lies within operator radius r of $I/2$, one query increases information about its finite unknown label by at most $4r^2$. A matrix packing and Fano's inequality then give the lower bound. In this fixed interior the operational distance has a dimension-free comparison with $\min\{1, \sqrt{N}\|E - F\|_{\text{op}}\}$. Reanalysing the binary Choi estimator of [5] consequently gives

$$(1.5) \quad M_{\text{int}}^*(d, N, \delta, 1/8) = \Theta(d^2 N \delta^{-2})$$

with absolute constants, achieved using independent one-call probes. At $N = 1$ this also gives the joint dimension-accuracy order $\Theta(d^2 \epsilon^{-2})$ for binary operator-norm learning. Thus existing independent-probe acquisition does admit a stronger future-loss analysis in the interior, whereas the projective boundary in (1.4) prevents that rate uniformly on the full body.

1.4. Deterministic descriptions and resources. Theorem 6.1 gives a deterministic finite rational codebook at the order in (1.2). Its greedy rule uses exact comparisons of Q_N^2 on a finite legal matrix grid. The operational separation in (1.1) controls its cardinality, and an inward rounding map gives coverage at every rank. The encoder and decoder reproduce the same dictionary from the public parameters, with no supplied custom dictionary. Every fixed-length word decodes to a legal effect. The required payload is

$$\frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N + 2) + d^2 \log_2(1/\delta) + O_d(1).$$

The construction is finite and exact; its enumeration cost is large. Optimal payload length does not imply efficient classical computation. The direct qubit code remains available with its own detailed resource accounting. Corollary 6.2 combines the common learner with this exact encoder, attaining the learning and payload orders simultaneously without additional calls to the unknown device. Corollary 7.6 preserves this conclusion under a prescribed independent-block cap. The finite-control theorem approximates all trusted preparations of our common learners with a pathwise error budget. The original loss, confidence, and call orders are retained; classical descriptions and trusted realization errors are charged explicitly.

1.5. Dimension-uniform operational descriptions. The interior learning result has a matching description theory. For $\mathfrak{I}_d = [I_d/4, 3I_d/4]$, all $d, N \geq 1$ and $0 < \delta \leq 2^{-13}$, Theorem 9.1 gives

$$(1.6) \quad \left(\frac{\sqrt{N}}{2048\delta} \right)^{d^2} \leq \mathcal{C}_N(\mathfrak{I}_d, \delta) \leq \left(\frac{25\sqrt{N}}{\delta} \right)^{d^2}.$$

The constants are absolute and the lower bound allows arbitrary legal binary centres. Theorem 9.2 realizes the upper bound by finite exact rational comparison and public reconstruction. Unlike entrywise transmission, its index does not incur a $d^2 \log d$ penalty. Its length is

$$A_d(N, \delta) + O(d^2), \quad A_d(N, \delta) = \frac{1}{2}d^2 \log_2 N + d^2 \log_2(1/\delta),$$

uniformly in dimension, horizon and accuracy.

If shared randomness and a worst-target failure probability η are allowed, Theorem 9.4 determines the optimal expected one-way prefix length as $(1 - \eta)A_d(N, \delta) + O(d^2)$, uniformly for $0 \leq \eta \leq 1/8$. The lower bound allows a randomized decoder and counts all nonpublic information. The upper bound uses a public target-independent abort event; its flag is not a hidden transmission. Finally, Theorem 9.5 gives one common finite-output learner with the joint optimal orders $M = \Theta(d^2 N \delta^{-2})$ and $B = A_d(N, \delta) + O(d^2)$ at failure $1/8$, using independent one-call acquisitions. This is a query and communication result; trusted tomography synthesis and exhaustive decoder computation are separate costs. The absence of the endpoint logarithm in (1.6) identifies the different interior law without changing the retained complete-body theorem.

1.6. Prior work and organization. Horizontal Kraus gauges, channel-extension metrology, and integration of local distinguishability have established antecedents [22, 7, 23, 24, 25, 26]. Inverse-Sylvester geometry and spectral matrix integration are likewise classical [27, 28]. The explicit closed-body midpoint comparison, its nested endpoint entropy, and learning under the future operational loss are the theorem objects studied here.

Mele–Bittel [5] and Zambrano–Ramos-Calderer–Kuang [29] already provide common estimators for unknown measurements in one-use losses. Their access, rank, dimension, and confidence statements are compared at theorem level in Section 13. The distinction in (1.3) is the dependence on the future horizon under the anisotropic loss d_N . The phase-estimation components have antecedents in [31, 32]; the conditional construction used here is proved directly.

The matrix comparison and entropy are followed by the qubit learning construction, its matrix extension, and the exact codebook theorem. The block-resource law, joint dimension bounds, uniform interior descriptions, and finite-control construction then identify how acquisition and preparation affect learning. The later qubit geometry and the final prerequisite appendix make the focused article self-contained. The global covering law and the matching description order use fixed dimension and small error. The full-body learner has the displayed unoptimized dimension factor, while the sharp joint dimension law is proved on the stated interior at fixed confidence. The exact matrix code is a finite exhaustive construction. All results concern ordered binary, input-consuming, classical-output measurements.

2. MATRIX GEOMETRY IN ARBITRARY INPUT DIMENSION

The binary effect body has a finite-use description that does not require eigenvector coordinates. Its regularization is the same at every spectral rank, and two-dimensional compressions supply lower witnesses for the full matrix problem. Throughout this section $d, N \geq 1$ are integers and

$$(2.1) \quad \mathfrak{E}_d = \{E \in \text{End}(\mathbb{C}^d) : E = E^*, 0 \preceq E \preceq I_d\}.$$

The measurement $\mathcal{M}_E$ has the ordered effects $E, I_d - E$ and no residual quantum output. The distance d_N has the same reference-assisted adaptive meaning as in Theorem 11.1. Write d_N^{na} for

its restriction to nonadaptive experiments, allowing entanglement across the calls and a retained reference. These experiments may depend on the known pair of hypotheses.

Put $\tau = N^{-1}$ and $V_E = E(I_d - E)$. On the real Hilbert space of Hermitian matrices with Hilbert–Schmidt inner product, define

$$(2.2) \quad \mathcal{A}_E(H) = V_E H + H V_E + \tau H,$$

$$(2.3) \quad g_{N,E}(H)^2 = N \langle H, \mathcal{A}_E^{-1}(H) \rangle_{\text{HS}}.$$

The operator $\mathcal{A}_E$ is strictly positive, including at a singular effect. For a pair E, F set

$$(2.4) \quad M = \frac{1}{2}(E + F), \quad H = F - E, \quad Q_N(E, F) = g_{N,M}(H).$$

In any orthonormal eigenbasis of M , with eigenvalues m_i and $v_i = m_i(1 - m_i)$, this is

$$(2.5) \quad Q_N(E, F)^2 = N \sum_{i,j=1}^d \frac{|H_{ij}|^2}{v_i + v_j + N^{-1}}.$$

Definition (2.4) makes the expression independent of all choices at repeated eigenvalues. We use it as a comparison modulus; a triangle inequality for Q_N is not required.

The inverse Sylvester operator is familiar from the Bures metric g^{B} on the positive definite cone [27]. Its normalization gives the algebraic identity

$$g_{N,E}(H)^2 = 2N g_{V_E + \tau I_d/2}^{\text{B}}(H, H).$$

Here H remains the effect tangent; the differential of the variance map $E \mapsto V_E$ is $H - EH - HE$.

Theorem 2.1 (Finite-use geometry of the matrix effect body). *For every $E, F \in \mathfrak{E}_d$,*

$$(2.6) \quad \frac{1}{8192d} \min\{1, Q_N(E, F)\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 8Q_N(E, F)\}.$$

The constants are uniform in the horizon and in both effects, without a spectral margin or a condition on eigenvalue multiplicities. The adaptive upper bound permits arbitrary retained references, common feedback, and bounded public stopping. Each lower witness uses inputs in one fixed subspace of dimension at most two. Moreover,

$$(2.7) \quad d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}.$$

The theorem includes every rank of projection and all of the intermediate singular strata. The variance $E(I_d - E)$, rather than the smallest eigenvalue of either outcome, determines the matrix weights. The factor N^{-1} makes the formula finite on the entire closed body. The proof first establishes a path inequality valid for every common adaptive tester, and then obtains the converse from one- and two-dimensional experiments.

2.1. A horizontal measurement dilation. Kraus-gauge bounds for channel estimation were developed in [22, 7, 23]. Their integration along channel paths appears in [25, Section III, equation (26)] and, with general adaptive bounds, in [26, Section IV.A]. We use an explicit horizontal lift for binary effects and control the finite-horizon regularization by a trace-norm remainder.

Lemma 2.2 (Horizontal derivative and regularized path bound). *Every continuously differentiable path $G : [0, 1] \rightarrow \mathfrak{E}_d$ satisfies*

$$(2.8) \quad d_N(\mathcal{M}_{G(0)}, \mathcal{M}_{G(1)}) \leq 4 \int_0^1 g_{N,G(t)}(G'(t)) dt.$$

Proof. We begin with an interior effect G , so that G and $I_d - G$ are invertible. Set $S = \sqrt{G(I_d - G)}$. For a Hermitian matrix K put $H_0 = SK + KS$. At the two factors $A_1 = \sqrt{G}$, $A_0 = \sqrt{I_d - G}$ prescribe derivatives

$$(2.9) \quad B_1 = \sqrt{I_d - G} K, \quad B_0 = -\sqrt{G} K.$$

They obey

$$(2.10) \quad \begin{aligned} A_1^* B_1 + B_1^* A_1 &= H_0, & A_0^* B_0 + B_0^* A_0 &= -H_0, \\ \sum_{y=0}^1 A_y^* B_y &= 0, & \sum_{y=0}^1 B_y^* B_y &= K^2. \end{aligned}$$

These tangent equations can be realized by genuine dilations. Let $G_1(s) = G + sH_0$, $G_0(s) = I_d - G - sH_0$ for s in a sufficiently small open interval, and put $R_y(s) = \sqrt{G_y(s)}$. The first line of (2.10) implies that

$$T_y = (B_y - R'_y(0))R_y(0)^{-1}$$

is anti-Hermitian. Indeed, multiplying $T_y + T_y^*$ on the left and right by $R_y(0)$ gives, with $R = R_y(0)$,

$$R(T_y + T_y^*)R = RB_y + B_y^*R - (RR'_y(0) + R'_y(0)R) = 0,$$

by differentiating $R_y(s)^2 = G_y(s)$. Thus $\tilde{A}_y(s) = \exp(sT_y)R_y(s)$ has derivative B_y at zero and satisfies $\tilde{A}_y(s)^* \tilde{A}_y(s) = G_y(s)$ exactly. A measurement isometry is

$$W(s)\psi = \sum_{y=0}^1 |y\rangle_C |y\rangle_L \tilde{A}_y(s)\psi.$$

Here C is the observed outcome, and L and the factor space of $\tilde{A}_y$ are environments. Tracing the environments implements $\mathcal{M}_{G(s)}$, also on inputs entangled with a reference. At zero, (2.10) gives

$$(2.11) \quad W^* \dot{W} = 0, \quad \dot{W}^* \dot{W} = K^2.$$

The horizontal Kraus condition has the antecedents [22, Theorems 4–5]. Its adaptive channel-extension form is $F_N \leq 4N\|\alpha\|$ when $\beta = 0$; see [23, equation (9)] and [24, equations (5)–(10)]. In the present lift $\alpha = K^2$ and $\beta = 0$. We give the cancellation explicitly for this binary measurement.

Fix a common tester with N slots. Purify its initial state and all intervening operations, and retain the measurement environments only for this comparison; neither the outcome copy nor the factor environment is accessible to the tester. The derivative of its final purification in the direction H_0 is the sum of N terms, each having one differentiated occurrence of W . The terms are mutually orthogonal. For two different differentiated slots, cancel the common isometries after the later slot; the remaining overlap contains $W^* \dot{W} \otimes I = 0$. Each term has norm at most $\|\dot{W}\|_{\text{op}} = \|K\|_{\text{op}}$, independently of the reference and memory dimensions. Hence the derivative of the purification has norm at most $\sqrt{N}\|K\|_{\text{op}}$. The pure-state trace-norm derivative bound, followed by contraction under the partial trace, bounds the observed output derivative by

$$(2.12) \quad 2\sqrt{N}\|K\|_{\text{op}}.$$

A publicly stopped tester is padded to N slots by calls on fixed dummy inputs, whose outcomes and environments are both discarded. The same purification argument then applies to its original output. Classical feedback is included by purifying the common controlled operations.

Now let H be an arbitrary Hermitian tangent at G , and set $\kappa = N^{-1/2}$. There is a unique Hermitian solution of

$$(2.13) \quad SK + KS + \kappa K = H.$$

Write $H = H_0 + R$, with $H_0 = SK + KS$ and $R = \kappa K$. Both are legitimate tangents at the interior effect G . The derivative of a fixed tester's output is linear in the inserted channel tangent. The part arising from H_0 is bounded by (2.12). The part arising from R is bounded by $2N\|R\|_{\text{op}}$. To verify the latter bound, the one-call tangent has two opposite reference blocks

$$B_\rho = \text{tr}_A[(R \otimes I)\rho], \quad -B_\rho.$$

Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|R\|_{\text{op}}$ for every input state. Insert this tangent into each of the N slots in turn; all subsequent common channels contract the Hermitian trace norm. This proves the bound for its sum. Thus the actual tangent H has output derivative bounded by

$$(2.14) \quad 2\sqrt{N}\|K\|_{\text{op}} + 2N\|R\|_{\text{op}} = 4\sqrt{N}\|K\|_{\text{op}}.$$

The splitting is used only for the differential at an interior effect; it does not postulate a physical mixture realizing the entire path.

In an eigenbasis of G , with $v_i = g_i(1 - g_i)$, equation (2.13) reads

$$K_{ij} = \frac{H_{ij}}{\sqrt{v_i} + \sqrt{v_j} + \kappa}.$$

Since $(\sqrt{v_i} + \sqrt{v_j} + \kappa)^2 \geq v_i + v_j + \tau$,

$$(2.15) \quad \sqrt{N}\|K\|_{\text{op}} \leq \sqrt{N}\|K\|_{\text{HS}} \leq g_{N,G}(H).$$

The integration of local distinguishability has direct antecedents: [25, Section III, equation (26), PDF version 3] treats the parallel path bound, and [26, Section IV.A, equations (14)–(22)] applies adaptive metrological estimates. Here the regularized binary Sylvester split and its residual produce the explicit integrand. For an interior path, integrate the resulting output derivative bound for each fixed tester, and then take the supremum. This proves (2.8) without exchanging a supremum with a derivative.

For a path in the closed body, apply the result to $G_\epsilon(t) = (1 - 2\epsilon)G(t) + \epsilon I_d$, where $0 < \epsilon < 1/2$. At fixed N , the form $g_{N,G}(H)$ is continuous on the whole body and bounded above by $N\|H\|_{\text{HS}}$. The integrals therefore converge as $\epsilon \downarrow 0$. The hybrid inequality $d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 2N\|E - F\|_{\text{op}}$ gives continuity of the distances at both endpoints. Passing to the limit proves the asserted closed-body statement. $\square$

2.2. The midpoint comparison.

Proof of Theorem 2.1. For the upper bound use $G(t) = (1 - t)E + tF$. The map $G \mapsto V_G = G - G^2$ is operator concave. For $0 < t \leq 1/2$, write $G(t) = (1 - 2t)E + 2tM$ and discard the positive multiple of V_E in the concavity inequality. Applying the same argument from the other endpoint gives

$$(2.16) \quad V_{G(t)} \succeq a(t)V_M, \quad a(t) = 2 \min\{t, 1 - t\}.$$

For a positive matrix D and a Hermitian matrix X ,

$$\langle X, (L_D + R_D)X \rangle_{\text{HS}} = 2 \text{tr}(DX^2) \geq 0.$$

Thus left-plus-right multiplication preserves the stated order on the real Hilbert space of Hermitian matrices. Since $a(t) \leq 1$, $\mathcal{A}_{G(t)} \succeq a(t)\mathcal{A}_M$. Inverting positive operators reverses their order, whence

$$g_{N,G(t)}(H) \leq a(t)^{-1/2}Q_N(E, F).$$

The scalar factor has integral 2 on $[0, 1]$. Lemma 2.2 yields $d_N \leq 8Q_N$; the trace-norm bound $d_N \leq 2$ gives the stated upper estimate.

For the lower bound choose any orthonormal eigenbasis e_i of M and put

$$(2.17) \quad q_{ij} = |H_{ij}| \sqrt{\frac{N}{v_i + v_j + \tau}}, \quad Q_N^2 = \sum_{i,j} q_{ij}^2.$$

Repeated input e_i gives Bernoulli probabilities $s = m_i - H_{ii}/2$, $t = m_i + H_{ii}/2$. Their variances satisfy

$$\max\{s(1-s), t(1-t)\} \leq s(1-s) + t(1-t) \leq 2m_i(1-m_i).$$

Lemma 11.2 therefore gives

$$(2.18) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{64} \min\{1, q_{ii}\}.$$

For $i \neq j$, restrict each input to $\mathcal{S} = \text{span}\{e_i, e_j\}$. The resulting channels have the compressed effects E_S, F_S , including on inputs entangled with a retained reference. Their midpoint is $M_S = \text{diag}(m_i, m_j)$. Write their spectral data as (p, q, u) and (p', q', v) in the notation of Section 11, with gaps r, r' and angle h when both gaps are positive. Put

$$V_E^S = p(1-p) + q(1-q), \quad V_F^S = p'(1-p') + q'(1-q').$$

For the compressed matrices the exact identity

$$V_{M_S} = \frac{1}{2}V_{E_S} + \frac{1}{2}V_{F_S} + \frac{1}{4}(F_S - E_S)^2$$

shows, on taking traces, that

$$(2.19) \quad v_i + v_j + \tau \geq \frac{1}{2}(V_E^S + \tau), \quad v_i + v_j + \tau \geq \frac{1}{2}(V_F^S + \tau).$$

The scalar part of $F_S - E_S$ has zero off-diagonal entry. Its traceless part has operator norm $|ru - r'v|/2$. Decomposing this vector along the smaller radius gives

$$(2.20) \quad \begin{aligned} |H_{ij}| &\leq \frac{1}{2}|r - r'| + \min\{r, r'\} \sin(h/2) \\ &\leq \frac{1}{2}(|p - p'| + |q - q'| + \min\{r, r'\}h). \end{aligned}$$

When one radius is zero, the angular term is zero and the same inequality holds without choosing its direction. Each endpoint variance is bounded by the appropriate V_E^S or V_F^S . Equations (2.19)–(2.20) therefore imply

$$(2.21) \quad q_{ij} \leq \frac{1}{\sqrt{2}} (T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\}h).$$

The nonadaptive assertion of Theorem 11.1 applied to this compressed pair yields

$$(2.22) \quad d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \geq \frac{1}{8192} \min\{1, q_{ij}\}.$$

Every such witness is a legitimate experiment for the original effects by a fixed isometric embedding of $\mathcal{S}$. Since $\max_{i,j} q_{ij} \geq Q_N/d$, taking the strongest of (2.18) and (2.22) proves the lower bound. For $d = 1$ only the diagonal argument is needed.

Finally, a one-use reference-assisted difference has output blocks B_ρ and $-B_\rho$, with $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Trace-norm duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An eigenstate of $E - F$ at an extremal eigenvalue attains equality. This proves (2.7). $\square$

Corollary 2.3 (Uniform comparison with nonadaptive experiments). *For all $E, F \in \mathfrak{E}_d$ and all $N \geq 1$,*

$$(2.23) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 65536d d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

Here nonadaptive experiments may entangle all inputs and retain a reference; they use no feedback between calls. The dimensional constant comes from the maximum of d^2 weighted entries and is not optimized.

Proof. If $Q_N \leq 1$, divide the upper estimate $8Q_N$ by the lower estimate $Q_N/(8192d)$. If $Q_N \geq 1$, the upper estimate is 2 and the lower estimate is $1/(8192d)$, which gives a smaller constant. The identical pair needs no division. $\square$

2.3. All spectral ranks and local forms.

Lemma 2.4 (Spectral noncancellation in every rank). *Let $\lambda_1(E) \geq \dots \geq \lambda_d(E)$ and $\lambda_1(F) \geq \dots \geq \lambda_d(F)$ be the ordered spectra. For every $1 \leq k \leq d$,*

$$(2.24) \quad B_N(\lambda_k(E), \lambda_k(F)) \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F).$$

For effects diagonal in the same ordered orthonormal basis,

$$(2.25) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \sum_{k=1}^d B_N(\lambda_k(E), \lambda_k(F)).$$

Proof. Suppose $\lambda_k(E) \geq \lambda_k(F)$. The span of the first k eigenvectors of E intersects the orthogonal complement of the first $k-1$ eigenvectors of F in a nonzero subspace. A unit vector in that intersection has success probabilities $s \geq \lambda_k(E) \geq \lambda_k(F) \geq t$ under the two hypotheses. The monotonicity in Lemma 11.2 implies $B_N(s, t) \geq B_N(\lambda_k(E), \lambda_k(F))$. Repeated use of this vector proves the first assertion; reverse the hypotheses for the opposite ordering. Multiplicities cause no change in the dimension argument.

For the upper bound, first measure the common eigenbasis and then produce a bit whose success probability is its corresponding eigenvalue. Supply N independent bits in each of the d rows before the experiment. Every common adaptive tester is a common channel of these row arrays, including its reference blocks and all publicly stopped histories. The tensor-product triangle inequality gives (2.25); unused row bits are discarded. $\square$

The following local comparison will permit volume estimates at every boundary stratum. It concerns matrices themselves and does not require a coordinate chart for eigenvectors.

Lemma 2.5 (Uniform local comparison of the variance forms). *Set $W_G = G(I_d - G) + \tau I_d/2$. For a Hermitian H , put $q = g_{N,G}(H)$. Then*

$$(2.26) \quad \|W_G^{-1/2} H W_G^{-1/2}\|_{\text{HS}} \leq 2q.$$

For either sign for which $G \pm H/2 \in \mathfrak{C}_d$, one has

$$(2.27) \quad \|W_G^{-1/2} (W_{G \pm H/2} - W_G) W_G^{-1/2}\|_{\text{op}} \leq q + \frac{3}{4}q^2.$$

In particular, for $q \leq 1/4$,

$$(2.28) \quad \frac{1}{2}W_G \preceq W_{G \pm H/2} \preceq 2W_G.$$

Thus, if either $Q_N(E, F) \leq 1/4$ or $g_{N,E}(F - E) \leq 1/4$, the midpoint and the fixed-endpoint quadratic forms are comparable within factors 2.

Proof. In an eigenbasis of W_G , with eigenvalues $w_i \geq \tau/2$,

$$\frac{w_i + w_j}{N w_i w_j} \leq 4.$$

Multiplying this inequality by $N|H_{ij}|^2/(w_i + w_j)$ and summing proves (2.26). Put $Z = W_G^{-1/2} H W_G^{-1/2}$. The exact expansion

$$W_{G \pm H/2} - W_G = \pm \frac{1}{2}(H - GH - HG) - \frac{1}{4}H^2$$

has normalized linear part $\pm((I_d - 2G)Z + Z(I_d - 2G))/4$, since G commutes with W_G . Its operator norm is at most $\|Z\|_{\text{op}}/2 \leq q$. The normalized quadratic part is $-ZW_G Z/4$. Since $\tau \leq 1$, one has

$\|W_G\|_{\text{op}} \leq 3/4$, and its norm is at most $3q^2/4$. This proves (2.27) and hence (2.28). The same order holds for left plus right multiplication, and inversion gives the quadratic-form comparison. Apply it first at $G = M$ or at $G = E$, respectively, to obtain the final assertion. $\square$

3. DESCRIPTION ENTROPY IN ARBITRARY MATRIX DIMENSION

The intrinsic comparison of Theorem 2.1 permits the full effect body to be treated without choosing eigenvector coordinates near repeated spectra. Throughout this section the input dimension $d \geq 1$ is fixed. Constants may depend on d , but not on the integer horizon $N \geq 1$ or on the accuracy parameter.

Write $\mathcal{C}_N(\mathfrak{E}_d, \delta)$ for the covering number under d_N , with centres in the same legal memoryless ordered binary interface, and write $\mathcal{C}_N^{\text{rat}}$ when the real and imaginary parts of every matrix entry of each centre are rational.

Theorem 3.1 (Entropy of the full effect body). *For every integer $d \geq 1$ there are constants $c_d, C_d, \delta_d > 0$ such that, for all integers $N \geq 1$ and all $0 < \delta \leq \delta_d$,*

$$\begin{aligned} c_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2} &\leq \mathcal{C}_N(\mathfrak{E}_d, \delta) \\ (3.1) \quad &\leq \mathcal{C}_N^{\text{rat}}(\mathfrak{E}_d, \delta) \\ &\leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2}. \end{aligned}$$

Consequently the optimum fixed-length reusable description has length

$$(3.2) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

The estimates include all support and spectral multiplicities. The rational-centre assertion here follows from a covering argument. Theorem 6.1 gives a deterministic finite exact encoder and decoder attaining the same order in every dimension. Theorem 12.2 retains its separate direct construction in dimension two. Computational cost is accounted for separately from the number of transmitted bits.

For $d = 1$, the result is the Bernoulli covering law $N^{1/2} \delta^{-1}$. For $d = 2$, it recovers Theorem 12.1. The next two cases have orders $N^{9/2} \log(N+2) \delta^{-9}$ and $N^8 [\log(N+2)]^2 \delta^{-16}$. The logarithmic power counts independent nested approaches of paired eigenvalues to the two support endpoints. We first establish a uniform measure estimate for actual operational balls, and then evaluate the total measure.

3.1. Operational balls and a regularized volume form. Set $n = d^2$, $\tau = N^{-1}$ and

$$W_E = E(I - E) + \frac{\tau}{2} I.$$

As in Section 2, let

$$(3.3) \quad g_{N,E}(H)^2 = N \langle H, (L_{W_E} + R_{W_E})^{-1} H \rangle_{\text{HS}}$$

on the real Hilbert space of Hermitian matrices. Let $G_{N,E}$ denote this positive quadratic form in Hilbert–Schmidt orthonormal coordinates, and define

$$(3.4) \quad \rho_N(E) = \sqrt{\det G_{N,E}}, \quad d\mu_N(E) = \rho_N(E) dE,$$

The real orthonormal basis consists of e_{ii} and, for $i < j$, $(e_{ij} + e_{ji})/\sqrt{2}$ and $i(e_{ij} - e_{ji})/\sqrt{2}$; dE is Lebesgue measure in these coordinates. The cutoff makes ρ_N positive and continuous on the whole closed body for each N .

Lemma 3.2 (Uniform volume of operational balls). *For fixed d there are $a_d, A_d, r_d > 0$ such that*

$$(3.5) \quad a_d r^{d^2} \leq \mu_N(B_{d_N}(E, r)) \leq A_d r^{d^2}$$

for every $N \geq 1$, every $E \in \mathfrak{E}_d$, and every $0 < r \leq r_d$. Balls are taken inside $\mathfrak{E}_d$.

Proof. We make explicit the two local consequences of Lemma 2.5 that will be used. If H is Hermitian and $q = g_{N,E}(H)$, diagonalizing W_E gives

$$(3.6) \quad \|W_E^{-1/2} H W_E^{-1/2}\|_{\text{HS}}^2 \leq 4q^2,$$

since its eigenvalues w_i are at least $\tau/2$ and

$$\frac{w_i + w_j}{N w_i w_j} = \tau(w_i^{-1} + w_j^{-1}) \leq 4.$$

For any legal $E + tH$, direct expansion yields

$$W_{E+tH} - W_E = \frac{t}{2}((I - 2E)H + H(I - 2E)) - t^2 H^2.$$

The matrices E and W_E commute, $\|I - 2E\|_{\text{op}} \leq 1$, and $\|W_E\|_{\text{op}} \leq 3/4$. Thus the norm of the normalized difference is at most $2|t|q + 3t^2q^2$. For $|t| \leq 1$ and q below an absolute constant, it follows that

$$(3.7) \quad \frac{1}{2}W_E \preceq W_{E+tH} \preceq 2W_E.$$

Loewner comparison passes to $L_W + R_W$ and, by inversion, to the quadratic forms. It also gives

$$(3.8) \quad 2^{-n/2} \rho_N(E) \leq \rho_N(E + tH) \leq 2^{n/2} \rho_N(E).$$

Apply this comparison from an endpoint to the midpoint and from the midpoint to either endpoint. Theorem 2.1 then provides constants $A, B, r_0 > 0$, depending only on d , such that

$$(3.9) \quad d_N(E, F) \leq r \implies g_{N,E}(F - E) \leq Ar,$$

$$(3.10) \quad g_{N,E}(F - E) \leq Br \implies d_N(E, F) \leq r,$$

for legal E, F and $0 < r \leq r_0$. Decrease r_0 so that Ar_0 lies in the small-form range of (3.7).

Let ω_n be the Euclidean volume of the unit ball in $\mathbb{R}^n$. The ellipsoid $g_{N,E}(H) \leq u$ has Lebesgue volume $\omega_n u^n / \rho_N(E)$. Equations (3.9) and (3.8) therefore imply

$$\mu_N(B_{d_N}(E, r)) \leq 2^{n/2} \omega_n (Ar)^n.$$

For the lower bound we construct a full legal ellipsoid, including when E lies on a support boundary. Choose a small constant $a > 0$ depending only on d , set $s = ar\tau$, and move towards the scalar interior point:

$$(3.11) \quad E_s = (1 - 2s)E + sI.$$

Take r_0 sufficiently small that $s \leq 1/4$. Then $sI \preceq E_s \preceq (1 - s)I$. Since the displacement commutes with E , its form is bounded by

$$g_{N,E}(E_s - E)^2 = N \sum_i \frac{s^2(1 - 2\lambda_i)^2}{2\lambda_i(1 - \lambda_i) + \tau} \leq d \frac{s^2}{\tau^2} = da^2 r^2.$$

Choose $a\sqrt{d} \leq B/4$. By (3.10), $d_N(E, E_s) \leq r/4$.

Let $X = X^*$ satisfy $g_{N,E_s}(X) \leq br$, with $b > 0$ to be chosen. Write ℓ_i for the eigenvalues of E_s and $w_i = \ell_i(1 - \ell_i) + \tau/2$. Since $\ell_i \geq s$ and $w_i \leq \ell_i + \tau/2$, we have

$$\begin{aligned} \|E_s^{-1/2} X E_s^{-1/2}\|_{\text{HS}}^2 &\leq g_{N,E_s}(X)^2 \max_{i,j} \frac{w_i + w_j}{N \ell_i \ell_j} \\ &\leq b^2 r^2 \left(\frac{2\tau}{s} + \frac{\tau^2}{s^2} \right) = b^2 \left(\frac{2r}{a} + \frac{1}{a^2} \right). \end{aligned}$$

The same bound holds with $I - E_s$ in place of E_s , because $1 - \ell_i \geq s$ and $w_i \leq 1 - \ell_i + \tau/2$. Taking $b \leq a/4$ and $ar_0 \leq 1$ makes both operator norms at most $1/2$. Consequently every point of the entire ellipsoid

$$\mathcal{A}_s = \{E_s + X : g_{N,E_s}(X) \leq br\}$$

is a legal effect. Choose also $b \leq B/4$. Equation (3.10) and the triangle inequality give $\mathcal{A}_s \subset B_{d_N}(E, r)$. The density comparison about E_s and the exact ellipsoid volume now imply

$$\mu_N(B_{d_N}(E, r)) \geq 2^{-n/2} \omega_n (br)^n.$$

All choices depend only on d . This proves the assertion on every boundary stratum as well as in the interior. $\square$

3.2. The total regularized volume.

Proposition 3.3 (Nested endpoint accumulation). *For fixed $d \geq 1$, uniformly over $N \geq 1$,*

$$(3.12) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} [\log(N+2)]^{[d/2]}.$$

Proof. In a spectral basis, the quadratic form has weight $N/(2w_i)$ on the i th diagonal direction and weight $N/(w_i + w_j)$ on each of the two Hilbert–Schmidt orthonormal real directions belonging to $i < j$. Hence

$$(3.13) \quad \rho_N(E) = 2^{-d/2} N^{d^2/2} \prod_i w_i^{-1/2} \prod_{i < j} (w_i + w_j)^{-1}.$$

We use the classical spectral change of variables [28]. For a simple spectrum, differentiation of $E = U \text{diag}(\lambda) U^*$ gives

$$U^*(dE)U = d \text{diag}(\lambda) + [U^* dU, \text{diag}(\lambda)].$$

Each complex off-diagonal pair therefore contributes $(\lambda_i - \lambda_j)^2$ to the real Jacobian. Integration over the compact unitary orbit and over permutations supplies a positive constant depending only on d . The repeated-spectrum set is Lebesgue null. This spectral change of variables, together with (3.13), yields

$$(3.14) \quad \mu_N(\mathfrak{E}_d) \asymp_d N^{d^2/2} I_d(\tau),$$

where

$$(3.15) \quad I_d(\tau) = \int_{[0,1]^d} \prod_i (v_i + \tau)^{-1/2} \prod_{i < j} \frac{(\lambda_i - \lambda_j)^2}{v_i + v_j + \tau} d\lambda, \quad v_i = \lambda_i(1 - \lambda_i).$$

Replacing $v_i + \tau/2$ by $v_i + \tau$ in the single-coordinate factors changes the integral by a factor between fixed positive constants depending only on d .

We prove

$$(3.16) \quad I_d(\tau) \asymp_d [\log(2 + \tau^{-1})]^{[d/2]}, \quad 0 < \tau \leq 1.$$

For the upper bound, partition each coordinate into $[0, 1/4]$, $[1/4, 3/4]$, and $[3/4, 1]$. Factors involving an eigenvalue in the middle interval are bounded above by constants, so these coordinates can be integrated out. For each of the remaining $l \leq d$ coordinates put $x_i = \lambda_i$ near zero and

$x_i = 1 - \lambda_i$ near one. On a sector $0 \leq x_1 \leq \dots \leq x_l \leq 1/4$, let $\epsilon_i \in \{1, -1\}$ record the endpoint and put

$$S_j = \sum_{i=1}^j \epsilon_i, \quad S_0 = 0, \quad y_j = x_j + \tau.$$

The individual factor is at most $Cy_j^{-1/2}$. For $i < j$, a same-endpoint pair factor is at most Cy_j , whereas an opposite-endpoint factor is at most C/y_j . The sector integrand is consequently bounded by

$$(3.17) \quad C_d \prod_{j=1}^l y_j^{a_j-1}, \quad a_j = \frac{1}{2} + \epsilon_j S_{j-1} = \frac{S_j^2 - S_{j-1}^2}{2}.$$

Its cumulative exponents satisfy

$$(3.18) \quad A_j = \sum_{i=1}^j a_i = \frac{S_j^2}{2} \geq 0.$$

Set $T = 1/4 + \tau$ and $L = \log(T/\tau)$. Substitute $y_j = T \exp(-z_j)$, where $z_1 \geq \dots \geq z_l \geq 0$, and then $u_j = z_j - z_{j+1}$ for $j < l$, $u_l = z_l$. The sector integral is bounded by

$$(3.19) \quad C_d \int_{u_j \geq 0, \sum_j u_j \leq L} \exp\left(-\sum_j A_j u_j\right) du.$$

The omitted factor T^{A_l} is bounded above uniformly in $0 < \tau \leq 1$. Each $A_j > 0$ is at least $1/2$, so its variable has an integrable exponential tail. Each vanishing A_j contributes at most one factor $1 + L$. A vanishing prefix sum requires an even j , and there are at most $\lfloor l/2 \rfloor$ such indices. Summing the finitely many sectors proves the upper estimate in (3.16). The case $l = 0$ contributes only a bounded constant.

For the lower bound let $m = \lfloor d/2 \rfloor$. Choose scales

$$(3.20) \quad \tau \leq t_1, \quad 8t_j \leq t_{j+1} \ (j < m), \quad t_m \leq 1/32,$$

and restrict $2m$ eigenvalues in pairs by

$$(3.21) \quad \lambda_j^- = a_j \in [t_j, 2t_j], \quad \lambda_j^+ = 1 - b_j, \quad b_j \in [t_j, 2t_j], \quad 1 \leq j \leq m.$$

If d is odd, restrict the remaining eigenvalue to $[3/8, 5/8]$; all its factors are bounded above and below by positive constants. For one pair the two individual spectral factors and its opposite-endpoint factor have product comparable to t_j^{-2} . For two pairs at scales $t_i \leq t_j/8$, the two same-endpoint factors are comparable to t_j and the two opposite-endpoint factors to t_j^{-1} . Their product is therefore bounded above and below by fixed constants. Thus the integrand on the box is comparable to $\prod_j t_j^{-2}$ and its volume to $\prod_j t_j^2$. Each box contributes at least $c_d > 0$.

Take the scales from $\tau, 8\tau, 8^2\tau, \dots$ up to $1/32$. The boxes for distinct increasing m -tuples are disjoint. Their number is a binomial coefficient bounded below by $c_d [\log(\tau^{-1})]^m$ once $\log(\tau^{-1})$ is sufficiently large depending on d . In the remaining bounded range, a fixed box of distinct eigenvalues strictly inside $(0, 1)$ has a positive uniform integral and supplies the lower estimate. For $m = 0$ this interior argument suffices on the full range. This proves (3.16) and, by (3.14), the proposition. $\square$

The balanced-prefix identity (3.18) accounts for all nested endpoint scales. A simultaneous approach of all eigenvalues to the endpoints detects only one possible critical scaling; the separated pairs in (3.21) show why the logarithmic multiplicity can be larger.

3.3. Covering, rational centres, and description length.

Proof of Theorem 3.1. Choose δ_d smaller than the radius in Lemma 3.2. A cover by L balls of radius δ satisfies

$$\mu_N(\mathfrak{E}_d) \leq LA_d \delta^{d^2},$$

which gives the required lower bound. Conversely, take a maximal δ -separated subset of $\mathfrak{E}_d$. Its balls of radius $\delta/3$ are pairwise disjoint, and the lower bound in (3.5) bounds its cardinality by

$$C_d \mu_N(\mathfrak{E}_d) \delta^{-d^2}.$$

It is a radius- δ cover by maximality. Such finite maximal sets exist: the effect body is compact, d_N is continuous by the hybrid bound, and $d_1(E, F) = 2\|E - F\|_{\text{op}}$ separates distinct effects. Proposition 3.3 now gives both estimates for legal centres.

Rational legal effects are dense. Indeed, first move towards $I/2$ by (3.11), and then approximate all independent real matrix coordinates by rational numbers while remaining strictly between 0 and I . For fixed N , continuity in the operational distance follows again from the hybrid bound. Replace every centre of a radius- $\delta/2$ cover by a rational legal effect at distance less than $\delta/2$. The resulting radius- δ cover has the same cardinality. This proves the rational-centre upper bound.

A fixed-length payload indexes the centres of a cover, and conversely each legal reusable program with a fixed payload supplies a centre. Taking base-two logarithms of (3.1) and rounding the index length gives (3.2). $\square$

4. LEARNING THE COMPLETE BINARY QUBIT EFFECT BODY

We now pass from supplied descriptions to an unknown device. Throughout this section the input dimension is two. Write

$$E = \frac{1}{2}((1+b)I + ru \cdot \sigma), \quad 0 \leq r \leq 1, \quad |b| + r \leq 1,$$

with the direction absent when $r = 0$. The bias, contrast, and direction are all unknown. The outputs are ordered and classical, and each call consumes its input without a residual quantum output. A learning procedure may prepare entangled input blocks, retain the unused input qubits, and use classical feedback between blocks. It has no access to a measurement environment. Its choices depend only on the public parameters and the observed record.

The integer N below is the horizon of a future comparison, measured by d_N . The number M of training calls is a separate resource. The output is a classical description of a legal effect. For a unit vector w set $P_w = (I + w \cdot \sigma)/2$ and write $\mathcal{M}_{p,q,w}$ for the measurement associated with $pP_w + q(I - P_w)$, where $p = (1 + b + r)/2$ and $q = (1 + b - r)/2$. We use the spectral quantities

$$V = p(1 - p) + q(1 - q), \quad K_N(p, q) = r\sqrt{N/(V + N^{-1})}.$$

Theorem 4.1 (Common learning at the finite-use scale). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every $N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common learning procedure returns an effect $\hat{E} \in \mathfrak{E}_2$ with*

$$(4.1) \quad \sup_{E \in \mathfrak{E}_2} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record it uses at most

$$(4.2) \quad M \leq CN\delta^{-2} \log(1/\eta)$$

calls, and every entangled input block has length at most N . Conversely, every common procedure satisfying (4.1), with at most M calls on every record, obeys

$$(4.3) \quad M \geq cN\delta^{-2} \log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. The constants are uniform over the closed effect body, including scalar, deterministic, and projective effects.

The budget in (4.2) is a deterministic upper bound for each completed record, including unsuccessful records. It counts calls during training; N specifies only the subsequent loss. The construction uses independent input blocks, retaining at most $N - 1$ unused input qubits after the next qubit has been supplied, and no quantum system between blocks. Its preparation model permits ideal trusted finite-state preparation and fresh fair random signs. Preparation cost, classical running time, and classical workspace are separate resources. The reusable payload is charged separately in Corollary 4.6. The proof below specifies a written statistical algorithm; the accompanying finite identity checks do not constitute a software implementation of its risk guarantee.

The proof constructs one decision tree for the entire parameter body. Its experiments are not selected from a known pair of hypotheses. The angular part uses multiscale phase estimation, a method with antecedents in [31, 32]. We give the phase and confidence bounds explicitly. A data-dependent amplitude threshold selects the block length without prior calibration of the spectrum.

4.1. Bernoulli errors and estimated axes. For $s, t \in [0, 1]$ put

$$(4.4) \quad \mathfrak{h}(s, t)^2 = (\sqrt{s} - \sqrt{t})^2 + (\sqrt{1-s} - \sqrt{1-t})^2.$$

Write $B_N(s, t)$ for the unhalved distance between the two N -fold Bernoulli product laws. We use the following elementary estimates at both support endpoints.

Lemma 4.2 (Endpoint estimation). *For every $s, t \in [0, 1]$,*

$$(4.5) \quad B_N(s, t) \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For each fixed $A > 0$ there is C_A such that, if $0 < \lambda \leq 1$ and

$$(4.6) \quad |s - t| \leq A(\sqrt{t(1-t)\lambda} + \lambda),$$

then $\mathfrak{h}(s, t) \leq C_A\sqrt{\lambda}$. If two estimates of an ordered pair (p, q) each have $\mathfrak{h}$ error at most ε , sorting the estimates preserves this bound up to an absolute constant.

Proof. The root fidelity is $1 - \mathfrak{h}^2/2$. Tensorization and the trace-fidelity inequality give

$$B_N(s, t) \leq 2\sqrt{1 - (1 - \mathfrak{h}(s, t)^2/2)^{2N}} \leq 2\sqrt{N} \mathfrak{h}(s, t).$$

For the first square root in (4.4), if $t \geq \lambda$ divide the difference by $\sqrt{t}$ in (4.6). If $t < \lambda$, both t and s are at most $C_A\lambda$. Apply the same argument to their complements. For sorting, set $\phi(t) = \arcsin \sqrt{t}$. Then

$$\mathfrak{h}(s, t) = 2 \sin(|\phi(s) - \phi(t)|/2).$$

The two distances are uniformly comparable on $[0, \pi/2]$. Sorting is nonexpansive for the sup norm against an already ordered pair of real numbers. Apply this fact in the ϕ coordinate. $\square$

If $\hat{t}$ is the empirical frequency in n Bernoulli trials, Bernstein's inequality gives, with failure at most $2e^{-L}$,

$$(4.7) \quad |\hat{t} - t| \leq \sqrt{2t(1-t)L/n} + \frac{2L}{3n}.$$

Thus Lemma 4.2 applies with $\lambda = L/n$, including at $t = 0, 1$.

For a unit estimate w of u , write

$$(4.8) \quad h = \angle(u, w), \quad \gamma = \frac{r}{2}(1 - u \cdot w).$$

The input states on the axes $+w$ and $-w$ have respective success probabilities

$$(4.9) \quad s_+ = p - \gamma, \quad s_- = q + \gamma, \quad p = \frac{1}{2}(1 + b + r), \quad q = \frac{1}{2}(1 + b - r).$$

If $\hat{x}$ estimates $x = ru$ with Euclidean error at most e , normalizing $\hat{x}$ gives

$$(4.10) \quad rh \leq \pi \min(r, e), \quad \gamma \leq \min(r, e^2/r) \quad (r > 0).$$

Indeed $\|u - w\| \leq 2e/r$, $h \leq (\pi/2)\|u - w\|$, and $\gamma = r\|u - w\|^2/4$. If $\hat{x} = 0$, choose a fixed direction; then $e \geq r$ gives the same bounds. At $r = 0$ both quantities vanish.

Lemma 4.3 (Product learning away from unit contrast). *Suppose $r \leq 7/8$. There is a product-input procedure, not requiring b, r, u as input, which uses eight groups of n calls and returns a legal effect F such that, for $1 \leq L \leq n$,*

$$(4.11) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq C(\sqrt{NL/n} + NL/n)$$

outside an event of probability at most Ce^{-L} .

Proof. First use the positive and negative eigenstates of each of the three Pauli axes, with n calls per state. Encode an observed outcome as $Z \in \{-1, 1\}$. The two means on axis j are $b + x_j$ and $b - x_j$. Estimate x_j by half their empirical difference. Put

$$w_0 = 1 - b^2, \quad \lambda = L/n.$$

The two variances on axis j sum to $2(w_0 - x_j^2)$. Bernstein's inequality and a union bound therefore give

$$(4.12) \quad e = \|\hat{x} - x\| \leq 3(\sqrt{w_0\lambda} + \lambda).$$

More explicitly, the coordinate error is at most $\sqrt{2w_0\lambda} + 4\lambda/3$; the sign observation has range length two. Taking the Euclidean norm gives (4.12), since $\sqrt{6} < 3$ and $4\sqrt{3}/3 < 3$.

Normalize to a direction w . Feasibility and $r \leq 7/8$ imply

$$(4.13) \quad w_0 \geq r(2 - r), \quad V = \frac{1}{2}(w_0 - r^2) \geq w_0/9.$$

Use (4.10) and Theorem 11.4. The square-root term in (4.12) contributes at most $C\sqrt{N\lambda}$ to $K_N(p, q)h$, and its additive term contributes at most $CN\lambda$. Consequently,

$$(4.14) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w}) \leq C(\sqrt{N\lambda} + N\lambda).$$

This calculation does not require a lower bound on r or w_0 .

It remains to control both endpoint probabilities in (4.9). For either $t = p$ or $t = q$ put $v_t = t(1 - t)$. Since

$$|p(1 - p) - q(1 - q)| = r|b| \leq r, \quad w_0 = 2V + r^2 \leq 4v_t + 3r,$$

we claim

$$(4.15) \quad \gamma \leq 9\sqrt{v_t\lambda} + 72\lambda.$$

Equation (4.12) gives $e^2 \leq 18(w_0\lambda + \lambda^2)$. If $r \leq \lambda$, use $\gamma \leq r$. If $r > \lambda$, use

$$\gamma \leq 72v_t\lambda/r + 72\lambda, \quad \gamma \leq r,$$

and $\min(r, 72v_t\lambda/r) \leq \sqrt{72v_t\lambda}$. This proves (4.15).

Make two fresh groups of n calls, along $+w$ and $-w$. Conditional on the previous record, these are Bernoulli trials with the parameters in (4.9). Their variances differ from v_t by at most γ . Equations (4.7) and (4.15) imply

$$|\hat{t}_{\text{raw}} - t| \leq C(\sqrt{v_t\lambda} + \lambda).$$

For the additional term use $\sqrt{\gamma\lambda} \leq C(\sqrt{v_t\lambda} + \lambda)$, which follows from $\sqrt{AB} \leq (A+B)/2$ with $A = \sqrt{v_t\lambda}$ and $B = \lambda$. Lemma 4.2 gives root-distance error $C\sqrt{\lambda}$ for both endpoints. Sort the two empirical parameters to obtain $\hat{p} \geq \hat{q}$ and return $F = \hat{p}P_w + \hat{q}(I - P_w)$. The aligned programme bound (11.9) and (4.5) give a spectral contribution at most $C\sqrt{N\lambda}$. Combine with (4.14). All concentration bounds for the fresh groups hold conditionally, so their failure probabilities may be added to those of the first six groups. $\square$

4.2. A common entangled experiment with unknown bias.

Lemma 4.4 (Two-plane phase observation). *Fix an orthonormal frame (e_1, e_2, w) known to the experimenter. For each $i = 1, 2$ and integer $m \geq 1$, there is an m -call experiment with an observation $Y \in \{-1, 1\}$ whose mean, for a chosen phase φ , is*

$$(4.16) \quad \mathbb{E}_E Y = \Re(e^{i\varphi} Z_i), \quad Z_i = [x \cdot w + ix \cdot e_i]^m, \quad x = ru.$$

Two phases estimate the complex number Z_i . For $0 < \zeta \leq 1$ and $0 < \epsilon \leq 1/2$, four settings, using the two planes, estimate both complex numbers with error at most ζ and failure at most ϵ , with at most $C\zeta^{-2} \log(C/\epsilon)$ independent blocks per setting.

Fix $a = 1/64$. If $r > 0$ and $h = \angle(u, w) \leq 2a/m$, then

$$(4.17) \quad .99r^m \leq |Z_i| \leq r^m, \quad \text{Arg } Z_i = m \arctan \frac{u \cdot e_i}{u \cdot w},$$

where the principal argument is used. If the amplitudes are bounded below by a fixed positive constant and ζ is sufficiently small, these observations produce a direction estimate w' with

$$(4.18) \quad \angle(u, w') \leq C\zeta/m.$$

Proof. Choose a computational basis along the axis perpendicular to the plane spanned by w, e_i , with phase convention such that the off-diagonal matrix element of $O = 2E - I = bI + ru \cdot \sigma$ is $r(u \cdot w + iu \cdot e_i)$. Prepare

$$|G_{\varphi, s}\rangle = \frac{|0\rangle^{\otimes m} + se^{i\varphi}|1\rangle^{\otimes m}}{\sqrt{2}},$$

where s is a fresh fair sign for this block. Conditional on the entire preceding record, the two signs have probability $1/2$ each; the random draw is independent of all earlier preparation randomness. Pass its qubits through the measurement and record $Y = s \prod_{j=1}^m Z_j$, where each output is encoded as $Z_j \in \{-1, 1\}$. Memorylessness makes the joint outcome law the product POVM on the entangled input; the qubits may be supplied sequentially while the unused ones are retained.

The expansion of $\langle G_{\varphi, s} | O^{\otimes m} | G_{\varphi, s} \rangle$ has diagonal term

$$\frac{1}{2}((b + ru \cdot f_i)^m + (b - ru \cdot f_i)^m),$$

where f_i is the chosen perpendicular axis. It is independent of s and disappears after multiplication by s and averaging. The off-diagonal terms give (4.16). At $x = 0$ the mean is zero and no direction is needed. In particular the experiment removes the unknown bias exactly. The phases $0, \pi/2$ give the real part and the negative imaginary part. More precisely, at each of the four settings use

$$(4.19) \quad B(\zeta, \epsilon) = \left\lceil 4\zeta^{-2} \log(8/\epsilon) \right\rceil$$

fresh independent blocks. Hoeffding's inequality bounds the error of each real sample mean by $\zeta/\sqrt{2}$ outside an event of probability $2\exp(-B\zeta^2/4)$. The union bound over the four means is at most ϵ , and each complex error is at most ζ . This bound holds conditional on every preceding record: the frame and the setting are then fixed, and the memoryless device acts on fresh blocks. Each estimate batch uses its prescribed number of blocks without stopping within the batch.

For the phase conclusions assume $r > 0$ and put $\theta_i = \arctan((u \cdot e_i)/(u \cdot w))$. Then $|m\theta_i| \leq 2a$ and

$$|Z_i| = r^m (1 - (u \cdot f_i)^2)^{m/2} \geq r^m \cos(h)^m \geq r^m (1 - 2a^2/m).$$

This proves (4.17), with no phase ambiguity. For the estimates set

$$\hat{\theta}_i = \text{Arg}(\hat{Z}_i)/m, \quad w' = \frac{w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2}{\|w + \tan(\hat{\theta}_1)e_1 + \tan(\hat{\theta}_2)e_2\|}.$$

The exact formula with θ_i recovers u . For $|\hat{Z}_i - Z_i| \leq |Z_i|/2$,

$$|\text{Arg}(\hat{Z}_i) - \text{Arg}(Z_i)| \leq 2|\hat{Z}_i - Z_i|/|Z_i|.$$

On $|\theta_i| \leq 1/4$ the displayed map from two angles to a unit vector is Lipschitz; a bound of 16 for angular distance against the largest coordinate error follows from differentiation of $\tan$ and normalization. This proves (4.18). Our fixed cone lies strictly inside the principal-argument interval, and the argument uses no unwrapping threshold from a separate phase-estimation theorem. $\square$

We make the reconstruction rule total. Fix a coordinate basis of $\mathbb{R}^3$ and a tie-breaking order. Given a unit vector w , choose the first coordinate vector minimizing its absolute inner product with w , normalize its projection onto $w^\perp$ to obtain e_1 , and put $e_2 = w \times e_1$. The projection has norm at least $\sqrt{2/3}$, so the frame is defined at every w . For complex estimates z_1, z_2 use the phase formula in the preceding proof only when both pass the guard

$$(4.20) \quad \Re z_i > 0, \quad 8|\Im z_i| \leq \Re z_i \quad (i = 1, 2).$$

Write $\mathcal{R}_m(w; z_1, z_2) = (\text{ok}, w')$ for the resulting update, and set $\mathcal{R}_m = (\text{fail}, w)$ otherwise. A passed guard gives $|\text{Arg } z_i| \leq \arctan(1/8) < 1/4$; the vector to be normalized has inner product one with w and cannot vanish. Zero complex estimates, the branch cut, and an out-of-cone phase therefore have prescribed outcomes before an argument or tangent is evaluated.

Lemma 4.5 (Selecting a block length from the observations). *Suppose $r \geq 5/8$ and an initial known direction w satisfies $\angle(u, w) \leq a = 1/64$. A common adaptive procedure selects a length m_* and a direction w_* , for $N \geq 1$ and $0 < \eta \leq 1/8$, using at most $CN \log(C/\eta)$ calls on every record. With failure at most $\eta/16$,*

$$(4.21) \quad \frac{1}{8} \min\{N, (1-r)^{-1}\} \leq m_* \leq N,$$

$$(4.22) \quad r^{m_*} \geq 1/5, \quad \angle(u, w_*) \leq a/m_*.$$

The reciprocal at $r = 1$ is interpreted as infinity. New observations at m_, w_* have both complex amplitudes at least $1/6$.*

Proof. Fix $\kappa = a/1024$, and let H be the largest power of two not exceeding N . At a tested length m , estimate both complex numbers in Lemma 4.4 to error at most κ with $B(\kappa, \epsilon_m)$ blocks per setting, where

$$(4.23) \quad \epsilon_m = \frac{\eta m}{32N}.$$

The following procedure stops at the first rejection; it never tests a larger length after that rejection. Here w_0 is the input direction and $\perp$ denotes an empty register.

SELECTBLOCK(N, η, w_0)

1. Set $m \leftarrow 1$, $w \leftarrow w_0$, and $A \leftarrow \perp$.
2. Obtain z_1, z_2 from the four fresh settings at length m and frame w , with budgets (4.23).
3. If $\min_i |z_i| < 1/4$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
4. Set $(s, v) \leftarrow \mathcal{R}_m(w; z_1, z_2)$. If $s = \text{fail}$, return A if $A \neq \perp$, and return $(1, w_0)$ otherwise.
5. Set $A \leftarrow (m, v)$. If $m = H$, return A .
6. Set $w \leftarrow v$, $m \leftarrow 2m$, and return to step 2.

Equality in the amplitude threshold is accepted, and equality in the second phase guard is allowed. All registers, decisions, and fallbacks are functions of the observed record.

We prove the assertions on the event that the successive complex estimates obey their bounds. Before a length m is tried the angular error is at most $2a/m$: this holds initially, and is preserved by each accepted update. If the amplitude test passes, each true amplitude $\rho = |Z_i|$ is at least $1/4 - \kappa > 1/5$. Since $|\text{Arg } Z_i| \leq 2a$,

$$\Re z_i \geq \rho(1 - 2a^2) - \kappa > 8(2a\rho + \kappa) \geq 8|\Im z_i|.$$

The strict inequality follows directly from $a = 1/64$, $\kappa = a/1024$, and $\rho \geq 1/4 - \kappa$. Thus the guard cannot reject such a stage on this event. The argument bound and the Lipschitz constant in Lemma 4.4 give an updated angular error at most

$$\frac{32\kappa}{(1/4 - \kappa)m} \leq \frac{a}{4m}.$$

Furthermore $r^m \geq 1/4 - \kappa$, whence $m(1 - r) \leq 2$.

On rejection, the amplitude test must have failed, and (4.17) gives $r^m < (1/4 + \kappa)/.99 < 1/3$. Since $r \geq 5/8$,

$$-\log r \leq (1 - r)/r \leq 2(1 - r), \quad m(1 - r) \geq (\log 3)/2.$$

The procedure returns the preceding accepted length $m/2$, which is therefore at least $(\log 3)/(4(1 - r))$. Length one is always accepted on the good event, since its true amplitudes exceed $.99(5/8)$. If the cap is reached, the selected length is $H \geq N/2$. These facts prove (4.21) and (4.22). At the updated direction the amplitude is at least $r^{m_*} \cos(a/m_*)^{m_*} > 1/6$.

For completeness, let $\mathcal{F}_m$ contain the record immediately before a possible stage m , let $T_m \in \mathcal{F}_m$ be the event that this stage is reached, and let G_m be its complex-error event. On every reached record the concentration bound gives $\mathbb{P}_E(G_m^c \mid \mathcal{F}_m) \leq \epsilon_m$. Consequently

$$(4.24) \quad \mathbb{P}_E\left(\bigcup_m (T_m \cap G_m^c)\right) \leq \sum_m \mathbb{E}_E[\mathbf{1}_{T_m} \mathbb{P}_E(G_m^c \mid \mathcal{F}_m)] \leq \sum_{m=1,2,4,\dots,H} \epsilon_m \leq \eta/16.$$

The sample-mean bound holds for every preceding record, even one whose frame has left the cone; the cone is needed only to turn those sample means into directional information. Independence between stage-failure events is not assumed. By Lemma 4.4, the number of calls is bounded by an absolute constant times

$$(4.25) \quad \sum_{m=1,2,4,\dots,H} m \log \frac{CN}{\eta m} \leq C' N \log(C'/\eta).$$

To see this, write $m = H/2^k$ and use the convergent sums of 2^{-k} and $k2^{-k}$, together with $1 \leq N/H < 2$. The public cap and these budgets bound the cost on every record, including records on which an estimate fails. Rejection can only shorten the sum. $\square$

TABLE 1. Confidence allocation for one execution of the learner.

Batch	Failure allowance	Conditioning and scope
Coarse Pauli vector	$\eta/8$	Unconditional; all six input groups jointly.
Product-branch Pauli vector	$\eta/8$	Given the coarse record; six fresh input groups jointly.
Dyadic stage m	$\eta m/(32N)$	Given the record before that stage; all reached stages together have allowance at most $\eta/16$.
Fine two-plane estimate	$\eta/8$	Given the complete search record, including the selected m_*, w_* .
Fresh endpoint groups	$\eta/16$ each	Given the final estimated axis; the two allowances sum to $\eta/8$, in either branch.

4.3. The learning bound and its converse. All statistical batches use fresh inputs and fresh randomization. Table 1 allocates their error probabilities. Each conditional bound is uniform over the preceding record. A row belonging to a branch not taken contributes no failure event.

The conservative sum of all rows is $9\eta/16 < \eta$. The conditional union calculation (4.24) applies to every row, so this sum requires no independence between the events attached to different batches.

Proof of Theorem 4.1. We choose constants in the following order. Fix the local cone $a = 1/64$ and $\kappa = a/1024$ as above. Choose a sufficiently small absolute constant $c_* > 0$ in the final phase tolerance below. Then choose the absolute multipliers in the Bernoulli sample counts sufficiently large. Finally reduce δ_0 , if needed, so that $\delta_0 \leq 1/4096$ and $c_*\delta_0 \leq a/100$. No optimized value of these constants is used. Set

$$(4.26) \quad L = \log(C_2/\eta), \quad n_0 = \lceil C_0 L \rceil, \quad n = \lceil C_1 N \delta^{-2} L \rceil,$$

where C_0, C_1, C_2 are absolute constants large enough for the concentration bounds and loss estimates below. The common procedure is the following.

- (1) Make the six coarse Pauli experiments, with n_0 calls per input, and form their half-difference vector $\hat{x}_0$.
- (2) If $\|\hat{x}_0\| < 3/4$, make six fresh Pauli groups of n calls per input and form $\hat{x}$. Set $w_f = \hat{x}/\|\hat{x}\|$, using the fixed third coordinate direction if $\hat{x} = 0$, and proceed to step 4.
- (3) Otherwise set $w_0 = \hat{x}_0/\|\hat{x}_0\|$ and run $\text{SELECTBLOCK}(N, \eta, w_0)$. At its returned (m_*, w_*) , obtain two fresh complex estimates with tolerance $\zeta_* = c_*\delta\sqrt{m_*/N}$ and failure allowance $\eta/8$. Apply $\mathcal{R}_{m_*}$ to them; take its updated direction as w_f on success and retain $w_f = w_*$ on failure.
- (4) Make n fresh calls along each of $+w_f$ and $-w_f$. If the empirical success frequencies are s_+, s_- , return

$$\hat{E} = \max(s_+, s_-)P_{w_f} + \min(s_+, s_-)(I - P_{w_f}).$$

The empirical frequencies already lie in $[0, 1]$. Ties in the coarse comparison use step 3; ties between endpoint frequencies give a scalar effect. Together with the frame, phase, and first-rejection conventions above, these rules define an output for every record, with no retry or additional calibration step.

We verify the risk. With failure at most $\eta/8$, the coarse half-difference estimate satisfies

$$\|\hat{x}_0 - x\| \leq \varepsilon_0, \quad \varepsilon_0 \leq a/100.$$

On this event the two decisions have the explicit overlap

$$\begin{aligned}\|\hat{x}_0\| < 3/4 &\implies r < 3/4 + 1/6400 < 7/8, \\ \|\hat{x}_0\| \geq 3/4 &\implies r \geq 3/4 - 1/6400 > 5/8.\end{aligned}$$

For the product branch, Lemma 4.3 gives error at most $\delta/2$ with the sample sizes in (4.26), after increasing C_1 . Its first six groups have joint allowance $\eta/8$ conditional on the coarse record. After their axis is fixed, each endpoint group has allowance $\eta/16$. These are the product-branch and endpoint rows of Table 1, and all samples here are fresh relative to the coarse test.

For the entangled branch, the normalized initial estimate has angular error at most $\pi\epsilon_0/r \leq 8\pi a/500 < a$ by (4.10). Apply Lemma 4.5. At the selected length and updated direction, make fresh two-plane observations with complex error tolerance

$$(4.27) \quad \zeta_* = c_*\delta\sqrt{m_*/N},$$

and failure allowance $\eta/8$, conditional on the full selection record. On the preceding good event, the two true amplitudes are at least $1/6$ and their phases have absolute value at most a . On the current complex-error event, $\zeta_* \leq a/100$ and the inequalities $|\sin t| \leq |t|$ and $\cos t \geq 1 - t^2/2$ show that both phase guards pass. The resulting direction w_f satisfies

$$(4.28) \quad h_f = \angle(u, w_f) \leq Cc_*\delta/\sqrt{Nm_*}.$$

The calls used in this step are at most

$$(4.29) \quad Cm_*\zeta_*^{-2}\log(C/\eta) \leq C'N\delta^{-2}\log(C/\eta).$$

Put $\nu = 1 - r$. Since $|b| \leq \nu$ and $r \geq 5/8$,

$$r\nu \leq V = \frac{1}{2}(1 - r^2 - b^2) \leq \nu.$$

Equation (4.21) implies

$$(4.30) \quad m_*(V + N^{-1}) \geq c_0 > 0$$

for an absolute c_0 . Thus (4.28) and Theorem 11.4 give

$$(4.31) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,w_f}) \leq Cc_*\delta.$$

The endpoint misalignment is also bounded:

$$(4.32) \quad \gamma = \frac{r}{2}(1 - u \cdot w_f) \leq rh_f^2/4 \leq Cc_*^2\delta^2/N.$$

The final fresh groups have the public size n in (4.26). Conditional on the record fixing w_f , their success probabilities are exactly $p - \gamma$ and $q + \gamma$, and each group obeys Bernstein's bound with allowance $\eta/16$. Conditional Bernstein bounds, Lemma 4.2, and $\mathfrak{h}(s, t)^2 \leq 2|s - t|$ give

$$(4.33) \quad \begin{aligned}\mathfrak{h}(\hat{p}_{\text{raw}}, p), \mathfrak{h}(\hat{q}_{\text{raw}}, q) &\leq C\sqrt{\log(C/\eta)/n} + C\sqrt{\gamma} \\ &\leq c'\delta/\sqrt{N}.\end{aligned}$$

The constant c' can be made as small as required by the preceding choices of c_* and the sample multiplier. Sort the estimates and return $\hat{E} = \hat{p}P_{w_f} + \hat{q}(I - P_{w_f})$. Equations (11.9) and (4.5) make the spectral loss at most $\delta/4$; choose c_* so that (4.31) contributes at most $\delta/4$. Table 1 now gives the stated risk.

Finally we check the budget for every record. The product branch uses exactly $6n_0 + 8n$ calls. The entangled branch uses at most

$$(4.34) \quad 6n_0 + 4 \sum_{m=1,2,4,\dots,H} mB(\kappa, \epsilon_m) + 4m_*B(\zeta_*, \eta/8) + 2n.$$

This is a pathwise upper bound: the sum includes even the stages not reached after the first rejection. Its dyadic term is $O(N \log(C/\eta))$ by (4.25), including the ceilings in (4.19). For every returned $1 \leq m_* \leq N$, even on a failed record,

$$4m_*B(\zeta_*, \eta/8) \leq 4N + 16c_*^{-2}N\delta^{-2} \log(64/\eta).$$

Here $m_*\zeta_*^{-2} = c_*^{-2}N\delta^{-2}$ exactly. Since $N \geq 1$, $\delta \leq \delta_0$, and $\eta \leq 1/8$, both branches satisfy (4.2). The prescribed fallbacks always leave a unit direction and a legal effect, and every block has at most $H \leq N$ inputs. This proves the upper bound with the same worst-record convention as the converse.

For the converse consider the scalar effects

$$(4.35) \quad E_0 = \tfrac{1}{2}I, \quad E_1 = (\tfrac{1}{2} + \Delta)I, \quad \Delta = 256\delta/\sqrt{N}.$$

Since $\delta \leq 1/4096$, both parameters lie in $[1/2, 9/16]$. The finite Bernoulli lower bound in Lemma 11.2 yields

$$(4.36) \quad \begin{aligned} d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_1}) &= B_N(1/2, 1/2 + \Delta) \\ &\geq \tfrac{1}{64} \min\{1, \Delta\sqrt{N/(5/4)}\} > 2\delta. \end{aligned}$$

For these devices every call gives a fresh Bernoulli bit independent of the input. On an entangled input, each conditional reference block is the same reduced state multiplied by the scalar outcome probability. Thus any adaptive learning record, including its final classical estimate, is a common stochastic processing of M independent Bernoulli bits and parameter-independent resources. Publicly stopped records may be padded. The relative entropy before processing is at most

$$(4.37) \quad MD(\text{Bern}(1/2) \parallel \text{Bern}(1/2 + \Delta)) = -\tfrac{M}{2} \log(1 - 4\Delta^2) \leq 3M\Delta^2.$$

The two radius- δ success balls are disjoint by (4.36). Testing whether the estimated effect belongs to the first ball therefore gives two errors at most η . For classical laws P, Q , Jensen's inequality gives root fidelity at least $e^{-D(P\parallel Q)/2}$. The trace-fidelity inequality then implies that the sum of testing errors is at least $\tfrac{1}{2}e^{-D(P\parallel Q)}$. Consequently

$$2\eta \geq \tfrac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2 \delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this proves (4.3). □

Corollary 4.6 (A learned joint description). *For rational δ in the range of Theorem 4.1, the learner may return one word of the rational joint code, with the same risk and call-count bounds. Its payload length is*

$$(4.38) \quad 2\log_2 N + \log_2 \log(N+2) + 4\log_2(1/\delta) + O(1), \quad 0 < \delta \leq \delta_0.$$

The decoding centre is a legal rational effect. This is a statement about device calls and payload length; classical arithmetic, workspace, and state-preparation costs are separate resources.

Proof. Use the learner with loss budget $\delta/2$. Its output is a finite computable expression determined by a finite record. In particular, the empirical frequencies are rational, and the guarded phase roots and normalized frames specify algebraic directions. Choose a positive rational $\varepsilon \leq \delta/(100N)$ and rational approximations (b_0, x_0) to its bias and Bloch vector with

$$|b_0 - \hat{b}| + \|x_0 - \hat{x}\| \leq \varepsilon.$$

Set $(b_1, x_1) = (b_0, x_0)/(1 + 2\varepsilon)$. Then

$$|b_1| + \|x_1\| \leq \frac{1 + \varepsilon}{1 + 2\varepsilon} \leq 1,$$

and the exact one-use formula (11.6) bounds the distance from the unrounded hypothesis by 3ε . The N -use hybrid inequality adds at most $3N\varepsilon$. Apply Theorem 12.2 to this rational estimated effect with public accuracy $\delta/2$. Its certified radius is $\delta/4$, so the total loss is less than δ . The capacity estimate in that theorem, at a constant rescaling of accuracy, gives (4.38). This postprocessing uses no additional device calls and receives no exact description of the unknown target. $\square$

The loss in Theorem 4.1 is a future adaptive distance. One-use measurement tomography is studied in [5, 29]; combining the stated one-use accuracy bounds only with the hybrid inequality gives a quadratic dependence on the future horizon. The argument here uses finite-use geometry and data-dependent entangled blocks to obtain the linear dependence in (4.2), with a matching scalar converse. Learning a stored programme from which a measurement is later retrieved, as in [30], has a different output resource. The present theorem gives a classical estimate over the entire ordered binary qubit body. Section 5 uses this common qubit procedure after a separate calibration to obtain the corresponding learning law on the full matrix effect body.

5. COMMON LEARNING IN ARBITRARY INPUT DIMENSION

The two-dimensional learner extends to the full effect body through a calibration step. The calibration must resolve the two support endpoints simultaneously. A bound in operator norm alone does not provide this information: after compression, a poorly aligned subspace can introduce noise at a scale larger than N^{-1} . The procedure below learns a basis in which this additional noise is bounded by the finite-horizon cutoff.

Throughout this section, $\mathcal{M}_E$ is the ordered binary measurement associated with $E \in \mathfrak{E}_d$. Training may use entangled input blocks and classical feedback, but it has no access to the measurement environment. Its output is a classical description of a legal effect. The future horizon N and the number of training calls remain separate resources.

Theorem 5.1 (Common learning on the matrix effect body). *There are absolute constants $c, C, \delta_0 > 0$ with the following property. For every $d, N \geq 1$, $0 < \delta \leq \delta_0$, and $0 < \eta \leq 1/8$, one common procedure returns a legal effect $\hat{E} \in \mathfrak{E}_d$ such that*

$$(5.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

On every record its number of calls satisfies

$$(5.2) \quad M \leq Cd^4N\delta^{-2}\log(d/\eta),$$

and every entangled input block has length at most N . Conversely, any common procedure satisfying (5.1) with at most M calls on every record satisfies

$$(5.3) \quad M \geq cN\delta^{-2}\log(1/\eta).$$

The converse permits arbitrary reference-assisted adaptive training and bounded public stopping. In particular, for each fixed d , the minimax number of calls is $\Theta_d(N\delta^{-2}\log(1/\eta))$ on the complete closed effect body. The displayed dependence on d in the upper bound is not asserted to be optimal.

We first give the calibration, including its finite budgets. All matrix orders below are Loewner orders. The use of a basis selected from earlier observations is always followed by fresh samples.

Lemma 5.2 (Simultaneous endpoint calibration). *Let $\tau = N^{-1}$ and $0 < \epsilon \leq 1/8$. There is a common product-input procedure returning $A \in \mathfrak{E}_d$ with at most $Cd^4N\log(d/\epsilon)$ calls on every*

record. With probability at least $1 - \epsilon$, it satisfies

$$(5.4) \quad \frac{1}{2}(A + \tau I) \preceq E + \tau I \preceq 2(A + \tau I),$$

$$(5.5) \quad \frac{1}{2}(I - A + \tau I) \preceq I - E + \tau I \preceq 2(I - A + \tau I),$$

$$(5.6) \quad \|E - A\|_{\text{op}} \leq \sqrt{\tau}.$$

The input states depend only on the public parameters and the preceding record; no spectral data about E are supplied.

Proof. Set $J = \lceil \log_2 N \rceil$, $H = 2^J$, and $t_j = 2^{-j}$ for $0 \leq j \leq J$. Thus $N \leq H < 2N$ unless $N = 1$, and $\tau/2 < t_J \leq \tau$. Start from $A_0 = I/2$. If $J = 0$, return A_0 : all three conclusions hold without samples.

We use

$$h_d = d + 4 \binom{d}{2} = 2d^2 - d, \quad K_d = (320d)^2, \quad \alpha = 1/16.$$

At stage $j \geq 1$, assign failure allowance and sample size

$$(5.7) \quad \epsilon_j = \frac{\epsilon 2^j}{4H}, \quad L_j = \log \frac{2h_d}{\epsilon_j}, \quad n_j = \left\lceil K_d t_j^{-1} L_j \right\rceil.$$

Choose an orthonormal eigenbasis $e_1, \dots, e_d$ of the current legal estimate A_{j-1} . The choice is deterministic and measurable, including at multiple eigenvalues. More explicitly, fix a public input basis. All calibration matrices have algebraic entries in that basis: the sample frequencies are rational, and spectral projections, clipping at rational endpoints, and normalization preserve algebraic entries. Order the distinct eigenvalues, project the public basis onto each eigenspace, and apply Gram–Schmidt in that fixed order, discarding exactly zero vectors. Exact algebraic arithmetic therefore supplies a finite choice at every stage and requires no oracle for arbitrary real spectral data. Make n_j independent calls on each of the h_d states

$$(5.8) \quad e_i, \quad \frac{e_i + e_j}{\sqrt{2}}, \quad \frac{e_i - e_j}{\sqrt{2}}, \quad \frac{e_i + ie_j}{\sqrt{2}}, \quad \frac{e_i - ie_j}{\sqrt{2}} \quad (i < j).$$

Here and below a unit vector denotes its pure input state. Estimate each diagonal entry of E by its empirical outcome frequency. Estimate the real part of each off-diagonal entry by half the frequency difference between the positive and negative real superpositions. Its imaginary part is half the frequency on $(e_i - ie_j)/\sqrt{2}$ minus half the frequency on $(e_i + ie_j)/\sqrt{2}$. These formulas define a Hermitian matrix B_j .

To define the procedure on every record, clip the eigenvalues of B_j to $[-\alpha t_j, 1 + \alpha t_j]$, obtaining $\tilde{B}_j$, and set

$$(5.9) \quad A_j = \frac{\tilde{B}_j + \alpha t_j I}{1 + 2\alpha t_j}.$$

The returned matrix is always a legal effect. In particular, a failed concentration event cannot affect the subsequent call budget or require an additional experiment.

We prove the invariant

$$(5.10) \quad \frac{1}{2}(A_j + t_j I) \preceq E + t_j I \preceq 2(A_j + t_j I), \quad \frac{1}{2}(I - A_j + t_j I) \preceq I - E + t_j I \preceq 2(I - A_j + t_j I)$$

on the successive good events. It holds for $j = 0$ because $0 \preceq E \preceq I$.

Fix a preceding good record and abbreviate $t = t_j$, $A = A_{j-1}$, $B = B_j$, $D = B - E$, and $q = L_j/n_j$. All expectations and probabilities in the next calculation are conditional on that

record. Bernstein's inequality and a union bound over the h_d groups show, with failure at most $2h_d e^{-L_j} = \epsilon_j$, that every empirical frequency has error at most

$$\sqrt{2p(1-p)q} + 2q/3.$$

For a pair of opposite superposition states, the sum of the outcome probabilities is $E_{ii} + E_{jj}$. Cauchy–Schwarz in each half-difference therefore gives the following two simultaneous bounds, also valid for $i = j$ after enlarging the constants:

$$(5.11) \quad |D_{ij}| \leq \sqrt{2(E_{ii} + E_{jj})q} + 2q, \quad |D_{ij}| \leq \sqrt{2(2 - E_{ii} - E_{jj})q} + 2q.$$

For an off-diagonal complex entry, apply the real estimate to the two quadratures and take their Euclidean norm. The factor $\sqrt{2}$ in (5.11) includes this operation.

Write a_i for the eigenvalues of A . The preceding invariant, whose regularization is $2t$, gives

$$E_{ii} \leq 2a_i + 2t, \quad 1 - E_{ii} \leq 2(1 - a_i) + 2t, \quad q \leq t/K_d.$$

Since

$$t(a_i + a_j + 2t) \leq 2(a_i + t)(a_j + t), \quad t \leq \sqrt{(a_i + t)(a_j + t)},$$

the first bound in (5.11) yields

$$(5.12) \quad \frac{|D_{ij}|}{\sqrt{(a_i + t)(a_j + t)}} \leq \frac{2\sqrt{2}}{\sqrt{K_d}} + \frac{2}{K_d} \leq \frac{5}{\sqrt{K_d}}.$$

Replacing a_i by $1 - a_i$ gives the same estimate for the complement. The Hilbert–Schmidt norm bounds the operator norm, so our choice of K_d implies

$$(5.13) \quad \begin{aligned} -\frac{1}{64}(A + tI) &\preceq D \preceq \frac{1}{64}(A + tI), \\ -\frac{1}{64}(I - A + tI) &\preceq D \preceq \frac{1}{64}(I - A + tI). \end{aligned}$$

The preceding invariant and $t_{j-1} = 2t$ also imply

$$A + tI \preceq 4(E + tI), \quad I - A + tI \preceq 4(I - E + tI).$$

Consequently

$$(5.14) \quad -\alpha(E + tI) \preceq D \preceq \alpha(E + tI), \quad -\alpha(I - E + tI) \preceq D \preceq \alpha(I - E + tI).$$

These inequalities place the spectrum of B inside $[-\alpha t, 1 + \alpha t]$. Thus the clipping in (5.9) does nothing on this event. On it,

$$A_j - E = \frac{D + \alpha t(I - 2E)}{1 + 2\alpha t}.$$

Since $-I \preceq I - 2E \preceq I$, equations (5.14) imply

$$\begin{aligned} -2\alpha(E + tI) &\preceq A_j - E \preceq 2\alpha(E + tI), \\ -2\alpha(I - E + tI) &\preceq A_j - E \preceq 2\alpha(I - E + tI). \end{aligned}$$

Adding the corresponding regularized endpoint gives factors $1 - 2\alpha = 7/8$ and $1 + 2\alpha = 9/8$. This proves the weaker factor-two invariant (5.10) for stage j .

The unweighted entry estimate also gives the operator-norm bound needed later. Because $E_{ii} + E_{jj} \leq 2$, we have

$$\|D\|_{\text{op}} \leq 4d\sqrt{t/K_d} = \sqrt{t}/80.$$

It follows that

$$(5.15) \quad \|A_j - E\|_{\text{op}} \leq \sqrt{t}/80 + \alpha t \leq \sqrt{t}.$$

At the final stage, adding $(\tau - t_J)I$ preserves the factor-two orders, and (5.15) gives (5.6).

The conditional failure allowances sum to less than $\epsilon/2$, since $\sum_{j=1}^J 2^j < 2H$. This proves the asserted probability bound. Finally, the deterministic call count is $h_d \sum_j n_j$. Writing $2^j = H/2^k$ yields

$$\sum_{j=1}^J 2^j L_j \leq H \sum_{k=0}^{J-1} 2^{-k} \left(\log \frac{8h_d}{\epsilon} + k \log 2 \right) \leq CH \log(d/\epsilon).$$

The ceiling terms contribute at most $h_d J \leq h_d H$. As $h_d K_d \leq Cd^4$ and $H < 2N$ when $N > 1$, the stated bound holds on every record. In particular, the number of dyadic stages does not introduce a factor $\log N$. $\square$

Lemma 5.3 (Calibrated variance and compression). *Suppose $A, E \in \mathfrak{E}_d$ satisfy (5.4)–(5.6). Set $W_G = G(I - G) + \tau I/2$. Then*

$$(5.16) \quad \frac{1}{12} W_A \preceq W_E \preceq 12 W_A.$$

If P is any orthogonal projection commuting with A and $C = PEP|_{P\mathbb{C}^d}$, then, on $P\mathbb{C}^d$,

$$(5.17) \quad \frac{1}{12} P W_A P \preceq W_C \preceq 14 P W_A P.$$

All constants are independent of the spectrum, its multiplicities, and the rank of P .

Proof. For a legal effect G , define the harmonic mean without a factor of two by

$$R_G = ((G + \tau I)^{-1} + (I - G + \tau I)^{-1})^{-1}.$$

The two factors commute because both are functions of G , so

$$(5.18) \quad R_G = \frac{G(I - G) + (\tau + \tau^2)I}{1 + 2\tau}, \quad \frac{1}{3}(G(I - G) + \tau I) \preceq R_G \preceq G(I - G) + \tau I.$$

The inequalities use only $0 < \tau \leq 1$. Inversion reverses Loewner order. Apply it to both endpoint comparisons, add, and invert again to get $R_E \asymp R_A$ with factors two. Equation (5.18) therefore compares $E(I - E) + \tau I$ and $A(I - A) + \tau I$ with factors six. Replacing the regularization τI by $\tau I/2$ proves (5.16).

The compression identity is exact:

$$(5.19) \quad C(I - C) = PE(I - E)P + PE(I - P)EP.$$

Because P commutes with A , $PE(I - P) = P(E - A)(I - P)$. Thus

$$0 \preceq PE(I - P)EP \preceq \|E - A\|_{\text{op}}^2 P \preceq \tau P.$$

The first term in (5.19), after adding $\tau P/2$, is $PW_E P$. Apply (5.16) and use $PW_A P \succeq \tau P/2$ to absorb the final τP with a factor two. This gives (5.17). $\square$

Proof of Theorem 5.1. For $d = 1$, Bernoulli sampling and Lemma 4.2 give the upper bound directly. Assume $d \geq 2$ and put $s = \binom{d}{2}$. Apply Lemma 5.2 with failure allowance $\eta/2$. Conditional on its good event, fix an orthonormal eigenbasis of the returned A , with eigenvalues a_i , and write

$$w_i = a_i(1 - a_i) + \tau/2.$$

For each $i < j$, let P_{ij} project onto $\text{span}(e_i, e_j)$ and let C_{ij} be the corresponding compression of E . An input qubit, including its entanglement with other input qubits or a retained reference, can be embedded isometrically into this two-dimensional subspace. The resulting observed channel is exactly $\mathcal{M}_{C_{ij}}$.

Run the common qubit learner of Theorem 4.1 on each compression, using fresh records, accuracy

$$(5.20) \quad \varepsilon = c_* \delta / d,$$

and failure allowance $\eta/(2s)$. The constant $c_* > 0$ will be chosen below. This returns legal 2×2 effects $\hat{C}_{ij}$. Every guarantee is conditional on the calibration record and is uniform over the resulting compression. A union bound therefore shows that, with total failure at most η , calibration succeeds and

$$(5.21) \quad d_N(\mathcal{M}_{C_{ij}}, \mathcal{M}_{\hat{C}_{ij}}) \leq \varepsilon \quad \text{for every } i < j.$$

We record explicitly how an operational error controls an endpoint form. Theorem 2.1 in dimension two and Lemma 2.5 imply that there are absolute $a_0, A_0 > 0$ such that, for legal $C, D \in \mathfrak{E}_2$,

$$(5.22) \quad d_N(\mathcal{M}_C, \mathcal{M}_D) \leq u \leq a_0 \implies g_{N,C}(D - C) \leq A_0 u.$$

Indeed, for $u < 1/16384$ the lower metric bound forces $Q_N(C, D) \leq 16384u$; decreasing a_0 makes this at most $1/4$, where the local comparison applies. Choose c_* small enough that $\varepsilon \leq a_0$ and lies within the qubit learner's accuracy range.

For a Hermitian matrix X on $\text{span}(e_i, e_j)$ put

$$\gamma_{ij}(X)^2 = N \left(\frac{|X_{ii}|^2}{2w_i} + \frac{|X_{jj}|^2}{2w_j} + \frac{2|X_{ij}|^2}{w_i + w_j} \right).$$

Equation (5.17), applied to left plus right multiplication and then inverted, gives

$$\gamma_{ij}(X) \leq \sqrt{14} g_{N,C_{ij}}(X).$$

Hence (5.21) and (5.22) imply

$$(5.23) \quad \gamma_{ij}(\hat{C}_{ij} - C_{ij}) \leq \sqrt{14} A_0 \varepsilon.$$

Assemble a Hermitian matrix B by taking each off-diagonal entry from its corresponding estimate $\hat{C}_{ij}$. For each diagonal entry choose once and for all one incident pair and use that pair's estimate. No consistency among different estimates of the same diagonal entry is assumed. In the diagonal coordinates of A , the form $g_{N,A}$ is the sum of the nonnegative coordinate terms displayed above. Each coordinate used in $B - E$ occurs in at least one of the pair forms, while unused duplicate diagonal terms can be discarded. Therefore

$$(5.24) \quad g_{N,A}(B - E)^2 \leq \sum_{i < j} \gamma_{ij}(\hat{C}_{ij} - C_{ij})^2 \leq 14sA_0^2\varepsilon^2.$$

The assembled matrix need not be legal. Define the estimate by minimizing its distance to the legal effect body in the known positive quadratic norm $g_{N,A}$. The minimum is attained and is unique. An exact minimizer F satisfies

$$g_{N,A}(F - E) \leq 2g_{N,A}(B - E),$$

because E is a feasible competitor. To obtain a finite procedure with a rational legal output, it suffices instead to choose $F \in \mathfrak{E}_d$ with rational entries such that

$$(5.25) \quad g_{N,A}(F - B) \leq \inf_{G \in \mathfrak{E}_d} g_{N,A}(G - B) + \varepsilon.$$

This changes the bound only to

$$(5.26) \quad g_{N,A}(F - E) \leq 2\sqrt{14s}A_0\varepsilon + \varepsilon \leq C_0d\varepsilon$$

for an absolute C_0 .

For completeness, (5.25) is an effective finite selection, not a query to the unknown effect. Since $W_A \succeq \tau I/2$, $g_{N,A}(X) \leq N\|X\|_{\text{HS}}$. A finite rational grid covering $\mathfrak{E}_d$ in Hilbert–Schmidt norm to $\varepsilon/(4N)$ therefore covers it in $g_{N,A}$ to $\varepsilon/4$. Such a legal grid is obtained by rounding after the inward move $G \mapsto (1 - 2r)G + rI$: entrywise rounding with Hilbert–Schmidt error at most $r/2$ preserves positivity of both outcomes, and the total rounding displacement is at most $C_d r$.

Concretely, choose positive rational $r \leq \varepsilon/[4N(\sqrt{d}+1)]$ and a rational entry mesh at most $r/(2d)$. The inward displacement is at most $r\sqrt{d}$, and the rounding displacement is at most $r/2$, so their sum is at most $\varepsilon/(4N)$. The grid and its rational entries are defined in the fixed public input basis, independently of the random eigenbasis used for calibration. Enumerate this bounded grid, retain legal matrices using exact positivity tests, and evaluate its finitely many objective values to tolerance $\varepsilon/4$. Selecting the smallest computed value proves (5.25) with slack. The calibration matrices are algebraic, and the qubit subroutine's outputs are finite computable expressions in its record, so these evaluations require no oracle for the unknown effect. Only the observed matrices, the public parameters, and finite classical calculations are used. Arithmetic and state preparation are not included in the device-call bound.

Finally, (5.16) gives

$$g_{N,E}(F-E) \leq \sqrt{12} g_{N,A}(F-E) \leq \sqrt{12} C_0 d \varepsilon.$$

Choose the absolute c_* in (5.20) so that this is at most $\delta/(8\sqrt{2})$, and reduce δ_0 if necessary so that it is also at most $1/4$. The endpoint-to-midpoint comparison in Lemma 2.5 and the upper bound in Theorem 2.1 then give

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 8Q_N(E, F) \leq 8\sqrt{2} g_{N,E}(F-E) \leq \delta.$$

This proves the risk bound.

The calibration costs at most $Cd^4N \log(d/\eta)$ calls. There are $s \leq d^2/2$ qubit learners, each with accuracy $c_*\delta/d$ and confidence $\eta/(2s)$. Their combined call count is at most $Cd^4N\delta^{-2} \log(d/\eta)$ by Theorem 4.1. All budgets hold on every record, all compressed entangled blocks have length at most N , and postprocessing uses no additional calls.

The confidence and resource allocation can be recorded as follows; the constants in the call bounds are absolute.

Stage	Failure allowance	Calls on every record
Endpoint calibration	$\eta/2$	$Cd^4N \log(d/\eta)$
Each of $s = \binom{d}{2}$ pair learners	$\eta/(2s)$	$CN(d/\delta)^2 \log(d/\eta)$
Legal rational selection	0	0

The conditional guarantees of the pair learners allow their allowances to be added after calibration, despite the dependence of the selected basis on its record.

For the lower bound take

$$E_0 = \frac{1}{2}I_d, \quad E_1 = (\frac{1}{2} + \Delta)I_d, \quad \Delta = 256\delta/\sqrt{N}.$$

Choose $\delta_0 \leq 1/4096$. The Bernoulli separation in (4.36) is unchanged by the input dimension, and the two effects are separated by more than 2δ under d_N . Every use of either scalar device produces a fresh Bernoulli bit independent of the input. Its conditional reference state is the same reduced state multiplied by that scalar outcome probability. Thus an arbitrary adaptive training record with at most M calls, padded if necessary, is a common processing of M independent Bernoulli bits and parameter-independent resources. Its relative entropy is at most $3M\Delta^2$ by (4.37). The two disjoint success balls turn the estimated effect into a test with both errors at most η . The same fidelity inequality used in the proof of Theorem 4.1 gives

$$2\eta \geq \frac{1}{2}e^{-3M\Delta^2}, \quad M \geq \frac{N}{3 \cdot 256^2\delta^2} \log \frac{1}{4\eta}.$$

For $\eta \leq 1/8$ this is (5.3). □

The common learner now has the same dimension and boundary scope as the metric and entropy theorems. Its calibration is a sequence of product experiments, while the compression learners provide the enhanced angular resolution when an estimated two-dimensional subspace is close

to opposite support endpoints. No pair of candidate hypotheses is supplied to either part of the procedure.

6. A FINITE RATIONAL CODE IN EVERY DIMENSION

The volume argument can be made into a single deterministic code whose decoder reconstructs its dictionary from the public parameters. The construction is finite and exact. Its purpose is to settle the constructive description problem; the enumeration below can be very large.

Theorem 6.1 (Deterministic rational matrix code). *Let $d, N \geq 1$ be integers and let $0 < \delta \leq 1$ be rational. There is a deterministic finite procedure, using rational arithmetic, that constructs an ordered list*

$$\mathcal{C}_{d,N,\delta} = (C_0, \dots, C_{M-1}) \subset \mathfrak{E}_d$$

of Gaussian-rational effects with the following properties. Every $E \in \mathfrak{E}_d$ has an assigned index $j(E)$ satisfying

$$(6.1) \quad d_N(\mathcal{M}_E, \mathcal{M}_{C_{j(E)}}) \leq \frac{11}{16}\delta.$$

For fixed d ,

$$(6.2) \quad M \leq C_d N^{d^2/2} [\log(N+2)]^{\lfloor d/2 \rfloor} \delta^{-d^2},$$

uniformly in N and δ . The encoder on a supplied Gaussian-rational effect and the decoder on its index both terminate using exact arithmetic. They require no matrix-dependent advice or preassigned list of centres. With (d, N, δ) public, the payload length is $\lceil \log_2 M \rceil$, and for $0 < \delta \leq \delta_d$ it is

$$(6.3) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

Here δ_d may be the small-error constant in Theorem 3.1. The procedure makes no assertion of polynomial running time in the payload length.

Proof. Set

$$(6.4) \quad K = \left\lceil \frac{32dN}{\delta} \right\rceil, \quad s = \frac{2d}{K}, \quad \eta = \frac{\delta}{16}.$$

Fix the order of independent real coordinates to be the d diagonal entries, followed by the real and imaginary parts of G_{ij} for $i < j$ in lexicographic order. Let $\mathcal{G}_{d,K}$ consist of the legal Hermitian matrices whose diagonal coordinates belong to $\{0, 1/K, \dots, 1\}$ and whose remaining coordinates belong to $\{-1, -1 + 1/K, \dots, 1\}$. List these matrices in the induced lexicographic order. This list is finite, and it contains zero.

Traverse the list once. Retain its first element. Retain a subsequent element G exactly when

$$(6.5) \quad Q_N(G, C)^2 > \eta^2 \quad \text{for every previously retained element } C.$$

The retained elements, in their order of appearance, form $\mathcal{C}_{d,N,\delta}$. Every grid element either is retained or has a retained predecessor C with $Q_N(G, C) \leq \eta$. No triangle inequality for Q_N is used.

We first verify that the whole procedure is exact. Legality of a Gaussian-rational Hermitian matrix is decidable by exact Schur elimination, applied to G and $I - G$. A negative pivot excludes positive semidefiniteness. A zero pivot in a positive semidefinite matrix forces the corresponding row and column of the remaining matrix to vanish; otherwise the test rejects. A positive pivot reduces the question to the exact Schur complement. These rules terminate after at most d pivots,

including on singular strata. For a legal rational pair, its midpoint A and variance $A(I - A)$ are Gaussian rational. The equation

$$A(I - A)X + XA(I - A) + N^{-1}X = G - C$$

has an invertible Gaussian-rational coefficient matrix. Solving it gives $Q_N(G, C)^2 = N \operatorname{tr}((G - C)X) \in \mathbb{Q}_{\geq 0}$. Consequently (6.5) is an exact rational comparison. It requires neither spectral coordinates nor a numerical eigenvalue threshold.

To encode E , first form the inward displacement

$$E_s = (1 - 2s)E + sI.$$

Since $s \leq 1/16$, its spectrum lies in $[s, 1 - s]$. Round each independent real coordinate of E_s to its nearest multiple of $1/K$, choosing the larger multiple at a tie, and call the resulting Hermitian matrix $G(E)$. A diagonal entry changes by at most $1/(2K)$ and the modulus of an off-diagonal entry changes by at most $\sqrt{2}/(2K)$. Thus

$$\|G(E) - E_s\|_{\text{op}} \leq \frac{d}{K}.$$

In particular,

$$(6.6) \quad \frac{d}{K}I \preceq G(E) \preceq \left(1 - \frac{d}{K}\right)I.$$

Every coordinate lies in the prescribed grid range, and hence $G(E) \in \mathcal{G}_{d,K}$. Moreover,

$$(6.7) \quad \|E - G(E)\|_{\text{op}} \leq s + \frac{d}{K} = \frac{3d}{K} \leq \frac{3\delta}{32N}.$$

Choose the first retained centre with $Q_N(G(E), C_j)^2 \leq \eta^2$. Such a centre exists by the completed greedy traversal. The hybrid inequality and Theorem 2.1 yield

$$\begin{aligned} d_N(\mathcal{M}_E, \mathcal{M}_{C_j}) &\leq 2N\|E - G(E)\|_{\text{op}} + 8Q_N(G(E), C_j) \\ &\leq \frac{3\delta}{16} + \frac{\delta}{2} = \frac{11\delta}{16}. \end{aligned}$$

This proves (6.1). If E is Gaussian rational, the prescribed rounding and every subsequent operation are exact finite rational calculations.

For two distinct retained centres, Theorem 2.1 gives

$$(6.8) \quad d_N(\mathcal{M}_C, \mathcal{M}_{C'}) \geq \frac{1}{8192d} \min\{1, Q_N(C, C')\} > \frac{\delta}{131072d},$$

where $\delta \leq 1$ was used. Let a_d, r_d be the constants of Lemma 3.2, and put

$$b_d = \min\left\{r_d, \frac{1}{393216d}\right\}, \quad r = b_d\delta.$$

The operational balls of radius r about distinct retained centres are disjoint, by (6.8) and the triangle inequality for d_N . Their measures are each at least $a_d(b_d\delta)^{d^2}$. Hence

$$Ma_d(b_d\delta)^{d^2} \leq \mu_N(\mathfrak{E}_d).$$

Proposition 3.3 proves (6.2).

Both parties reconstruct this same finite list from (d, N, δ) . The encoder transmits the binary representation of j in $\lceil \log_2 M \rceil$ bits. The decoder reconstructs the list and returns C_j for $j < M$; every unused word of the same length is assigned the legal effect $I/2$. The codeword contains no additional centre coordinates. Equation (6.2) gives the upper bound in (6.3); the covering lower bound of Theorem 3.1 gives the matching lower bound, since this list is a radius- δ cover by (6.1). The description of the fixed reconstruction algorithm contributes only a constant. As usual,

if the parameters are not public, their explicitly chosen header representation must be charged separately. $\square$

Real input and finite precision. The covering statement concerns the entire real effect body. To interpret its encoder computationally on a real input, a representation of that input must be supplied. A certified rational approximation of each independent real coordinate to error at most $1/(8K)$ is sufficient. Apply the inward displacement to this approximation and then use the same exact nearest-grid rounding. Relative to the true E_s , each independent coordinate then has error at most $5/(8K)$; each complex off-diagonal error is at most $5\sqrt{2}/(8K) < 1/K$. Therefore (6.6)–(6.7) and the full error bound remain valid. The exact supplied-matrix implementation takes Gaussian-rational input; no physical-device learning operation is implicit in that interface. Computational scope. The ambient grid contains at most

$$(6.9) \quad (K+1)^d(2K+1)^{d(d-1)}$$

candidate coordinate tuples. The literal algorithm performs exact legality tests and, for each legal tuple, comparisons with the retained list. Retaining the list uses $O(Md^2)$ rational entries, and the decoder repeats the same construction. These are finite bounds, but their dependence on the requested horizon and precision prevents an efficiency assertion in the payload length. The exact kernel `matrix_codec.py` implements this specified enumeration, greedy test, rounding, and replay. A resource cutoff is reported as an incomplete construction and never as a codebook. Regression checks of small complete cases and of matrix kernels certify those computations; they do not assert that a full large-dimensional dictionary has been enumerated.

Corollary 6.2 (Learning into an optimal-order description). *For every fixed $d \geq 1$, every integer $N \geq 1$, every rational $0 < \delta \leq \min\{1, \delta_0, \delta_d\}$, and every $0 < \eta \leq 1/8$, there is one learner, common to all $E \in \mathfrak{E}_d$, which returns a fixed-length word whose deterministic public decoder produces a legal effect C with*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

The training budget on every record is at most $Cd^4N\delta^{-2}\log(d/\eta)$, and each entangled block has at most N calls. The word has length

$$(6.10) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For fixed d , both the training order $N\delta^{-2}\log(1/\eta)$ and the payload order in (6.10) are optimal. Payload optimality is among fixed-length words with a fixed public decoder into legal memoryless ordered binary effects. Encoding after learning uses no additional calls to the unknown device.

Proof. Apply Theorem 5.1 with accuracy $\delta/2$. It returns a legal effect F with rational entries in the public input basis. Encode F using Theorem 6.1 with accuracy $\delta/2$. On the learning success event, the triangle inequality gives

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \frac{\delta}{2} + \frac{11\delta}{32} = \frac{27\delta}{32} < \delta.$$

The second operation is a deterministic computation on the recorded estimate. It uses no device calls and does not increase the failure probability. Replacing δ by $\delta/2$ changes the training constant and adds only $O_d(1)$ to the payload length.

The scalar subfamily converse in Theorem 5.1 applies to this learning task as well. For the payload converse, a B -bit word and fixed decoder have at most 2^B possible decoded effects. Every E has a strictly positive success probability, so at least one of these effects lies within distance δ of E . The decoded family is therefore a radius- δ cover of $\mathfrak{E}_d$. The lower bound in Theorem 3.1 gives (6.10) with the opposite inequality up to $O_d(1)$. $\square$

7. THE COST OF LIMITING INDEPENDENT QUANTUM BLOCKS

The training law changes when the number of calls in one independent quantum experiment is prescribed. The relevant restriction concerns access to the next queried input. Quantum systems retained from completed experiments may still be stored and measured together. This distinction allows a direct comparison with tomography based on independently prepared probe–reference pairs.

Definition 7.1 (Independent blocks with retained outputs). A training protocol has independent blocks of length at most b if it has the following form. At a block boundary it may apply an arbitrary common quantum instrument to all its retained systems. The resulting classical history h , including its classical random choices and the number of calls already charged, determines whether to stop, or else a length $1 \leq t(h) \leq b$ and a fresh $t(h)$ -use tester. Conditional on h , this tester starts in a parameter-independent state tensor-separated from all retained systems. It may use arbitrary references, entangled inputs, and adaptive control within its $t(h)$ calls.

From its preparation until its last call, the entire fresh tester, including its reference and unused inputs, is isolated from the retained systems. Those systems are available again at the next block boundary, when the new outputs may be joined to them and processed collectively. Thus classical feedback may select the next tester, while no retained quantum register can be coupled into a block that still has calls to make. A block length is declared before that block and charged in full; an internal early exit is padded within its declared length. The sum of declared lengths is at most M on every record. Stopping at block boundaries may depend on the full classical history. The final estimate may use an arbitrary joint measurement of all retained systems. Classical record spaces are standard Borel and quantum registers are separable.

The retained quantum systems belong to the tester; the unknown measurement still consumes its input and returns only its ordered binary classical label. The definition permits an unbounded amount of retained quantum storage and classical computation. It also permits all probes to be prepared in advance when their states factor into the prescribed independent blocks: their preparation can be deferred without changing the experiment. In particular, independent one-call Choi preparations followed by a collective measurement belong to the case $b = 1$. The distinction between classical input control and quantum memory on the output side has an antecedent in the channel-discrimination model of [9, Section II.2]. The familiar quadratic information gain within a block and additive gain across independent blocks also occur in the metrological bounds of [33, 34]. We give a finite-budget fidelity argument because a local information bound alone does not supply the confidence dependence needed here.

Write $M_b^*(d, N, \delta, \eta)$ for the least every-record call budget of a protocol in Definition 7.1 whose legal estimate satisfies

$$(7.1) \quad \sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta.$$

The future distance d_N retains its unrestricted adaptive definition and its unhalved trace-norm normalization. The restriction on b applies only to training.

Theorem 7.2 (Sharp tradeoff for independent training blocks). *There are absolute constants $c, C, \delta_* > 0$ such that, for every $d \geq 2$, $N \geq 1$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$, and $0 < \eta \leq 1/8$,*

$$(7.2) \quad c \frac{N^2}{b\delta^2} \log \frac{1}{\eta} \leq M_b^*(d, N, \delta, \eta) \leq C d^4 \frac{N^2}{b\delta^2} \log \frac{d}{\eta}.$$

The upper bound is attained by one procedure common to the entire closed effect body. The lower bound allows classical feedback, arbitrary quantum storage and collective processing at block

boundaries, and every-record bounded stopping. Consequently, for each fixed $d \geq 2$,

$$(7.3) \quad M_b^*(d, N, \delta, \eta) = \Theta_d \left(\frac{N^2}{b\delta^2} \log \frac{1}{\eta} \right).$$

In dimension $d = 1$ the corresponding order is $\Theta(N\delta^{-2} \log(1/\eta))$, independently of b .

The endpoints $b = 1$ and $b = N$ differ by a factor of order N . The claim is uniform in the block cap, horizon, accuracy, and confidence; the upper factor d^4 is not asserted to be sharp in growing dimension.

Lemma 7.3 (Changing the future horizon). *For all effects E, F and integers $N, b \geq 1$,*

$$(7.4) \quad d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \left\lceil \frac{N}{b} \right\rceil d_b(\mathcal{M}_E, \mathcal{M}_F).$$

Proof. Fix an N -use tester and partition its calls into consecutive groups of at most b . Replace the device in one group at a time. For each replacement the common prefix prepares a reference-assisted input to a tester with at most b calls, including all memory carried from the prefix. The common suffix is a channel on that tester's output and cannot increase trace distance. Each replacement therefore costs at most d_b . Sum these costs and take the supremum over the original tester. Public stopping is included by padding to N calls with discarded dummy experiments. $\square$

Upper bound in Theorem 7.2. Put $q = \lceil N/b \rceil$. Apply Theorem 5.1 with future horizon b , accuracy δ/q , and failure allowance η . Its construction uses fresh independent blocks of length at most b , with classical records between them, so it satisfies Definition 7.1. Lemma 7.3 converts its success event into (7.1). On every record its call count is at most

$$Cd^4bq^2\delta^{-2} \log(d/\eta) \leq 4Cd^4 \frac{N^2}{b\delta^2} \log(d/\eta),$$

because $1 \leq b \leq N$. Choose δ_* no larger than the accuracy constant in Theorem 5.1. $\square$

For positive, possibly subnormalized operators define root fidelity by

$$\mathfrak{f}(A, B) = \|\sqrt{A}\sqrt{B}\|_1.$$

We use its tensor-product identity, its additivity on direct sums, and its monotonicity under a common channel. In particular, for a common instrument $(\mathcal{J}_z)_z$,

$$(7.5) \quad \sum_z \mathfrak{f}(\mathcal{J}_z(A), \mathcal{J}_z(B)) \geq \mathfrak{f}(A, B).$$

These facts hold for subnormalized operators by homogeneity. For states, the Bures angle $\arccos \mathfrak{f}$ is a contractive metric; see [1, 25].

Lemma 7.4 (Fidelity with adaptive choices of fresh blocks). *Fix a two-dimensional subspace and, for $\phi \in \mathbb{R}$, set*

$$(7.6) \quad E_\phi = \frac{1}{2}(I_2 + \cos \phi \sigma_1 + \sin \phi \sigma_2) \oplus \frac{1}{2}I_{d-2},$$

with the second summand absent when $d = 2$. If $b|\theta| \leq 1$, the final states Γ_0, Γ_θ of any protocol in Definition 7.1, applied to E_0, E_θ with every-record budget M , satisfy

$$(7.7) \quad \mathfrak{f}(\Gamma_0, \Gamma_\theta) \geq \exp(-Mb\theta^2/4).$$

Furthermore, if $N|\theta| \leq \pi$, then

$$(7.8) \quad d_N^{\text{na}}(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2 \sin(N|\theta|/2).$$

Proof. For qubits, (7.8) follows from [16, Corollary 1 and Theorem 2]. We give the calculation to specify the normalization and the fidelity estimate needed below.

Let $U_\theta = e^{-i\theta\sigma_3/2} \oplus I_{d-2}$. Then $E_\theta = U_\theta E_0 U_\theta^*$ and $\mathcal{M}_{E_\theta} = \mathcal{M}_{E_0} \circ \text{Ad}_{U_\theta^*}$. If $|\theta| \leq \pi$, every purified input ψ , including an arbitrary reference, satisfies

$$|\langle \psi, (U_\theta^* \otimes I) \psi \rangle| \geq \text{Re} \langle \psi, (U_\theta^* \otimes I) \psi \rangle \geq \cos(|\theta|/2).$$

The last inequality follows from the spectrum of $(U_\theta + U_\theta^*)/2$. Applying the common measurement and discarding a purification can only increase fidelity. Thus one call changes the output Bures angle by at most $|\theta|/2$, uniformly over its input and reference.

For a common adaptive tester, apply the triangle inequality before each query: the angle between the two current states is contracted by the same channel, and changing the channel on the second state adds at most $|\theta|/2$. The interposed controls are common channels and also contract the angle. Induction from the common initial state proves that every t -use tester has output states ρ_0, ρ_θ satisfying

$$(7.9) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq \cos(t|\theta|/2) \quad \text{when } t|\theta| \leq \pi.$$

This argument permits arbitrary adaptive quantum memory inside the tester. Its output includes all internal outcomes before any postselection. For $t \leq b$ and $b|\theta| \leq 1$, $\cos x \geq e^{-x^2}$ for $0 \leq x \leq 1/2$ gives

$$(7.10) \quad \mathfrak{f}(\rho_0, \rho_\theta) \geq e^{-t^2\theta^2/4} \geq e^{-at}, \quad a = b\theta^2/4.$$

We next retain the whole classical history, including all branch lengths and stopping decisions. At a boundary write the two classical-quantum states as direct sums of subnormalized retained states A_h, B_h . Their traces include the respective probabilities of h . Let $m(h)$ be the charged calls in that history, and consider

$$(7.11) \quad \Psi = \sum_h e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h).$$

Initially $\Psi = e^{-aM}$, since the initial resources are the same. A common boundary instrument preserves $m(h)$ along each of its children, so (7.5) shows that it cannot decrease Ψ . The history is kept until the end; in particular, records with different charged call counts are never merged in this calculation.

At a continuing history h , the prescribed fresh tester has the same specification under both hypotheses. Conditional on h , its output states ρ_0^h, ρ_θ^h are normalized and are tensor-separated from the old retained state. Immediately after this block the contribution of this history is therefore

$$\begin{aligned} & e^{-a(M-m(h)-t(h))} \mathfrak{f}(A_h \otimes \rho_0^h, B_h \otimes \rho_\theta^h) \\ & \geq e^{-a(M-m(h)-t(h))} e^{-at(h)} \mathfrak{f}(A_h, B_h) = e^{-a(M-m(h))} \mathfrak{f}(A_h, B_h). \end{aligned}$$

The subsequent joint instrument is covered by (7.5). A stopped history is carried unchanged. Since each continuing block charges at least one call, all histories are terminal after at most M such generations. At that time all weights in (7.11) are at most one. Additivity of root fidelity on the terminal histories therefore gives

$$\mathfrak{f}\left(\bigoplus_h A_h, \bigoplus_h B_h\right) = \sum_h \mathfrak{f}(A_h, B_h) \geq \Psi \geq e^{-aM}.$$

Any final collective processing or discarding of history again increases fidelity. This proves (7.7) without asserting independence of the unconditional records or imposing a common support condition on their laws.

For general measurable classical outcomes, choose a measure dominating the two history laws and interpret A_h, B_h as their operator-valued densities. Direct-sum additivity becomes the corresponding integral identity, and the same proof uses integrals in (7.5) and (7.11). These identities follow equivalently by finite coarse-graining of the classical record and refinement. The charged call count has only the values $0, \dots, M$ and is retained in every such coarse-graining.

Finally, use the first two coordinates for a parallel N -input state

$$\frac{|0\rangle^{\otimes N} + e^{i\alpha}|1\rangle^{\otimes N}}{\sqrt{2}}, \quad \alpha = N\theta/2 + \pi/2.$$

The parity of the ordered output signs has mean $\cos(N\phi - \alpha)$ under E_ϕ , hence means $-\sin(N\theta/2)$ and $\sin(N\theta/2)$ under the two hypotheses. Its unhalved distance is $2|\sin(N\theta/2)|$. For $N|\theta| \leq \pi$, apply (7.9) with $t = N$ to any future adaptive tester and use $\|\rho - \sigma\|_1 \leq 2\sqrt{1 - \mathbf{f}(\rho, \sigma)^2}$. The resulting upper bound is $2\sin(N|\theta|/2)$, equal to the parallel witness. This proves (7.8). $\square$

Lower bound in Theorem 7.2. Reduce δ_* further so that $\delta_* \leq 1/4$, and set $\theta = 4\delta/N$ in (7.6). Then $b|\theta| \leq 1$, while (7.8) gives

$$d_N(\mathcal{M}_{E_0}, \mathcal{M}_{E_\theta}) = 2\sin(2\delta) > 2\delta.$$

The two radius- δ success balls are disjoint. Thus any estimate satisfying (7.1) determines a test with both error probabilities α, β at most η : choose the first hypothesis precisely when the estimate lies in its radius- δ ball. The classical root fidelity of that test is

$$\sqrt{(1-\alpha)\beta} + \sqrt{\alpha(1-\beta)} \leq 2\sqrt{\eta}.$$

Fidelity is nondecreasing under the map from the training output to this test. Lemma 7.4 consequently implies

$$e^{-Mb\theta^2/4} \leq 2\sqrt{\eta}, \quad M \geq \frac{N^2}{8b\delta^2} \log \frac{1}{4\eta} \geq \frac{N^2}{24b\delta^2} \log \frac{1}{\eta},$$

where the last inequality uses $\eta \leq 1/8$. This proves the asserted lower bound in every $d \geq 2$.

For $d = 1$ every effect is scalar. Bernoulli endpoint estimation in Lemma 4.2 gives the upper order with independent one-call samples. The scalar converse in Theorem 5.1 gives the matching lower order even with unrestricted adaptive quantum training. Thus increasing b has no effect in dimension one. $\square$

Corollary 7.5 (The limitation of independent one-call acquisition). *Fix $d \geq 2$. Every learner that acquires its data using fresh one-call probes and fresh references, with classical feedback and arbitrary subsequent collective processing as in Definition 7.1, requires $\Omega_d(N^2\delta^{-2}\log(1/\eta))$ calls for (7.1). The complete closed effect body admits the same risk with $O_d(N\delta^{-2}\log(1/\eta))$ calls when independent blocks of length up to N are permitted.*

Proof. Use $b = 1$ and $b = N$ in Theorem 7.2. $\square$

This conclusion applies to the acquisition interfaces of [5, 29]. The channel learner in [5, Section III.2 and Algorithm I.2] prepares independent Choi copies before their collective processing. The measurement tomography procedure in [29, Section 3] queries separately prepared known inputs. Both are contained in $b = 1$ above. Any estimator retaining these interfaces and required to satisfy the present worst-case future- N loss therefore incurs the quadratic horizon order, including when its final estimator is collective. The block lower bound settles this interface comparison directly; it does not rely on substituting δ/N into a sufficient one-use accuracy guarantee.

Corollary 7.6 (Learning with bounded blocks into an optimal-order word). *For fixed $d \geq 2$, $1 \leq b \leq N$, rational $0 < \delta \leq \min\{1, \delta_*, \delta_d\}$, and $0 < \eta \leq 1/8$, there is a common learner in*

Definition 7.1 that returns a fixed-length word whose deterministic public decoder produces a legal effect C with

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E \{d_N(\mathcal{M}_E, \mathcal{M}_C) > \delta\} \leq \eta.$$

Its every-record training budget is at most $Cd^4N^2(b\delta^2)^{-1} \log(d/\eta)$ and its payload has length

$$(7.12) \quad \frac{d^2}{2} \log_2 N + \lfloor d/2 \rfloor \log_2 \log(N+2) + d^2 \log_2(1/\delta) + O_d(1).$$

For each fixed d , both the training order in this block model and the payload order among fixed public decoders into legal effects are optimal. Encoding uses no additional device calls.

Proof. Run the learner of Theorem 7.2 at accuracy $\delta/2$. The upper construction inherits the legal rational output of Theorem 5.1. Apply the exact codec of Theorem 6.1 at accuracy $\delta/2$. On the learning success event its decoded effect obeys

$$d_N(\mathcal{M}_E, \mathcal{M}_C) \leq \delta/2 + 11\delta/32 = 27\delta/32 < \delta.$$

The codec is a deterministic computation on the recorded rational estimate. Theorem 6.1 gives (7.12). The training converse is Theorem 7.2. For the payload converse, positive success probability for every E forces the decoded family to cover $\mathfrak{E}_d$ by radius- δ balls; apply Theorem 3.1. As before, reconstruction time, workspace, and state-preparation complexity are separate from the two resource counts. $\square$

8. DIMENSION AND ACCURACY IN THE UNIFORMLY NOISY INTERIOR

The block converse concerns projective boundary effects. A different obstruction is present in the full-dimensional interior: an effect close to a fair coin supplies little information per call, even when the learner retains coherent quantum memory. We quantify this obstruction before analysing a one-use estimator under the future loss. Entropies and relative entropies in this section use natural logarithms. The quantum information identities used below are in [1]; the particular conditional bound is proved here.

Lemma 8.1 (Information supplied by a weak binary measurement). *Let X have a finite alphabet. Conditional on $X = x$, an input register A and a retained register R have an arbitrary joint state ρ_{AR}^x . Apply the consuming binary measurement with effect $E_x = I/2 + H_x$, where $\|H_x\|_{\text{op}} \leq r < 1/2$, to A , and call its classical outcome Y . Then*

$$(8.1) \quad I(X : YR)_{\text{out}} \leq I(X : AR)_{\text{in}} + \log(1 + 4r^2) \leq I(X : AR)_{\text{in}} + 4r^2.$$

The input states need not be independent of x or of one another.

Proof. Fix x , abbreviate $\rho = \rho_R^x$, and let Δ be the partial trace of $(H_x \otimes I)\rho_{AR}^x$ over A . Positivity of $rI \pm H_x$ gives

$$-r\rho \preceq \Delta \preceq r\rho, \quad \omega_{YR}^x = |1\rangle\langle 1| \otimes (\rho/2 + \Delta) + |0\rangle\langle 0| \otimes (\rho/2 - \Delta).$$

On the support of ρ , write $\Delta = \rho^{1/2}K\rho^{1/2}$ with $\|K\|_{\text{op}} \leq r$. Set $\sigma = (I_Y/2) \otimes \rho$. The support inclusion $\text{supp } \omega^x \subseteq \text{supp } \sigma$ holds, and direct multiplication gives

$$\begin{aligned} \text{tr}[(\omega^x)^2 \sigma^{-1}] &= 1 + 4 \text{tr}(\Delta^2 \rho^{-1}) \\ &= 1 + 4 \text{tr}(\rho K^2) \leq 1 + 4r^2. \end{aligned}$$

All inverses in this calculation are support inverses. For density matrices a, b with the same support inclusion, $D(a||b) \leq \log \text{tr}(a^2 b^{-1})$ follows directly from scalar Jensen: in eigenbases of a and b , use the probability weights $a_i |\langle u_i, v_j \rangle|^2$ and apply the concavity of $\log$ to a_i/b_j . Consequently

$$D(\omega_{YR}^x || (I_Y/2) \otimes \rho_R^x) \leq \log(1 + 4r^2).$$

Average over x . Subtracting $D(\omega_{YR} \| (I_Y/2) \otimes \omega_R)$ from the average on the left gives exactly $I(X : Y | R)_{\text{out}}$. Relative entropy is nonnegative. The retained marginal is unchanged by the call, so data processing gives $I(X : R)_{\text{out}} = I(X : R)_{\text{in}} \leq I(X : AR)_{\text{in}}$. Adding these two inequalities proves the claim. For a separable retained register, apply finite-rank compression to R and pass to the limit. The bound is uniform, and mutual information with the finite classical register X is continuous under trace-norm convergence. $\square$

The same calculation applies to any finite classical-output measurement uniformly close to a scalar one. More precisely, if $p_y > 0$, $\sum_y p_y = 1$, and $(1-r)p_y I \preceq E_{x,y} \preceq (1+r)p_y I$, its per-call information increment is at most $\log(1+r^2)$. Indeed, the corresponding collision expression is $1 + \sum_y p_y^{-1} \text{tr}(\Delta_y^2 \rho^{-1}) \leq 1 + r^2$; the linear terms cancel because $\sum_y \Delta_y = 0$. We use only the binary case in what follows.

Theorem 8.2 (A joint dimension and future-loss lower bound). *There is an absolute $c > 0$ such that, for every $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 < \eta \leq 1/8$, a common learner with*

$$\sup_{I_d/4 \preceq E \preceq 3I_d/4} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\widehat{E}}) > \delta\} \leq \eta$$

requires at least

$$(8.2) \quad M \geq cd^2 N \delta^{-2}$$

calls in its worst record. The learner may retain arbitrary quantum memory, entangle its training calls, use coherent adaptive controls, and stop publicly. Its estimate may be any legal binary effect; it need not lie in the interior subfamily.

Proof. Put $r = 2048\delta/\sqrt{N} \leq 1/4$. The real vector space of Hermitian matrices has dimension $p = d^2$. A maximal finite $r/4$ -separated set $\{H_x\}_{x=1}^L$ in its operator-norm ball of radius r covers that ball by translates of the radius- $r/4$ ball. Comparing Lebesgue volumes gives $L \geq 4^p$. Such a finite maximal set exists by compactness. Set $E_x = I/2 + H_x$, so every E_x lies between $I/4$ and $3I/4$.

For $x \neq z$, take a unit eigenvector of $H_x - H_z$ achieving its operator norm and repeat this input N times. The two outcome laws are Bernoulli products with parameters $s, t \in [1/4, 3/4]$ and $|s - t| \geq r/4$. Since

$$T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + N^{-1}}} \geq \frac{1}{2} \sqrt{N} |s - t| \geq 256\delta,$$

Lemma 11.2 and $256\delta \leq 1$ give

$$(8.3) \quad d_N(\mathcal{M}_{E_x}, \mathcal{M}_{E_z}) \geq 4\delta.$$

The witness may depend on the pair; it is used only to establish separation, not as a common learning rule.

Choose X uniformly from this packing. The learner's initial resources are independent of X . Common intervening channels cannot increase their mutual information with X . Lemma 8.1 applies to each call even when its input and memory already depend on X through previous calls. Pad a stopped experiment by ignored dummy calls to its worst-record budget M . Its final quantum and classical registers therefore have mutual information at most $4Mr^2$ with X .

From the returned effect choose a nearest member of the finite packing, with a fixed tie rule. This is a statistical testing reduction; no computation of an adaptive optimum is required from the learner. The selection is measurable because the hybrid bound makes d_N continuous in each effect. The disjointness in (8.3) makes the chosen member correct whenever the learning loss is at most δ . Fano's inequality and data processing yield

$$4Mr^2 \geq (1 - \eta) \log L - \log 2 \geq \frac{3}{4} d^2 \log 2.$$

Substituting r proves (8.2). $\square$

Corollary 8.3 (Joint dimension and accuracy in one-use learning). *At failure probability at most $1/8$, the minimax number of calls needed to learn an arbitrary ordered binary effect to operator-norm error ϵ is*

$$\Theta(d^2\epsilon^{-2}), \quad d \geq 1, \quad 0 < \epsilon \leq 2^{-14},$$

with absolute constants. The lower bound permits arbitrary coherent adaptive training. The upper bound uses independent Choi probes and collective processing of their outputs.

Proof. For binary measurements $d_1(\mathcal{M}_E, \mathcal{M}_F) = 2\|E - F\|_{\text{op}}$, including retained references; see [5, Lemmas III.7–III.8]. Theorem 8.2 with $N = 1$ and $\delta = 2\epsilon$ gives the lower bound. The upper bound is [5, Corollary III.9], at $\eta = 1/8$. Its stated condition $\eta > 4\exp(-4d^2)$ holds for every $d \geq 1$. Thus the lower argument couples dimension and accuracy, while the upper algorithm is the existing one-use estimator. $\square$

Lemma 8.4 (Dimension-free interior comparison). *If $I_d/8 \preceq E, F \preceq 7I_d/8$, then*

$$(8.4) \quad \frac{1}{128} \min\{1, \sqrt{N}\|E - F\|_{\text{op}}\} \leq d_N^{\text{na}}(\mathcal{M}_E, \mathcal{M}_F) \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 4\sqrt{N}\|E - F\|_{\text{op}}\}.$$

The constants are independent of d and N .

Proof. The same repeated eigenvector input and Lemma 11.2 give the lower bound. For the upper bound take $G(t) = E + t(F - E)$ and $H = F - E$. Along this path $S(t) = \sqrt{G(t)(I - G(t))} \succeq I/4$. The unregularized Sylvester equation $SK + KS = H$ has solution

$$K = \int_0^\infty e^{-uS} H e^{-uS} du, \quad \|K\|_{\text{op}} \leq 2\|H\|_{\text{op}}.$$

The horizontal calculation (2.12) bounds the derivative of each fixed common tester's output by $2\sqrt{N}\|K\|_{\text{op}}$. Integrate on $[0, 1]$ and then take the supremum over testers. This proves the upper bound using the same horizontal Kraus and adaptive path methods as Lemma 2.2; see also [22, 23, 24, 26]. $\square$

Theorem 8.5 (Sharp learning on a fixed full-dimensional interior). *For the family $I_d/4 \preceq E \preceq 3I_d/4$, the minimax training cost for future- N loss δ and failure at most $1/8$ is*

$$(8.5) \quad \Theta(d^2 N \delta^{-2}), \quad d, N \geq 1, \quad 0 < \delta \leq 2^{-13}.$$

The constants are absolute. The upper procedure uses fresh one-call probes and permits collective final processing. The lower bound permits arbitrary coherent adaptive training. Thus the same order holds for every fresh-block cap $1 \leq b \leq N$.

Proof. Theorem 8.2 supplies the lower bound on this very family. For the upper bound apply the binary estimator of [5, Corollary III.9] at operator-norm accuracy $\epsilon = \delta/(8\sqrt{N})$ and failure $1/8$. It uses $O(d^2 N \delta^{-2})$ calls and returns a legal effect F . On its successful event $\|E - F\|_{\text{op}} \leq \epsilon \leq 1/8$, so both endpoints lie in $[I/8, 7I/8]$. Lemma 8.4 then gives $d_N(E, F) \leq \delta/2$. Independent input/reference pairs prepare the Choi copies; all coherent processing acts on completed outputs. This is a fresh one-call experiment in Definition 7.1. $\square$

This establishes a stronger future-loss analysis of an existing tomography estimator on a fixed interior. On the complete effect body, Corollary 7.5 supplies the different quadratic-horizon obstruction for the same acquisition class. The interior comparison is dimension-free, and the interior learning order is joint in d, N, δ ; neither assertion changes the dimension dependence or boundary range of the global comparison and covering theorems. In particular, for $d \geq 2$,

$1 \leq b \leq N$, $0 < \delta \leq \min\{2^{-13}, \delta_*\}$ and $0 < \eta \leq 1/8$, the full-body fresh-block learner has the two simultaneous lower bounds

$$M \geq c\delta^{-2} \left(d^2 N + \frac{N^2}{b} \log(1/\eta) \right),$$

after decreasing an absolute c , while its constructive upper bound remains $Cd^4(N^2/b)\delta^{-2}\log(d/\eta)$. The displayed d^4 upper factor is not matched by this lower bound.

9. JOINT DIMENSION BOUNDS FOR OPERATIONAL DESCRIPTIONS

The dimension-free comparison of Lemma 8.4 allows a description theorem whose constants are uniform in the matrix dimension. This complements, rather than replaces, the closed-body law of Theorem 3.1. Put

$$\mathfrak{I}_d = \{E = E^* : I_d/4 \preceq E \preceq 3I_d/4\}, \quad p = d^2.$$

Throughout this section, centres are legal ordered binary effects and the operational distance is the unhalved distance d_N of the same consuming, classical-output interface. In particular, the future tester may retain a reference, use coherent adaptive controls, and stop at a public bounded time. All logarithms in description lengths have base two.

Theorem 9.1 (Uniform interior operational entropy). *Let $d, N \geq 1$ and $0 < \delta \leq 2^{-13}$. The least cardinality $\mathcal{C}_N(\mathfrak{I}_d, \delta)$ of a d_N -cover of $\mathfrak{I}_d$, allowing centres anywhere in $\mathfrak{E}_d$, satisfies*

$$(9.1) \quad \left(\frac{\sqrt{N}}{2048\delta} \right)^p \leq \mathcal{C}_N(\mathfrak{I}_d, \delta) \leq \left(\frac{25\sqrt{N}}{\delta} \right)^p.$$

The upper bound is attained by a public finite exact rational construction when δ is rational. For real δ , a volumetric net gives the stated upper bound. Consequently the optimal deterministic fixed-length description into legal effects is

$$(9.2) \quad B_{\text{int}}^*(d, N, \delta) = \frac{d^2}{2} \log_2 N + d^2 \log_2(1/\delta) + O(d^2),$$

where the remainder is uniform in all three parameters.

Proof. The real vector space of Hermitian matrices has dimension p , and $\mathfrak{I}_d$ is precisely the translate of its operator-norm ball of radius $1/4$. Set $s = 512\delta/\sqrt{N}$. A maximal subset $\{E_1, \dots, E_L\}$ of $\mathfrak{I}_d$ whose distinct points have operator distance strictly greater than s is finite and its closed s -balls cover $\mathfrak{I}_d$. Translation-invariant Lebesgue measure and homogeneity therefore give

$$Ls^p \geq (1/4)^p, \quad L \geq \left(\frac{\sqrt{N}}{2048\delta} \right)^p.$$

The unit-ball volume cancels; no dimension-dependent volume estimate is used. Lemma 8.4 separates any two selected effects in d_N by at least 4δ , since $512\delta \leq 1/16$. A closed operational ball of radius δ contains at most one selected effect, irrespective of its centre. This uses the triangle inequality for d_N , not a property of a comparison modulus.

For the upper bound choose a maximal operator-norm $\delta/(4\sqrt{N})$ -separated subset of $\mathfrak{I}_d$. It covers the latter at that operator radius and hence at operational radius δ . The disjoint open balls of half that radius lie inside the concentric ball of radius $1/4 + \delta/(8\sqrt{N})$. Their number is at most $(1 + 2\sqrt{N}/\delta)^p$, which is smaller than the asserted bound. The rational construction with a specified finite stopping rule is proved next. A fixed public B -bit decoder has at most 2^B outputs, and indexing a smallest finite cover suffices. Taking logarithms, with one bit for rounding to an integer, proves (9.2). $\square$

9.1. An exact public interior dictionary. Only the public parameters (d, N, δ) determine the dictionary. Let $\delta \in \mathbb{Q} \cap (0, 1]$ and set

$$(9.3) \quad k = \lceil \sqrt{N} \rceil, \quad K = \left\lceil \frac{128dk}{\delta} \right\rceil, \quad t = \frac{\delta}{16k}.$$

The integer k is computed by an integer square-root test. Enumerate Hermitian matrices with diagonal coordinates in $\{0, 1/K, \dots, 1\}$ and real and imaginary off-diagonal coordinates in $\{-1, -1 + 1/K, \dots, 1\}$. Order coordinates as in Theorem 6.1 and retain only the matrices in $[I_d/8, 7I_d/8]$. From this ordered finite stream construct a list $\mathcal{D}$ greedily, accepting G precisely when $\|G - C\|_{\text{op}} > t$ for every previously accepted C . For a Hermitian rational difference H , the complementary test is

$$(9.4) \quad \|H\|_{\text{op}} \leq t \iff tI_d - H \succeq 0 \text{ and } tI_d + H \succeq 0.$$

Thus exact rational positive-semidefinite tests settle the comparison, including equality, without computing eigenvalues. The Schur test of Theorem 6.1 applies to both inequalities.

Theorem 9.2 (Finite exact interior coding). *For every $d, N \geq 1$ and rational $0 < \delta \leq 1$, the preceding construction terminates and defines a common encoder and decoder. For a supplied Gaussian-rational $E \in \mathfrak{J}_d$, the decoded centre $C \in [I_d/8, 7I_d/8]$ satisfies*

$$(9.5) \quad d_N(E, C) \leq 5\delta/16.$$

The same conclusion holds for a real $E \in \mathfrak{J}_d$ supplied with a Gaussian-rational Hermitian approximation A certified to satisfy $\|A - E\|_{\text{op}} \leq d/K$. The payload is an index of length $\lceil \log_2 |\mathcal{D}| \rceil$, and

$$(9.6) \quad |\mathcal{D}| \leq (1 + 12k/\delta)^p \leq (25\sqrt{N}/\delta)^p.$$

All words of that length have legal decodings. No nonpublic matrix coordinates or target-dependent dictionary are free advice.

Proof. Round every real coordinate of A to the nearest multiple of $1/K$, with ties towards positive infinity, and preserve conjugate entries. Call the result G . In every row the diagonal rounding error is at most $1/(2K)$ and each off-diagonal error has modulus at most $1/(\sqrt{2}K)$. The Hermitian row-sum bound gives $\|G - A\|_{\text{op}} \leq d/K$. Therefore

$$\|G - E\|_{\text{op}} \leq 2d/K \leq \delta/(64k) \leq 1/64.$$

For rational E take $A = E$; the same, slightly looser, bound applies. Weyl's inequality puts G in $[I_d/8, 7I_d/8]$. Its diagonal and off-diagonal coordinates also lie in the enumerated ranges, so it occurs in the legal finite stream.

If G is retained, select $C = G$. Otherwise the strict greedy rule supplies an earlier retained centre with $\|G - C\|_{\text{op}} \leq t$; select the first such centre. Hence

$$\|E - C\|_{\text{op}} \leq \frac{\delta}{64k} + \frac{\delta}{16k} = \frac{5\delta}{64k}.$$

Both endpoints lie in the range of Lemma 8.4, which proves (9.5). Legality and both selection tests use finite exact rational arithmetic.

Distinct dictionary elements have operator distance greater than t , and every element lies within operator radius $3/8$ of $I_d/2$. The open operator balls of radius $t/2$ about them are disjoint and lie in the ball of radius $3/8 + t/2$. Cancelling the common unit-ball volume gives $(1 + 3/(4t))^p = (1 + 12k/\delta)^p$. Since $k \leq 2\sqrt{N}$ and $\sqrt{N}/\delta \geq 1$, the second inequality in (9.6) follows.

The decoder reconstructs the entire ordered list from the public parameters before interpreting its index. Unused fixed-length words return $I_d/2$. The list is nonempty because the rounded centre $I_d/2$ occurs. This proves termination and total legal decoding. A rational approximation

to a real input is supplied data under the displayed certificate, not an arbitrary exact-real oracle provided by the theorem. $\square$

The raw enumeration has at most $(K+1)^d(2K+1)^{d(d-1)}$ tuples. Exact comparisons against the retained list and the bit lengths of Schur-complement entries are additional reconstruction costs. They are not controlled by (9.2). The reference implementation rejects a resource-limited incomplete enumeration without emitting a codebook or payload. Thus the theorem is a finite exact construction, not a polynomial-time coding result. Volumetric packing and rational nets are standard mechanisms; the dimension-uniform operational comparison is what transfers them to the present future-use loss.

9.2. Randomized prefix descriptions. The fixed-length conclusion has a counterpart which permits shared randomness and counts expected transmitted length. This distinguishes an index-cardinality converse from a converse for a randomized communication protocol.

Definition 9.3 (One-way prefix description). The encoder is supplied $E \in \mathfrak{I}_d$. A public random variable R , independent of E , is available to both parties. Conditionally on each public seed, the possible finite binary messages W form a prefix-free set. Private randomization is allowed, and the decoder returns a legal effect C from (R, W) and its independent private randomness. No timing, stopping signal, nonpublic header, or other side channel is uncharged. The cost is $\sup_E \mathbb{E}_E |W|$ and the risk is $\sup_E \mathbb{P}_E \{d_N(E, C) > \delta\}$. The public parameters d, N, δ, η are not part of the message.

Theorem 9.4 (Randomized expected description length). *Let $d, N \geq 1$, $0 < \delta \leq 2^{-13}$ and $0 \leq \eta \leq 1/8$. Write*

$$A_d(N, \delta) = d^2 \log_2(\sqrt{N}/\delta), \quad h_2(\eta) = -\eta \log_2 \eta - (1 - \eta) \log_2(1 - \eta).$$

Here $0 \log_2 0 = 0$. The infimum $\mathcal{B}_\eta$ of the costs in Definition 9.3 with risk at most η satisfies

$$(9.7) \quad (1 - \eta)(A_d(N, \delta) - 11d^2) - h_2(\eta) \leq \mathcal{B}_\eta \leq (1 - \eta)(A_d(N, \delta) + d^2 \log_2 25 + 1).$$

In particular, $\mathcal{B}_\eta = (1 - \eta)A_d(N, \delta) + O(d^2)$, with absolute uniform constants. For rational δ the achievable nonempty messages are supplied by the finite exact interior dictionary.

Proof. Use the 4δ -separated packing of Theorem 9.1 and let X be uniform on its L labels, independently of R . A decoded effect within δ of the true packing point identifies its label uniquely. Choose a nearest label with a fixed tie rule otherwise. This produces $\hat{X}$ with error probability at most η .

For completeness, if J indicates a classification error, the chain rule gives

$$H(X | \hat{X}, R) \leq H(J) + \mathbb{P}(J = 1) \log_2(L - 1) \leq h_2(\eta) + \eta \log_2 L.$$

The final inequality uses $\eta \leq 1/8$ and $L \geq 4$. Since X and R are independent, data processing gives

$$(1 - \eta) \log_2 L - h_2(\eta) \leq I(X; \hat{X} | R) \leq I(X; W | R) \leq H(W | R).$$

The decoder's private randomness is independent conditionally on (R, W) and does not increase this information. For each public seed, Kraft's inequality and nonnegativity of relative entropy give $H(W | R = r) \leq \mathbb{E}(|W| | R = r)$; integrating proves the same conditional inequality. These statements also apply to countable messages of finite expected length by truncation; infinite cost already satisfies the lower bound. Worst-target expected length is at least its average over the packing. The packing cardinality in (9.1) now proves the first inequality. This is the usual Fano–Kraft argument, written here to include all the stated randomization and message conventions.

For achievability, use a public coin independent of the target. With probability η transmit the empty word and decode to $I_d/2$. With the complementary probability use the deterministic index code of Theorem 9.2, or an interior net for real δ . For each seed the permitted set is either the singleton empty word or the fixed-length index set, and hence is prefix-free. The decoder knows which case occurs from R , so there is no free transmitted abort flag. Failure can occur only on the public abort event. Equations (9.5) and (9.6), with integer length rounding, give the upper bound. The model permits the specified Bernoulli public coin; the zero-failure finite exact dictionary does not require it. $\square$

9.3. Learning and transmitting at the joint rates.

Theorem 9.5 (Jointly optimal learned interior descriptions). *In the ideal-control query model of Theorem 8.5, let $d, N \geq 1$ and let $0 < \delta \leq 2^{-13}$ be rational. There is one common learner on $\mathfrak{J}_d$ with a finite binary output and a fixed public deterministic decoder such that*

$$(9.8) \quad \sup_{E \in \mathfrak{J}_d} \mathbb{P}_E\{d_N(E, C) > \delta\} \leq 1/8, \quad M = O(d^2 N \delta^{-2}), \quad B = A_d(N, \delta) + O(d^2).$$

The constants are absolute. The acquisition uses independent one-call input/reference pairs, followed by collective processing of completed outputs. No device call is used for encoding or replay. Any learner with these error and decoder conventions has $M = \Omega(d^2 N \delta^{-2})$ and $B \geq A_d(N, \delta) - 11d^2$.

Proof. Apply the binary estimator of [5, Corollary III.9] at operator accuracy $\epsilon = \delta/(64k)$ and failure $1/8$. It returns a legal effect F with $\|F - E\|_{\text{op}} \leq \epsilon$ on its good event, using $O(d^2 k^2 \delta^{-2})$ calls. Let T be the spectral clipping of F to $[I/4, 3I/4]$. The good event and Weyl's inequality imply $\|T - F\|_{\text{op}} \leq \epsilon$, whence

$$\|T - E\|_{\text{op}} \leq 2\epsilon, \quad d_N(E, T) \leq 8\sqrt{N}\epsilon \leq \delta/8.$$

This argument does not assume that scalar clipping is operator-Lipschitz for a noncommuting pair.

Round the coordinates of T and encode their grid point by the finite dictionary of Theorem 9.2. Its proof gives $d_N(T, C) \leq 5\delta/16$ even though the displayed estimator may initially be real-valued. The resulting map from the original classical outcome to a dictionary index is measurable: spectral clipping is continuous, coordinate rounding is Borel with specified ties, and the final list is finite. Composing the final tomography POVM with this finite partition defines a common finite-output measurement. It needs neither an unknown-effect-dependent measurement nor an exact-real side channel to the decoder. Equivalently, certified finite coordinate approximations can be used with the rounding slack of the coding theorem when supplied by a classical implementation. On the good event the total operational error is at most $7\delta/16 < \delta$. Since $k^2 \leq 4N$, the call bound and the payload upper bound follow.

For the call converse, a decoded word is also an estimator, so Theorem 8.2 applies. For the length converse, positive success probability for every target implies that the finite range of the fixed deterministic decoder is a δ -cover. The first inequality of (9.1) applies even when those centres lie outside $\mathfrak{J}_d$. $\square$

The last theorem is a query and transmitted-length statement. It inherits the ideal trusted tomography operations of [5]; the measurable finite readout above does not assert an efficient gate synthesis for them. The separate finite-control theorem for our full-body learners remains unchanged. The interior description bounds neither remove the endpoint logarithms on the complete body nor optimize its d^4 learning upper bound. Rather, on the same full-dimensional interior, they identify both training and description costs with constants uniform in dimension, horizon and accuracy.

10. LEARNING WITH FINITE TRUSTED CONTROL SPECIFICATIONS

The learning theorems count unknown-device calls. We now remove exact physical preparation from their hypotheses, while keeping their loss and call orders. A trusted preparation instruction has a finite classical description and a certified trace-norm error; a trusted control instruction has a finite description and a certified diamond-norm error. These are accuracy assumptions on the available controls. They assert neither a universal hardware implementation nor a bound on preparation time, gate count, or classical workspace. All trace norms in this section are unhalved.

Lemma 10.1 (An adaptive control-error budget). *Fix a finite classical decision tree of experiments, with bounded numbers of calls and trusted instructions on every record. Its unknown channel is unchanged. At a scheduled trusted event j , conditional on the preceding classical history h , replace a channel $\mathcal{C}_{j,h}$ by a channel $\tilde{\mathcal{C}}_{j,h}$ such that*

$$(10.1) \quad \|\tilde{\mathcal{C}}_{j,h} - \mathcal{C}_{j,h}\|_{\diamond} \leq e_j(h), \quad \sum_{j \text{ on } h_*} e_j(h_{<j}) \leq e \quad \text{for every complete record } h_*.$$

For a fresh state preparation, its trace-norm error suffices in place of the diamond norm: conditional on h , the new state is tensor-separated from every carried system in both experiments. The final classical–quantum states satisfy

$$(10.2) \quad \|\tilde{\rho}_{\text{out}} - \rho_{\text{out}}\|_1 \leq e.$$

In particular, the laws of any unchanged classical output have total variation distance at most $e/2$.

Proof. Include each observed outcome in the classical history and pad stopped branches by identities. Quantum registers of different sizes can be placed in a finite direct sum. Each event is then a channel on the complete classical–quantum state. A quantum instrument means its complete channel $\rho \mapsto \sum_x |x\rangle\langle x| \otimes \mathcal{C}_x(\rho)$: if errors are instead supplied separately for its completely positive branches, their sum is the event’s error charge.

At each event, insert the ideal preceding state between the actual and ideal updates. Contractivity of the actual channel and the diamond-norm bound give

$$D_j \leq D_{j-1} + \sum_h p_{j-1}(h) e_j(h),$$

where D_j is the unhalved trace-norm difference and p_{j-1} is the ideal history law. Unknown-channel calls and unchanged classical calculations have charge zero. Summing the inequality gives the ideal expectation of the pathwise sum in (10.1), which is at most e . A replacement preparation channel has diamond distance equal to the trace distance of the prepared states, including when another register is retained. Finally, classical readout and deterministic postprocessing are contractions, and classical total variation is one half of the trace norm. No continuity of the decision rules or of the final loss is used. $\square$

We check that the learners to which this lemma is applied have finite, computable control specifications. Take integer dimensions and horizons and rational positive accuracy and confidence targets. All absolute proof constants may be fixed once as sufficiently conservative rational numbers. For sample budgets, replace each $\log x$, $x > 1$ rational, by

$$\ell(x) = \min\{k \in \mathbb{N} : 2^k \geq x\}.$$

Here $\log x \leq \ell(x) \leq 1 + \log x / \log 2$. Thus the concentration bounds and call orders are preserved, while ceilings of the resulting rational sample budgets are decidable by integer arithmetic. This convention avoids requiring an exact comparison of a transcendental expression with an integer.

Calibration frequencies and all complex empirical means are rational. The ordered spectral projections, fixed-order Gram–Schmidt procedure, clipping, normalized frames, and tie rules of Sections 4–5 use algebraic arithmetic. The notation for phase recovery does not require a

transcendental oracle. For a nonzero empirical z passing (4.20), put $q = z/|z|$ and select the root v of $v^m = q$ with largest real part. It is unique: the passed guard places $\text{Arg } q$ strictly between $-\pi$ and π , so its principal m th root is strictly closer in angle to 1 than every other root. All roots are algebraic, and comparison of their real parts is decidable. Consequently

$$\tan(\text{Arg}(z)/m) = \frac{\Im v}{\Re v}$$

is algebraic; its denominator is positive under the guard. Zero, amplitude, frame-selection, and phase-guard equalities are therefore decidable exactly. The finite rational legalization grid in Theorem 5.1 is in the public input basis; its positivity tests are exact, and its finitely many algebraic objectives can be evaluated with the prescribed slack. The same holds for the exact rational codec. Every record consequently has a finite, terminating classical computation and a legal output.

There are finitely many records at fixed public parameters: the device outputs and preparation signs have finite alphabets and their numbers are bounded on every path. This does not discretize the unknown effect or the retained quantum state. Within a block the state still lies in its full continuous quantum state space. Only the experiment's classical instructions and observations have finite descriptions.

Finite descriptions can also specify controls with any required positive accuracy. If a computable unit vector $\psi \in \mathbb{C}^D$ is prescribed and $0 < \epsilon \leq 1$, compute a nonzero Gaussian rational vector z with $\|z - \psi\|_2 \leq \epsilon/8$ and set $\phi = z/\sqrt{\sum_k |z_k|^2}$. Then

$$(10.3) \quad \|\phi\langle\phi| - \psi\langle\psi|\|_1 \leq 2\|\phi - \psi\|_2 \leq 4\|z - \psi\|_2 \leq \epsilon/2.$$

Coordinate accuracy $O(\epsilon/\sqrt{D})$ suffices, so z and its normalization rule have a description of $O(D \log(D/\epsilon))$ bits. The normalization is specified algebraically; rational normalized amplitudes are not assumed. A trusted preparation within $\epsilon/2$ of this finitely specified target has total error at most ϵ from the ideal state.

For completeness, finite specifications can preserve the exact channel constraints as well. Approximate a known algebraic Stinespring isometry V by a Gaussian rational matrix A with $\|A - V\|_{\text{op}} \leq r < 1/2$, and specify $W = A(A^*A)^{-1/2}$. Then $W^*W = I$ and $\|W - V\|_{\text{op}} \leq 2r$: every singular value of A differs from one by at most r . The channels obtained from these isometries differ in diamond norm by at most $4r$, by expansion of $W\rho W^* - V\rho V^*$ and partial-trace contractivity. Keeping a finite outcome flag gives the same construction for instruments. Their actual trusted realizations are charged separately in (10.1). Independent rounding of Kraus entries without restoring the channel constraints would not suffice.

Theorem 10.2 (Finite-control form of the learning guarantees). *The common matrix learner of Theorem 5.1 and the learner with training-block cap $1 \leq b \leq N$ of Theorem 7.2 admit the preceding finite trusted-control model in their stated ranges. They have the original success target*

$$\sup_{E \in \mathfrak{E}_d} \mathbb{P}_E\{d_N(\mathcal{M}_E, \mathcal{M}_{\hat{E}}) > \delta\} \leq \eta$$

with every-record call orders, respectively,

$$(10.4) \quad Cd^4 N \delta^{-2} \log(d/\eta), \quad Cd^4 \frac{N^2}{b} \delta^{-2} \log(d/\eta).$$

Their original block caps and exact legality of the returned classical effect are preserved. The same transfer applies when the classical output is the public codeword. In dimension one the order remains $CN\delta^{-2} \log(1/\eta)$ independently of b .

Proof. Use the corresponding ideal learner at the same accuracy δ and confidence $\eta/2$, with the finite sample-budget convention above. Fix a computable public upper bound M_* on its calls on

every record. The explicit constructions use fresh product states or fresh GHZ states embedded into a known two-dimensional subspace. They can be implemented by one preparation event per block, hence at most M_* such events. Given any preceding record, compute that record's prescribed state to trace accuracy η/M_* , using (10.3) and the trusted realization allowance. Lemma 10.1 gives final output total variation at most $\eta/2$. For each fixed unknown E , apply it to the set of returned effects whose original loss exceeds δ . The ideal probability is at most $\eta/2$, so the actual probability is at most η . Discontinuous decisions, unsuccessful records, and exact equality branches are included in this comparison.

Classical postprocessing is identical in the two experiments; it does not round the returned effect or test a perturbed loss. Thus legality is exact on every actual record and no additional δ -dependent loss allowance is needed. Halving η and upper-rounding sample budgets change only the absolute constants in (10.4). $\square$

Each approximate preparation above supplies a state on the same $m \leq b$ probe registers as its ideal block; preparation workspace is discarded or reset before the experiment and no quantum system is passed to the next block. Approximation therefore does not enlarge entangled input width or introduce coherent coupling from old memory into fresh probes. In a realization with S additional scheduled control instructions, use a public bound on the total number of trusted events and charge every instruction in (10.1). Per-event norm precision is then of order $\eta/(M_* + S)$. For an m -probe state the direct coordinate description has $D = d^m$ and size $O(D \log(D(M_* + S)/\eta))$ bits. The precision per coordinate is logarithmic in $(M_* + S)/\eta$, with its dimension dependence explicitly charged; this is not a polynomial preparation bound. Real-valued target tolerances may be replaced by supplied rational targets between one half and the requested values, preserving all orders and the requested guarantees. No representation of an arbitrary real input, and no oracle for the unknown effect, is assumed.

11. THE FULL BINARY EFFECT BODY

The spectral boundary of a biased measurement has two distinct endpoints. A vanishing eigenvalue of one effect need not give a coherent angular scale: the other outcome can still destroy coherence. We therefore retain both endpoint coordinates and determine their joint finite-use geometry.

Let

$$(11.1) \quad \mathcal{E} = \left\{ (a, x) \in \mathbb{R} \times \mathbb{R}^3 : 0 \leq a - |x| \leq a + |x| \leq 2 \right\}, \quad E_{a,x} = \tfrac{1}{2}(aI + x \cdot \sigma).$$

We identify a point (a, x) of $\mathcal{E}$ with its effect $E_{a,x}$. Write $\mathcal{M}_{a,x}$ for the measurement with ordered effects $E_{a,x}, I - E_{a,x}$ and no residual quantum output. Its maximum and minimum eigenvalues, spectral gap, and bias are

$$p = \tfrac{1}{2}(a + |x|), \quad q = \tfrac{1}{2}(a - |x|), \quad r = p - q = |x|, \quad b = p + q - 1.$$

Thus $0 \leq q \leq p \leq 1$ and $|b| + r \leq 1$. When $r > 0$ put $u = x/r$, so that $E_{a,x} = pP_u + q(I - P_u)$, where $P_u = (I + u \cdot \sigma)/2$. Directions are not identified under outcome exchange.

For an integer $N \geq 1$ set $\tau = N^{-1}$ and define

$$(11.2) \quad T_N(s, t) = |s - t| \sqrt{\frac{N}{\max\{s(1-s), t(1-t)\} + \tau}}, \quad 0 \leq s, t \leq 1,$$

$$V(p, q) = p(1-p) + q(1-q) = \tfrac{1}{2}(1 - b^2 - r^2),$$

$$(11.3) \quad K_N(p, q) = r \sqrt{\frac{N}{V(p, q) + \tau}}.$$

For two effects with data (p, q, u) and (p', q', v) let $h \in [0, \pi]$ be the angle between u, v when both spectral gaps are positive. If either gap is zero, set the following angular term equal to zero, without choosing a direction at the repeated eigenvalue:

$$(11.4) \quad H_N(E, F) = T_N(p, p') + T_N(q, q') + \min\{K_N(p, q), K_N(p', q')\} h.$$

Theorem 11.1 (Finite-use geometry of binary qubit effects). *For every $E, F \in \mathcal{E}$ and $N \geq 1$,*

$$(11.5) \quad \frac{1}{8192} \min\{1, H_N(E, F)\} \leq d_N(\mathcal{M}_E, \mathcal{M}_F) \leq \min\{2, 3H_N(E, F)\}.$$

The distance permits all common reference-assisted adaptive testers with at most N calls, including feedback and bounded public stopping, and uses the unhalved trace norm. The constants are independent of both spectra, both directions, and the horizon. In particular,

$$(11.6) \quad d_1(\mathcal{M}_{a,x}, \mathcal{M}_{a',x'}) = |a - a'| + |x - x'|.$$

The lower comparison is witnessed by pair-dependent nonadaptive tests. It does not require a common estimator of all four target coordinates.

The denominator in (11.3) involves the two outcome variances. Replacing it by the distance to the nearest spectral face would give an incorrect angular scale on a face such as $q = 0, 0 < p < 1$. The proof below supplies both a scalar-overlap dilation upper bound and explicit finite-distance lower witnesses.

11.1. The spectral endpoints. Put

$$B_N(s, t) = \|\text{Bern}(s)^{\otimes N} - \text{Bern}(t)^{\otimes N}\|_1.$$

Lemma 11.2 (A Bernoulli estimate at both support endpoints). *For $s, t \in [0, 1]$,*

$$(11.7) \quad \frac{1}{64} \min\{1, T_N(s, t)\} \leq B_N(s, t) \leq \min\{2, 3T_N(s, t)\}.$$

For a fixed one of the two parameters, B_N increases as the other parameter moves away from it on either side.

Proof. Assume $s < t$, put $\Delta = t - s$, and write $v = \max\{s(1 - s), t(1 - t)\}$. The identical pair is immediate. A coupling of individual bits gives $B_N \leq 2N\Delta$. The root fidelity $F = \sqrt{st} + \sqrt{(1 - s)(1 - t)}$ satisfies

$$\begin{aligned} 1 - F &= \frac{\Delta^2}{2} \left(\frac{1}{(\sqrt{s} + \sqrt{t})^2} + \frac{1}{(\sqrt{1 - s} + \sqrt{1 - t})^2} \right) \\ &\leq \frac{\Delta^2(1 + \Delta)}{2t(1 - s)} \leq \frac{\Delta^2}{2v}, \end{aligned}$$

whenever $v > 0$. For the last inequality, both endpoint variances are at most $t(1 - s)/(1 + \Delta)$: after cancellation the required inequalities are $s(1 + \Delta) \leq t$ and $(1 - t)(1 + \Delta) \leq 1 - s$. They follow respectively from $\Delta(1 - s) \geq 0$ and $t\Delta \geq 0$. The product-fidelity inequality gives $B_N \leq 2\sqrt{N}\Delta/\sqrt{v}$. For $z = Nv$, the inequality $\min\{2, 2/\sqrt{z}\}\sqrt{z + 1} \leq 2\sqrt{2} < 3$ combines the two upper bounds. At $v = 0$ use the coupling bound.

For the lower bound, complement both bits if necessary to have $s + t \leq 1$. If $t > 3/4$, then $\Delta \geq 2t - 1 > 1/2$ and $B_N \geq B_1 = 2\Delta > 1$, which suffices. Otherwise $0 < t \leq 3/4$. Set

$$k = \min \left\{ N, \max \left\{ 1, \left\lfloor \frac{1}{2t} \right\rfloor \right\} \right\}, \quad m = \lfloor N/k \rfloor.$$

Then $k \geq \frac{1}{4} \min\{N, t^{-1}\}$, $m \geq N/(2k)$, and $(k-1)t \leq 1/2$. The event of no success in a block of k trials has probabilities $e_s = (1-s)^k$ and $e_t = (1-t)^k$, with

$$e_s - e_t \geq k\Delta(1-t)^{k-1} \geq k\Delta/2.$$

There is a common stochastic processing of this binary event whose success probability is $d + ce$, where

$$c = \frac{1}{\max\{e_s + e_t, 2 - e_s - e_t\}} \geq \frac{1}{2}, \quad d = \frac{1}{2}(1 - c(e_s + e_t)).$$

Indeed $d \geq 0$, $d+c \leq 1$, and the processed pair has parameters $(1 \pm c(e_s - e_t))/2$. Apply Lemma A.3 to m independent blocks and discard the unused calls. Contraction under this processing gives

$$B_N(s, t) \geq \frac{1}{16\sqrt{2}} \min\{1, \Delta\sqrt{N \min(N, t^{-1})}\}.$$

Since $v \geq t(1-t) \geq t/4$, $T_N(s, t) \leq 2\Delta\sqrt{N \min(N, t^{-1})}$. This proves the stated lower constant.

For monotonicity, the likelihood ratio of the two binomial count laws is monotone in the count. For a fixed larger parameter their total variation is the maximum, over upper count intervals, of the larger-parameter probability minus the smaller-parameter probability. The latter probabilities decrease when the smaller parameter decreases. Fixing the smaller parameter gives the analogous statement with lower count intervals. Endpoint cases follow by continuity of the finite sums. $\square$

Lemma 11.3 (Spectral projection without cancellation). *For arbitrary directions and spectra,*

$$(11.8) \quad B_N(p, p') \leq d_N(\mathcal{M}_E, \mathcal{M}_F), \quad B_N(q, q') \leq d_N(\mathcal{M}_E, \mathcal{M}_F).$$

If the two directions agree, then

$$(11.9) \quad d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p',q',u}) \leq B_N(p, p') + B_N(q, q').$$

Consequently, with $d = d_N(\mathcal{M}_E, \mathcal{M}_F)$,

$$(11.10) \quad \begin{aligned} d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) &\leq 3d, \\ d_N(\mathcal{M}_{p',q',u}, \mathcal{M}_{p',q',v}) &\leq 3d. \end{aligned}$$

Proof. Suppose first $p \geq p'$. Repeatedly input a maximum-eigenvalue eigenstate of E . Its success probabilities under the two hypotheses are p and some $t \leq p'$. The monotonicity in Lemma 11.2 gives $B_N(p, p') \leq B_N(p, t) \leq d$. If $p < p'$, interchange the hypotheses. For $q \leq q'$, use a minimum-eigenvalue eigenstate of E ; the other success probability is at least q' . Reverse the hypotheses if $q > q'$. These are separate admissible witnesses; no single input is asserted to witness both endpoint changes.

For aligned directions, first make the common projective measurement $\{P_u, I - P_u\}$ and, according to its outcome, generate a bit with success probability p or q . This implements exactly the measurement channel, including on inputs entangled with a reference: the reference blocks are the corresponding partial matrix elements weighted by the displayed success probabilities. Supply in advance N independent p -bits and N independent q -bits; at a call consume the next unused bit from the selected array. Every fixed adaptive tester is a common channel of these two arrays. The triangle inequality for tensor products therefore bounds its output distance by $B_N(p, p') + B_N(q, q')$. This proves (11.9) with arbitrary references, feedback, and stopping; all unqueried row bits are discarded. Combining it with (11.8) bounds every aligned spectral change by $2d$. The triangle inequality, aligning at u or at v , then proves (11.10). $\square$

11.2. Angular geometry with both eigenvalues retained.

Theorem 11.4 (Angular transition for a biased measurement). *For fixed $0 \leq q \leq p \leq 1$, $N \geq 1$, and directions at angle $h \in [0, \pi]$,*

$$(11.11) \quad \frac{1}{512} \min\{1, K_N(p, q)h\} \leq d_N(\mathcal{M}_{p,q,u}, \mathcal{M}_{p,q,v}) \leq \min\{2, \sqrt{2}K_N(p, q)h\}.$$

More precisely, put

$$A = \sqrt{p(1-p)} + \sqrt{q(1-q)}, \quad z = (p-q) \sin(h/2).$$

When $A > 0$,

$$(11.12) \quad d_N \leq \min \left\{ 2, 2Nz, 2\sqrt{1 - \left(\frac{A^2}{A^2 + z^2} \right)^N} \right\}.$$

When $A = 0$, the bound $\min\{2, 2Nz\}$ remains valid.

Proof. The one-use distance is $2z$: the effect difference is traceless with operator norm z , and the same reference trace-norm duality as in Lemma A.2 applies. Replacing the calls one at a time gives the adaptive bound $2Nz$.

We next construct a dilation with scalar cross overlap. Common unitary conjugation puts the two directions in one real meridian. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_t = \exp(tJ), \quad t = h/2, \quad E = \text{diag}(p, q), \quad F = R_t E R_t^\top.$$

A measurement dilation is

$$W_E \psi = |+\rangle_C |+\rangle_L \sqrt{E} \psi + |-\rangle_C |-\rangle_L \sqrt{I-E} \psi.$$

The square-root vectors belong to a further two-dimensional environment. Tracing both environments leaves the ordered classical output and implements the measurement even with a reference. Put

$$f = \sqrt{pq} + \sqrt{(1-p)(1-q)}.$$

For any ϕ , apply the common rotation $R_{\phi-t}$ to the two-dimensional environment of the dilation W_F . The cross overlap is

$$\begin{aligned} W_E^\dagger (I \otimes R_{\phi-t}) W_F &= (\sqrt{E} R_\phi \sqrt{E} + \sqrt{I-E} R_\phi \sqrt{I-E}) R_{-t} \\ &= (\cos \phi I + f \sin \phi J) R_{-t}. \end{aligned}$$

Here the rotation acts as the identity on the outcome registers. When $f > 0$, choose

$$D = \sqrt{f^2 \cos^2 t + \sin^2 t}, \quad \cos \phi = \frac{f \cos t}{D}, \quad \sin \phi = \frac{\sin t}{D}.$$

Then the displayed overlap is cI , where $c = f/D \in [0, 1]$. This construction also covers $t = 0$ and $t = \pi/2$. To express it in the spectral coordinates, put $T = \sqrt{p(1-q)} + \sqrt{q(1-p)}$. Direct multiplication gives

$$fT = A, \quad (1 - f^2)T^2 = (p - q)^2.$$

If $A > 0$ then $f, T > 0$, and therefore

$$(11.13) \quad c^2 = \frac{f^2}{f^2 \cos^2 t + \sin^2 t} = \frac{A^2}{A^2 + (p - q)^2 \sin^2 t}.$$

Fix any common adaptive tester. Purify its initial state and all intermediate operations, and retain the measurement environments in this proof. The environments are never accessed by the tester. At each call the scalar identity for the cross overlap multiplies the inner product of the two purifications by c , independently of the current input and reference. The common intervening

isometries preserve the inner product. A tester that stops early is padded to N calls on dummy inputs, with their outcomes ignored in its final output. Thus two purifications of that final output have inner product c^N . Trace-norm contraction and the pure-state formula give $d_N \leq 2\sqrt{1 - c^{2N}}$, proving (11.12). This is a pairwise dilation comparison; it does not require a processor for the entire effect body or an unknown-input environment-seizing procedure.

Since $V \leq A^2 \leq 2V$ and $1 - (1 + y)^{-N} \leq Ny$ for $y \geq 0$, the two nontrivial upper bounds imply

$$d_N \leq \min\{Nrh, \sqrt{N}rh/\sqrt{V}\} \leq \sqrt{2}K_N(p, q)h$$

when $V > 0$. The final inequality follows by putting $z = NV$ and using $\min\{1, z^{-1/2}\}\sqrt{z+1} \leq \sqrt{2}$. If $V = 0$, the only possibilities are scalar deterministic effects or a projective measurement, and the hybrid bound suffices.

For the lower bound, two explicit witnesses suffice. First suppose $r > 0$ and $h > 0$. Input a pure state in the direction $u - v$. Its success probabilities are

$$\frac{1 + b + r \sin(h/2)}{2}, \quad \frac{1 + b - r \sin(h/2)}{2}.$$

Their difference is $r \sin(h/2)$. The larger of their Bernoulli variances is at most $1 - b^2$: after complementing both output laws we may assume $a = 1 + b \leq 1$; both success probabilities lie in $[0, a]$, so their variances are at most $a \leq a(2 - a) = 1 - b^2$. Lemma 11.2, together with $\sin(h/2) \geq h/\pi$, now gives

$$(11.14) \quad d_N \geq \frac{1}{64\pi} \min \left\{ 1, rh \sqrt{\frac{N}{1 - b^2 + \tau}} \right\}.$$

The repeated pure input is a nonadaptive test.

For the second witness, choose a half-turn of the input qubit about an axis normal to the plane of u, v . It sends both Bloch directions to their negatives. With probability 1/2 use the original measurement, and with probability 1/2 apply this half-turn before the call and swap the two recorded outcomes afterwards. Discard the random choice. The resulting measurement has effects $(I \pm ru \cdot \sigma)/2$ or $(I \pm rv \cdot \sigma)/2$ under the respective hypotheses: the bias has been removed and the visibility is r . The same random preprocessing and postprocessing are used under both hypotheses and on every call. Any test for this unbiased pair is consequently an admissible test for the original pair. Theorem A.1 gives

$$(11.15) \quad d_N \geq \frac{1}{64} \min \left\{ 1, rh \sqrt{\frac{N}{1 - r^2 + \tau}} \right\}.$$

The witness is the finite entangled-block test of Lemma A.4, with this common processing. If the directions coincide or are antipodal, the required half-turn is chosen about any perpendicular axis.

Finally, feasibility $|b| + r \leq 1$ implies

$$\min\{1 - b^2, 1 - r^2\} \leq 2(1 - b^2 - r^2) = 4V.$$

For example, write $x = \max\{|b|, r\}$ and $y = \min\{|b|, r\}$; then $x \leq 1 - y$, $0 \leq y \leq 1/2$, and $x^2 + 2y^2 \leq (1 - y)^2 + 2y^2 \leq 1$, which is the asserted inequality. Hence the larger of the two coefficients in (11.14) and (11.15) is at least $K_N(p, q)/2$. Their stronger lower bound is at least $(128\pi)^{-1} \min\{1, K_N(p, q)h\}$, which implies (11.11). If $r = 0$ or $h = 0$, the two channels coincide and the assertion is immediate. $\square$

Proof of Theorem 11.1. For positive gaps use a path whose angular leg has the spectrum with the smaller angular coefficient. Lemma 11.3, followed by Lemma 11.2 and Theorem 11.4, gives

$$d_N(\mathcal{M}_E, \mathcal{M}_F) \leq 3T_N(p, p') + 3T_N(q, q') + \sqrt{2} \min\{K_N(p, q), K_N(p', q')\}h \leq 3H_N(E, F).$$

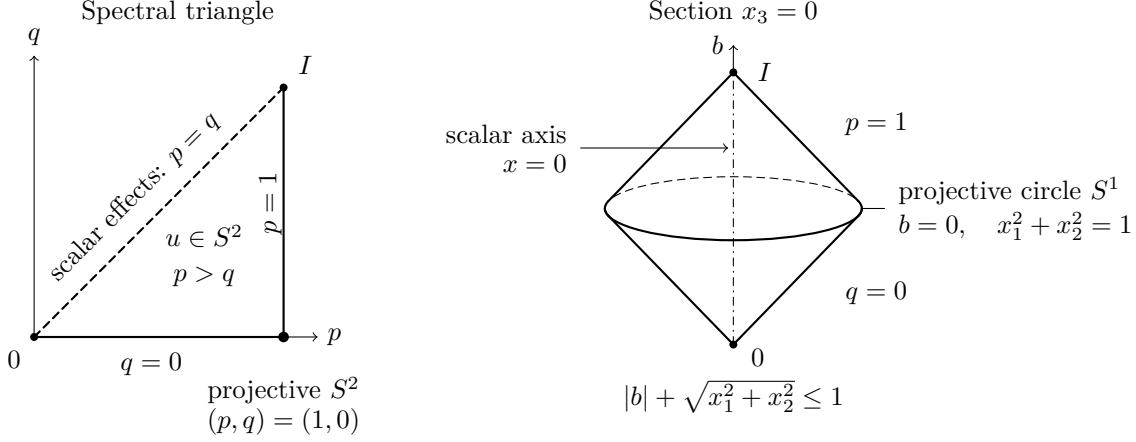


FIGURE 1. The four-dimensional effect body $|b| + |x| \leq 1$. Each point of the spectral triangle with $p > q$ carries a direction $u \in S^2$; on the scalar diagonal all directions represent the same effect. The boundary pieces $q = 0$ and $p = 1$ meet at the projective stratum $(p, q) = (1, 0)$, which carries the full sphere S^2 . The right panel is a projection of the three-dimensional section $x_3 = 0$; its equatorial circle is the intersection of that sphere with the section.

If one gap vanishes, that scalar effect may be aligned with the other direction at no cost, so the same estimate holds with angular term zero.

For the lower bound, put $A = T_N(p, p')$, $B = T_N(q, q')$, and $C = \min\{K_N(p, q), K_N(p', q')\}h$. The spectral projection lemma gives $d_N \geq 64^{-1} \min\{1, A\}$ and $d_N \geq 64^{-1} \min\{1, B\}$. Its angular projections and Theorem 11.4 give $d_N \geq 1536^{-1} \min\{1, C\}$. The largest of these three bounds is at least $4608^{-1} \min\{1, A + B + C\}$, which proves the weaker displayed constant $1/8192$. At a scalar effect only the two spectral bounds are needed. To check the assertion about lower witnesses, let d_N^{na} be the supremum over nonadaptive tests. The two spectral lower bounds hold with d_N^{na} , and the aligned spectral upper bound is consequently at most $2d_N^{\text{na}}$. Apply the triangle inequality to the output of a nonadaptive angular witness on the three-channel path. Its actual-pair leg is at most d_N^{na} , and its aligned spectral leg is at most $2d_N^{\text{na}}$. The same lower comparison thus holds with d_N^{na} in place of d_N .

It remains to verify the one-use identity. For a reference-assisted input state ρ , the difference has output blocks B_ρ and $-B_\rho$, where $B_\rho = \text{tr}_A[(E - F) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B_\rho\|_1 \leq \|E - F\|_{\text{op}}$. An extremal eigenstate of $E - F$ attains equality. Thus $d_1 = 2\|E - F\|_{\text{op}}$; the two eigenvalues of $E - F$ are $((a - a') \pm |x - x'|)/2$. This proves (11.6). $\square$

Remark 11.5 (Boundary faces and repeated eigenvalues). On $p = q = t$, the measurement is the input-independent coin $\text{Bern}(t)$ and all directions represent the same effect. Its exact finite-use distance along this scalar interval is $B_N(t, t')$. On the face $q = 0$, $0 < p < 1$, the angular coefficient is

$$K_N(p, 0) = p \sqrt{\frac{N}{p(1-p) + N^{-1}}}.$$

For a fixed p in this interval it has square-root-horizon order, despite the zero eigenvalue. At the projective edge $(p, q) = (1, 0)$ it equals N . At the two deterministic apices $p = q = 0$ and $p = q = 1$, the angular coefficient vanishes while the spectral moduli retain their finite-use support cutoff. These cases are part of the same theorem, without a spectral margin imposed on the target.

12. THE DESCRIPTION ENTROPY OF ALL BINARY QUBIT EFFECTS

Let

$$\mathfrak{E} = \{E = E^* \in \mathbb{C}^{2 \times 2} : 0 \leq E \leq I\}.$$

An effect E specifies the ordered measurement with effects $E, I - E$. Write $C_N(\mathfrak{E}, \delta)$ for its covering number under d_N , allowing arbitrary legal memoryless centres of this interface, and write $C_N^{\text{rat}}(\mathfrak{E}, \delta)$ when the centres have rational Cartesian coordinates. Both eigenvalues and the eigenspaces are charged. The spectral representation $E = pP_u + q(I - P_u)$, $p \geq q$, represents a scalar effect only once, independently of u .

Theorem 12.1 (The full effect body). *There are absolute constants $c, C, \delta_0 > 0$ such that, for every integer $N \geq 1$ and $0 < \delta \leq \delta_0$,*

$$(12.1) \quad cN^2 \log(N+2) \delta^{-4} \leq C_N(\mathfrak{E}, \delta) \leq C_N^{\text{rat}}(\mathfrak{E}, \delta) \leq CN^2 \log(N+2) \delta^{-4}.$$

Consequently the optimum fixed-length reusable payload is

$$(12.2) \quad 2 \log_2 N + \log_2 \log(N+2) + 4 \log_2(1/\delta) + O(1).$$

The constants are uniform over the whole closed effect body, including scalar, deterministic and projective effects. For rational accuracy input, rational target data admit an exact rational encoder attaining this order.

The logarithm records accumulation near the projective stratum, where both outcome probabilities can approach their support endpoints. We prove the upper bound by a finite rational construction and the lower bound by separated families of projective-corner boxes.

12.1. A rational spectral and angular code. For $N \geq 1$ and rational $0 < \delta \leq 1/4$, put

$$(12.3) \quad B = \left\lceil \sqrt{64N/\delta^2} \right\rceil, \quad s_i = \begin{cases} \frac{i^2}{B^2 + i^2}, & 0 \leq i \leq B, \\ 1 - \frac{(2B-i)^2}{B^2 + (2B-i)^2}, & B < i \leq 2B. \end{cases}$$

Thus $0 = s_0 < s_1 < \dots < s_{2B} = 1$. For $0 \leq j < i \leq 2B$ define

$$(12.4) \quad K_{ij}^2 = \frac{N(s_i - s_j)^2}{s_i(1-s_i) + s_j(1-s_j) + N^{-1}}, \quad A_{ij} = \max \left\{ 1, \left\lceil \sqrt{256K_{ij}^2/\delta^2} \right\rceil \right\}.$$

Use the six signed stereographic charts

$$(12.5) \quad u_a = \epsilon \frac{1 - |z|^2}{1 + |z|^2}, \quad u_b = \frac{2z_b}{1 + |z|^2} \quad (b \neq a), \quad a \in \{1, 2, 3\}, \quad \epsilon \in \{-1, 1\},$$

with $z \in \{-A_{ij}, \dots, A_{ij}\}^2/A_{ij}$ for the spectral pair (s_i, s_j) . Every chart word decodes to $s_i P_u + s_j(I - P_u)$. A diagonal pair $i = j$ has the single word $s_i I$. The exact number of words, including redundant chart words, is

$$(12.6) \quad L_{N,\delta} = (2B+1) + \sum_{0 \leq j < i \leq 2B} 6(2A_{ij}+1)^2.$$

Theorem 12.2 (Exact rational realization). *The words in (12.6) form a radius- $\delta/2$ cover of $\mathfrak{E}$ by legal rational effects, and*

$$(12.7) \quad L_{N,\delta} \leq CN^2 \log(N+2) \delta^{-4}$$

for an absolute C . If

$$E = \frac{1}{2}((1+b)I + x \cdot \sigma), \quad b \in \mathbb{Q}, \quad x \in \mathbb{Q}^3, \quad |b| + |x| \leq 1,$$

then a covering word is computable by rational arithmetic and integer comparisons. Its payload is one fixed-length index in the set of (12.6).

Proof. Write $f(t) = t^2/(1+t^2)$, $0 \leq t \leq 1$. For a probability in $[0, 1/2]$, round its inverse coordinate t to the nearest multiple of $1/B$; for a probability in $[1/2, 1]$, use the reflected chart $1 - f(t)$. Choose the multiple of smaller magnitude at ties. This defines a nondecreasing quantizer on $[0, 1]$, so separately quantizing $p \geq q$ preserves their order.

The square-root vector associated with $f(t)$ is $(t, 1)/\sqrt{1+t^2}$; its derivative has norm $1/(1+t^2) \leq 1$. The reflected chart has the same bound. Hence the Euclidean square-root distance between each probability and its quantization is at most $1/(2B)$. If H is this distance, the product-fidelity inequality gives unhalved distance at most $2\sqrt{N}H$ between the corresponding N -fold Bernoulli laws. The common two-coin programme in Lemma 11.3 therefore gives a spectral error at most

$$2\sqrt{N}/B \leq \delta/4.$$

This argument includes adaptive testers and entangled references.

If the quantized eigenvalues coincide, this completes the encoding. Otherwise choose a signed chart whose distinguished coordinate has maximal absolute value in the target direction. Its inverse coordinate belongs to $[-1, 1]^2$. Rounding its two coordinates to the grid of denominator A_{ij} gives angular error at most $2\sqrt{2}/A_{ij}$, since geodesic distance is bounded by the length of the mapped straight segment and the differential of stereographic projection has norm at most two. The angular upper bound in Theorem 11.4 gives error at most

$$\sqrt{2}K_{ij} \frac{2\sqrt{2}}{A_{ij}} \leq \delta/4.$$

The triangle inequality proves the covering assertion. Formula (12.5) preserves the unit sphere exactly, so every decoded effect is legal and rational.

We verify exact computability even when the input eigenvalues are irrational. Put $Q = |x|^2 \in \mathbb{Q}$. The eigenvalues are $(1 + b \pm \sqrt{Q})/2$. Each decision in the spectral rounding compares one of these numbers with a rational threshold $f((k + 1/2)/B)$ or its reflection. Such a comparison is the sign of a rational number plus or minus $\sqrt{Q}$; checking signs before squaring makes it an exact rational comparison. If $Q = 0$, the two quantized eigenvalues coincide. Otherwise choose the smallest index a maximizing $|x_a|$ and put $\epsilon = \text{sgn}(x_a)$. The inverse chart is

$$z_b = \frac{x_b}{\sqrt{Q} + |x_a|} \quad (b \neq a).$$

Its comparison with any rational rounding threshold again reduces to the sign of a rational number plus a rational multiple of $\sqrt{Q}$. The integers B, A_{ij} are computed by integer squaring and rational comparison. Thus no irrational normalization is required. The decoder uses only the displayed rational formulae.

It remains to count the words. Set $m_i = \min(i, 2B - i)$ and $U = B^2/N$. Directly from (12.3),

$$\frac{m_i^2}{4B^2} \leq s_i(1 - s_i) \leq \frac{m_i^2}{B^2}, \quad K_{ij}^2 \leq \frac{4NB^2}{m_i^2 + m_j^2 + U}.$$

Each value of m_i occurs at most twice. Since $(2A_{ij} + 1)^2 \leq C(1 + K_{ij}^2/\delta^2)$, we obtain

$$(12.8) \quad L_{N,\delta} \leq CB^2 + \frac{CNB^2}{\delta^2} \sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U}.$$

Here $U \geq 1$. The part with $\max(a, b) \leq \sqrt{U}$ is bounded by an absolute constant. Each successive dyadic square ring contributes at most another absolute constant. There are at most $C \log(N + 2)$

such rings, because $B/\sqrt{U} = \sqrt{N}$. Consequently

$$\sum_{a,b=0}^B \frac{1}{a^2 + b^2 + U} \leq C \log(N+2).$$

Finally $B^2 \leq CN/\delta^2$, so (12.8) proves (12.7). The support cutoff in the denominator is retained throughout the count, including the endpoint grid words. $\square$

12.2. Packing the projective corner.

Proof of Theorem 12.1. The upper bound follows from Theorem 12.2. For a real δ , choose a rational δ' with $\delta \leq \delta' \leq 2\delta$ and use its radius- $\delta'/2$ code; decrease δ_0 to at most $1/8$. For the lower bound first let $N \geq 16$. For every

$$t = 8^j/N \leq 1/16, \quad j \geq 0,$$

consider the effects with

$$(12.9) \quad p = 1 - \alpha, \quad q = \beta, \quad t \leq \alpha, \beta \leq 2t, \quad u = u(z), \quad z \in [-1/4, 1/4]^2,$$

where $u(z)$ is the chart in (12.5) with $a = 3, \epsilon = 1$. On this square, spherical distance is bounded above and below by absolute positive multiples of Euclidean distance in z . Throughout (12.9), the eigenvalue gap is at least $3/4$ and

$$p(1-p) \asymp q(1-q) \asymp t, \quad p(1-p) + q(1-q) + N^{-1} \asymp t.$$

The finite-distance comparison of Theorem 11.1 therefore supplies an absolute $c_1 > 0$ such that any two points in this box satisfy

$$(12.10) \quad d_N \geq c_1 \min \left\{ 1, \sqrt{N/t} |(\alpha, \beta, z) - (\alpha', \beta', z')|_\infty \right\}.$$

Choose an absolute sufficiently large A and then an absolute sufficiently small δ_0 . In each of the four coordinates take a grid of spacing $A\delta\sqrt{t/N}$. Since $Nt \geq 1$, this spacing is at most half the width of either spectral interval when $\delta \leq \delta_0$. The resulting points have pairwise distance at least 4δ by (12.10), and their number is at least

$$c \left(\frac{t}{\delta\sqrt{t/N}} \right)^2 \left(\frac{1}{\delta\sqrt{t/N}} \right)^2 = cN^2\delta^{-4}.$$

Points from different boxes are also uniformly separated. Indeed, if $t' \geq 8t$, then $\alpha' \geq t'$ and $\alpha \leq 2t \leq t'/4$. The spectral modulus of the corresponding upper eigenvalues is bounded below by an absolute positive multiple of $\sqrt{Nt'} \geq 1$. The spectral noncancellation bound in Theorem 11.1 gives $d_N \geq c_2 > 0$. Shrinking δ_0 ensures that this is at least 4δ . There are at least $c \log(N+2)$ permitted boxes. Their union is therefore a 4δ -packing of cardinality $cN^2 \log(N+2)\delta^{-4}$.

For $1 \leq N < 16$, use instead a fixed spectral rectangle $5/8 \leq p \leq 3/4$, $1/4 \leq q \leq 3/8$ and the same angular chart. Theorem 11.1 gives a uniform local Euclidean lower bound, and a grid of spacing $C\delta$ has at least $c\delta^{-4}$ points. The factor $N^2 \log(N+2)$ is bounded on this finite range. This proves the same lower estimate for all $N \geq 1$.

A radius- δ ball about any legal centre contains at most one point of a 4δ -packing, by the triangle inequality. Hence the packing proves the lower bound with the stated centre convention. Taking binary logarithms and rounding the index length proves (12.2). $\square$

12.3. The source of the logarithm. The spectral and angular weights also give the following integral form of the same accumulation. Put $v_p = p(1 - p)$, $v_q = q(1 - q)$ and $\tau = N^{-1}$. Then

$$(12.11) \quad \int_{0 \leq q \leq p \leq 1} \frac{N^2(p - q)^2 dp dq}{(v_p + v_q + \tau)\sqrt{(v_p + \tau)(v_q + \tau)}} \asymp N^2 \log(N + 2).$$

To verify this, near $p = 1, q = 0$ put $\alpha = 1 - p, \beta = q$. After removing the factor N^2 , the integrand is comparable to

$$\frac{1}{(\alpha + \beta + \tau)\sqrt{(\alpha + \tau)(\beta + \tau)}}.$$

For $s = \alpha + \beta$, integration over $0 \leq \alpha \leq s$ is bounded above by an absolute constant times $1/(s + \tau)$. For $s \geq 2\tau$, restriction to $s/4 \leq \alpha \leq 3s/4$ gives the matching lower bound. Integration in s proves logarithmic growth. Near the scalar endpoint $p = q = 0$, the factor $(p - q)^2$ is at most $(p + q)^2$, while $v_p + v_q \asymp p + q$; the remaining integral is uniformly bounded. Complementation gives the same conclusion near $p = q = 1$. Away from these endpoint neighbourhoods, $v_p + v_q$ is bounded below and the factors $v_p^{-1/2}, v_q^{-1/2}$ are integrable. Bounded N is absorbed into the constants. Thus the extra logarithm is contributed by the projective corner. The finite covering construction and packing above also include the collapsed scalar coordinates, where the spectral representation ceases to be a chart.

13. MEASUREMENT DISCRIMINATION, BLOCK RESOURCES, AND LEARNING

The present binary interface retains the ordered classical outcome and an arbitrary external reference, but no postmeasurement quantum system from the device. This fixes the comparison with the measurement-discrimination literature. The projective theorem of Puchała and collaborators [16, Corollary 1 and Theorem 2] gives an exact multiple-use formula with ancillary access. It is used in the retained observable-readout theorem. The restricted ancillary interface in Fiurášek–Mičuda [19] leads to a different comparison of adaptive and parallel probes.

Krawiec–Pawela–Puchała [20, Section 4] construct a pair of nine-outcome qutrit rank-one POVMs that admit perfect adaptive discrimination with two calls and no perfect discrimination by any finite parallel experiment. Datta–Biswas–Saha–Augusiak [21, Theorems 3–5] study entanglement-assisted single-use discrimination, including perfect constructions and a qubit entanglement advantage. Their qubit construction has three outcomes. These are direct antecedents for general POVM discrimination with a retained reference. They do not supply a full binary-effect family covering law. Our comparison (2.23) permits an adaptive advantage and controls its size uniformly in N for fixed input dimension.

The differential argument in Lemma 2.2 belongs to the established horizontal-gauge and channel-extension method [22, 7, 23, 24]. Yuan–Fung give the parallel path estimate [25, Section III, equation (26), PDF version 3]; Sieniawski–Demkowicz–Dobrzański integrate adaptive bounds [26, Section IV.A]. The binary Sylvester choice here gives the specific regularized variance $M(I - M)$, which can be integrated explicitly and matched by compressed nonadaptive tests. The inverse-Sylvester expressions of Bures geometry [27] and its spectral integration [28] are related methods. Our form acts on the effect displacement H with variance $V = M(I - M)$; it is not the ordinary Bures pullback under $M \mapsto V$, whose differential would be $H - MH - HM$. The operational-ball lemma and the nested endpoint integral are both required for Theorem 3.1.

13.1. One-use tomography and its acquisition resources. Mele–Bittel [5, Theorem I.1] give channel tomography with $O(d_{\text{in}}d_{\text{out}}k\epsilon^{-2})$ calls at fixed confidence. Their binary specialization [5, Lemmas III.7–III.8 and Corollary III.9] returns a legal effect, with operator error at most ϵ , using

$$M = 1024d^2\epsilon^{-2} + O(d\epsilon^{-2}\log(1/\eta)), \quad 4e^{-4d^2} < \eta < 1, \quad 0 < \epsilon < 1.$$

The construction prepares independent maximally entangled input/reference pairs, then processes the Choi copies collectively [5, Algorithm I.2, Figure 1 and Section III.2]. Consequently it belongs to the fresh-block class with $b = 1$. Calling all invocations one parallel block, as the source does, does not entangle the inputs of distinct invocations. Coherent purification and collective output tomography are allowed by our $b = 1$ class. For all $0 < \eta \leq 1/8$, the amplification in [5, Remark III.4, equation (165)] gives the sufficient budget $O(d^2 \epsilon^{-2} \log(1/\eta))$ with the same acquisition restriction.

Zambrano–Ramos–Calderer–Kueng [29, Definition 1 and Theorem 2] use worst-input one-use outcome total variation. For L outcomes, their global-2-design protocol has sufficient budget

$$M \geq \frac{8(d^2 + \epsilon(d^2 + 1)/6)}{\epsilon^2} \log \frac{2^{L+1} 9^{2d}}{\eta}.$$

Known inputs are probed separately; least squares and projection return a legal POVM. This is also $b = 1$, with no retained reference. For $L = 2$ their loss is $\|E - F\|_{\text{op}}$, whereas ours has $d_1(E, F) = 2\|E - F\|_{\text{op}}$. Their Theorem 9 gives a $\Omega((d^3 + d^2 L)\epsilon^{-2})$ lower bound at constant confidence for nonadaptive single-copy measurements on known input states. This restriction stays attached to that lower bound.

13.2. What stronger future-loss analysis can establish. The hybrid inequality

$$d_N(E, F) \leq 2N\|E - F\|_{\text{op}}$$

transfers both one-use guarantees to sufficient budgets of order $O_d(N^2 \delta^{-2} \log(1/\eta))$. By itself this remains an upper-bound conversion. Theorem 7.2 and Corollary 7.5 now give the distinct converse: for every fixed $d \geq 2$, every fresh $b = 1$ acquisition procedure that learns the entire ordered binary effect body under d_N requires

$$M \geq cN^2 \delta^{-2} \log(1/\eta).$$

The statement permits classical adaptive choices, stored reference outputs, their collective processing, and public bounded stopping. It therefore applies to the acquisitions in both cited tomography protocols, including any reconstruction from the same acquired systems. No reanalysis of those acquisitions can give the uniform full-body $O_d(N \delta^{-2} \log(1/\eta))$ guarantee.

There is nevertheless a useful positive reanalysis on a promised interior class. Theorem 8.5 applies Mele–Bittel’s binary estimator with $\epsilon = \delta/(8\sqrt{N})$ and $\eta = 1/8$ to

$$\frac{1}{4}I \preceq E \preceq \frac{3}{4}I.$$

The horizontal path estimate gives $d_N(E, F) \leq 4\sqrt{N}\|E - F\|_{\text{op}}$ when $\|E - F\|_{\text{op}} \leq 1/8$. Thus $O(d^2 N \delta^{-2})$ fresh single-call probes suffice at this confidence. Theorem 8.2 proves the matching $\Omega(d^2 N \delta^{-2})$ lower bound even for arbitrary coherent adaptive training. This is a joint dimension, horizon and accuracy law on the displayed interior class at constant confidence. The projective boundary supplies the additional full-body $b = 1$ obstruction. The two conclusions answer different quantifiers in the estimator-reanalysis question.

13.3. The full-body resource law and its antecedents. For $d \geq 2$, $1 \leq b \leq N$, $0 < \delta \leq \delta_*$ and $0 < \eta \leq 1/8$, Theorem 7.2 states

$$c \frac{N^2}{b} \delta^{-2} \log(1/\eta) \leq M_b^*(d, N, \delta, \eta) \leq C d^4 \frac{N^2}{b} \delta^{-2} \log(d/\eta).$$

Here M_b^* is the worst-case call budget for a common classical estimate in the fresh-block model of Section 7. A block may use its own reference and adaptive quantum controls; only the classical record can determine the next fresh probe. Stored quantum outputs may be processed jointly but cannot become quantum inputs to a later block. The bound holds on every record. At $b = N$

it recovers the retained fixed-dimensional learning order; at $b = 1$ it proves the direct quadratic-horizon obstruction. The scalar case $d = 1$ instead has order $N\delta^{-2}\log(1/\eta)$, independently of b .

Result and acquisition	Loss and parameter range	Training-call order
Mele–Bittel, independent Choi probes; $b = 1$	Entire binary body; one-use operator error ϵ ; amplified $0 < \eta \leq 1/8$	$O(d^2\epsilon^{-2}\log(1/\eta))$
Zambrano et al., known individual probes; $b = 1$	Binary specialization of global designs; one-use operator error ϵ ; $0 < \eta < 1$	$O(d^2\epsilon^{-2}[d + \log(1/\eta)])$
Theorem 8.5, independent Choi probes; $b = 1$	$\frac{1}{4}I \preceq E \preceq \frac{3}{4}I$; future d_N error δ ; $\eta = 1/8$	$\Theta(d^2N\delta^{-2})$; lower allows fully coherent training
Theorem 7.2, fresh blocks of size $\leq b$	Entire binary body; future d_N error δ ; $d \geq 2$, $1 \leq b \leq N$, $0 < \eta \leq 1/8$	$\Theta_d((N^2/b)\delta^{-2}\log(1/\eta))$; explicit upper factor d^4

All new learning rows have an absolute small-error cap. The full-body dimension dependence is not sharp; neither this table nor the block theorem proves a growing-dimensional minimax law for that body.

The metrological block scaling is established background. Hyllus et al. [33, Observation 1, equation (9)] and Tóth [34, Observation 3, equation (6)] prove, for b -producible qubit probes under linear phase encoding,

$$F_Q \leq \lfloor M/b \rfloor b^2 + (M - b\lfloor M/b \rfloor)^2 \leq Mb.$$

Our block parameter also permits sequential quantum controls inside a fresh block; it is not a general measure of multipartite entanglement in the final state. The finite-pair fidelity proof supplies the confidence factor without an unbiasedness assumption. Salek–Hayashi–Winter [9, Section II.2 and Section V.2] already formulate classical feed-forward with arbitrary quantum memory at the output and no quantum memory at the input, and analyze asymptotic discrimination exponents. Wilde et al. [14, Proposition 21 and Appendix B] give adaptive divergence converses and fidelity-to-error inequalities. Section 7 checks the finite-budget conditional fidelity argument, variable block lengths and stopping explicitly. The all-effect upper bound follows by blocking the future tester and invoking the retained common learner at the shorter horizon. The asserted increment is this common future-loss minimax statement and its application to specified tomography acquisitions; the Mb metrological scale is not new.

A different literature bounds how many already supplied state copies can be measured jointly [36, 37]. Those output measurement restrictions differ from the fresh-query restriction here, which allows an unrestricted final joint measurement. Mele–Bittel’s proposed blockwise Choi processing [5, Section I.5] concerns that other resource.

Recent channel-learning lower bounds also require careful domain comparison. Oufkir–Girardi [6] and Chen–Zhang–Yu [35, Theorems 1.1 and 4.2] treat fully coherent channel learning in one-use diamond loss. The latter gives $\Omega(rd_{\text{in}}d_{\text{out}}\epsilon^{-2})$ when $rd_{\text{out}} \geq 2d_{\text{in}}$ at constant confidence. A two-dimensional quantum output need not be classical, so this general-channel theorem is not imported as a binary-qc lower bound. Mele–Bittel’s actual binary converses are their projector packing [5, Proposition IV.17] and the separate diagonal-coin bound $\Omega(d\epsilon^{-2}\log(d/\eta))$ [5, Corollary IV.20]. The self-contained near-fair binary information argument in Lemma 8.1 yields the growing-dimension lower bound used here. Its relation to other channel hard-instance constructions remains a specific priority comparison, not a claim that such lower-bound methods are new.

13.4. Retained calibration and exact descriptions. Multiscale phase recovery is established [31, 32]. The qubit learner proves its own safe-angle induction, bias cancellation, unknown-noise

block selection and conditional confidence allocation. The matrix learner calibrates both support endpoints, controls the variance created by data-selected compressions, and assembles a legal estimate. These supply one procedure for the entire closed binary effect body, including rank changes and repeated eigenvalues. Pair-dependent tests in the metric proof retain their different role.

The deterministic rational construction in Theorem 6.1 supplies a finite exact selection rule and optimal-order payload. Exhaustive reconstruction time, workspace and state preparation remain separate from call count and payload. No efficient high-dimensional encoder is inferred. The closed-body entropy constants remain fixed-dimensional and its matching entropy and payload orders have a small-error range. In contrast, Theorem 9.1 has explicit dimension-uniform constants on $[I/4, 3I/4]$. Its norm-ball volume argument and the Fano–Kraft proof of Theorem 9.4 use standard mechanisms. Their role is to derive the precise future-loss description statements, not to claim priority for volume comparison or information inequalities. The learned-description theorem credits the Mele–Bittel estimator and retains its ideal trusted operations; it adds finite-output coarsening and a public dictionary without extra device calls. Multiple-outcome measurements, residual quantum outputs, sharp full-body growing-dimensional learning, and efficient exact reconstruction require additional arguments.

APPENDIX A. A NOISE-UNIFORM READOUT TRANSITION

The dependence on the readout basis can change scale when noise approaches an extremal measurement. We determine that transition in a fixed one-dimensional family, without a positive spectral margin uniform in the noise. The proof is a finite-distance argument; no conclusion is inferred merely from a Fisher-information calculation.

Let σ_1, σ_2 be the first two Pauli matrices. For $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ and a public visibility $0 \leq \lambda \leq 1$, put

$$(A.1) \quad E_{\theta,\lambda}^{\pm} = \frac{1}{2}(I \pm \lambda(\cos \theta \sigma_1 + \sin \theta \sigma_2)), \quad \mathcal{M}_{\theta,\lambda}^{\pm}(X) = \text{tr}(E_{\theta,\lambda}^{\pm} X).$$

Both outcome labels are retained. The output quantum system is one-dimensional. In particular, θ and $\theta + \pi$ are not identified by an outcome relabelling. Let $h(\theta, \phi) \in [0, \pi]$ be circular distance and set

$$(A.2) \quad \eta = 1 - \lambda^2, \quad K_{N,\lambda} = \lambda \sqrt{N \min\{N, \eta^{-1}\}}, \quad N \geq 1,$$

where $\eta^{-1} = +\infty$ at $\lambda = 1$. Thus $K_{N,1} = N$ and $K_{N,0} = 0$. The visibility may depend arbitrarily on N and the requested error; it is not a target coordinate charged in this theorem.

Write $\mathcal{C}_{N,\lambda}(\delta)$ for the least number of legal memoryless binary instruments of this interface whose d_N -balls of radius δ cover all targets in (A.1). Centres need not belong to the displayed family. Define $\mathcal{C}_{N,\lambda}^{\text{rat}}(\delta)$ by restricting centres to family members with rational $\cos \theta, \sin \theta$; rational visibility then gives rational effects. A constant in the following statement is independent of all three parameters N, λ, δ .

Theorem A.1 (Noise-uniform finite-use coding). *For all $N \geq 1$, $0 \leq \lambda \leq 1$, and all pairs of angles,*

$$(A.3) \quad \frac{1}{64} \min\{1, K_{N,\lambda} h(\theta, \phi)\} \leq d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min\{2, K_{N,\lambda} h(\theta, \phi)\}.$$

There are absolute $c, C > 0$ such that, for $0 < \delta \leq 2^{-10}$,

$$(A.4) \quad c \left(1 + \frac{K_{N,\lambda}}{\delta}\right) \leq \mathcal{C}_{N,\lambda}(\delta) \leq \mathcal{C}_{N,\lambda}^{\text{rat}}(\delta) \leq C \left(1 + \frac{K_{N,\lambda}}{\delta}\right).$$

Consequently the optimum fixed-length reusable payload is

$$(A.5) \quad \log_2 \left(1 + \frac{K_{N,\lambda}}{\delta} \right) + O(1).$$

The upper construction is an exact rational algorithm on rational visibility, tolerance and unit-circle input coordinates. At $\lambda = 0$ the family is a singleton and its optimum payload is zero.

The constants and the additive remainder are uniform up to the erased and projective endpoints. In contrast, they do not assert an optimal leading constant, a large-error coherent covering law, or a coding law in which an unknown visibility is also encoded.

A.1. A two-point programme with its noise dependence retained.

Lemma A.2 (An adaptive upper modulus). *Let $h = h(\theta, \phi)$ and $a = \lambda \sin(h/2)$. If $\eta > 0$, then*

$$(A.6) \quad d_N(\mathcal{M}_{\theta,\lambda}, \mathcal{M}_{\phi,\lambda}) \leq \min \left\{ 2, 2Na, 2\sqrt{1 - \left(\frac{\eta}{\eta + a^2} \right)^N} \right\}.$$

For $\eta = 0$ the bound $\min\{2, 2Na\}$ remains valid.

Proof. The difference of the two positive-outcome effects has operator norm a . On a bipartite input state, the two output blocks of the difference are B and $-B$, where $B = \text{tr}_A[(E_{\theta,\lambda}^+ - E_{\phi,\lambda}^+) \otimes I]\rho$. Duality against Hermitian contractions gives $\|B\|_1 \leq a$. An eigenstate of the effect difference attains equality. Hence $d_1 = 2a$, including reference assistance. Replacing the calls one at a time and using trace-norm contraction proves $d_N \leq 2Na$ for every common adaptive tester.

Choose angle representatives with difference h and let u, v be the unit vectors along their midpoint and its oriented perpendicular in the equatorial plane. The two Bloch vectors are $m \pm av$, where $m = \lambda \cos(h/2)u$. Put $b = (1 - \|m\|^2)^{1/2} = (\eta + a^2)^{1/2}$. The two Bloch vectors $m \pm bv$ have norm one, so they define two legal binary projective instruments, say $\mathcal{A}_+$ and $\mathcal{A}_-$. The target pair is exactly the two mixtures of these same instruments with programme laws

$$p_{\pm} = ((1 \pm t)/2, (1 \mp t)/2), \quad t = a/b.$$

This construction depends on the pair, which is permissible in a metric upper bound. It is not one fixed processor for the whole circle.

For a fixed tester, first supply N independent programme bits and, at each requested call, apply the indicated $\mathcal{A}_+$ or $\mathcal{A}_-$. All subsequent operations, including the reference, feedback and a stopping rule, are a common channel of these bits. Unused bits are discarded. The root fidelity of p_+, p_- is $\sqrt{1 - t^2}$. Purifying the classical laws and using contraction gives

$$\|p_+^{\otimes N} - p_-^{\otimes N}\|_1 \leq 2\sqrt{1 - (1 - t^2)^N}.$$

This also follows directly from the classical fidelity inequality. Substitution proves (A.6). Finally $1 - (1 - t^2)^N \leq Nt^2$ and $a \leq \lambda h/2$ imply

$$d_N \leq \min\{N\lambda h, \sqrt{N/\eta} \lambda h, 2\}.$$

The limiting projective bound is supplied by the hybrid estimate, not by division by zero. $\square$

Classical simulation by common extremal channels and the effect of noise on the Heisenberg scale are established methods [7]. Environment/programme-state processing under adaptive use is also an established framework [13, 14, 15]. The point needed here is the exact two-point expression (A.6) and its uniform conversion to a family cover, rather than a new claim about the origin of those methods.

A.2. Finite entangled blocks and a binary product estimate.

Lemma A.3 (Symmetric Bernoulli products). *Let $P_{\pm}$ be Bernoulli laws with success probabilities $(1 \pm a)/2$, where $0 \leq a \leq 1$. For every integer $m \geq 1$,*

$$(A.7) \quad \|P_+^{\otimes m} - P_-^{\otimes m}\|_1 \geq \frac{1}{2} \min\{1, a\sqrt{m}\}.$$

Proof. Retain the largest odd number $r = 2j + 1 \leq m$ of coordinates, so $r \geq m/2$. For their majority event write $g_r(a) = P_+(\text{majority}) - P_-(\text{majority})$. Differentiating the finite binomial sum and cancelling consecutive terms gives

$$g'_r(a) = r \binom{2j}{j} 4^{-j} (1 - a^2)^j.$$

For $j \geq 1$, the bound $\binom{2j}{j} 4^{-j} \geq 1/(2\sqrt{j})$ follows by induction: the ratio of consecutive left sides is $(2j+1)/(2j+2) \geq \sqrt{j/(j+1)}$. For $0 \leq a \leq r^{-1/2}$, Bernoulli's inequality gives $(1 - a^2)^j \geq 1 - j/r \geq 1/2$. Thus $g'_r(a) \geq \sqrt{r}/(2\sqrt{2})$ on that interval. The case $r = 1$ is immediate. Integration from zero and monotonicity yield $g_r(a) \geq (2\sqrt{2})^{-1} \min\{1, a\sqrt{r}\}$. Unhalved distance is at least twice this event difference; using $r \geq m/2$ proves the stated bound. $\square$

Lemma A.4 (A noise-uniform pairwise witness). *The lower inequality in (A.3) is witnessed by nonadaptive entangled blocks, each of at most N qubits, followed by classical parity and majority processing.*

Proof. A block of k inputs is prepared in $(|0\rangle^{\otimes k} + e^{i\alpha}|1\rangle^{\otimes k})/\sqrt{2}$. Multiplying the k output signs of (A.1) has mean $\lambda^k \cos(k\theta - \alpha)$. This identity follows by applying $\bigotimes_{i=1}^k (E^+ - E^-)$ to the two off-diagonal terms of the block density matrix. For the two hypotheses choose the one common phase $\alpha = k(\theta + \phi)/2 + \pi/2$. Their parity laws then have opposite biases of magnitude

$$(A.8) \quad a_k = \lambda^k |\sin(kh/2)|.$$

Repeating $m = \lfloor N/k \rfloor$ independent blocks and ignoring remaining calls is an admissible tester. Its unhalved output distance is at least $\frac{1}{2} \min\{1, \sqrt{m} a_k\}$ by Lemma A.3. The phase and block size may depend on the pair, but never on the unknown hypothesis or a private outcome.

Assume $\lambda > 0$ and put $T = \min\{N, \eta^{-1}\}$, $k_0 = \lfloor T \rfloor$. Then $T/2 \leq k_0 \leq T$. For every $1 \leq k \leq k_0$,

$$(A.9) \quad \lambda^k \geq \lambda/2.$$

Indeed, when $0 < \eta < 1$, the inequality $\log(1 - \eta) \geq -\eta/(1 - \eta)$ and $k - 1 \leq \eta^{-1} - 1$ give $(k - 1) \log \lambda \geq -1/2$. At $\eta = 0$ the assertion is immediate.

If $h \leq 1/k_0$, choose $k = k_0$. Since $\sin(kh/2) \geq kh/\pi$ and $m \geq N/(2k)$, we obtain

$$\sqrt{m} a_k \geq \frac{\lambda \sqrt{N k_0}}{2\pi \sqrt{2}} h \geq \frac{K_{N,\lambda}}{4\pi} h.$$

This gives a lower constant $1/(8\pi)$. If $k_0 = 1$, use $k = 1$ for all $h \leq \pi$ instead. Now $T < 2$ unless $T = 1$, and $\sqrt{N} \lambda \sin(h/2) \geq K_{N,\lambda} h/(\pi \sqrt{2})$; this is stronger than required.

It remains that $k_0 \geq 2$ and $h > 1/k_0$. Here $\lambda \geq 1/\sqrt{2}$. Choose $k = \max\{1, \lfloor 1/h \rfloor\}$. Then $k \leq k_0$ and $1/2 \leq kh \leq \pi$. Equations (A.8) and (A.9) give $a_k \geq 1/(4\pi \sqrt{2})$. Even one such block has distance at least $1/(8\pi \sqrt{2}) > 1/64$. The three cases prove the claim. For $\lambda = 0$ both sides of the desired inequality vanish. $\square$

A.3. Covering and exact rational realization.

Proof of Theorem A.1. The metric assertion follows from the preceding lemmas. For a lower cover choose points on the circle with circular separation at least $256\delta/K_{N,\lambda}$ whenever that quantity is at most π . An equally spaced packing has at least a fixed constant times $K_{N,\lambda}/\delta$ points. Every two points have d_N -distance at least 4δ , because $256\delta \leq 1/4$. No radius- δ ball, even about an arbitrary legal instrument, can contain two such points. If the required angular separation exceeds π , the trivial lower bound one suffices, with a smaller absolute constant. This proves the first inequality in (A.4).

For the upper bound use the two rational semicircle charts

$$(A.10) \quad z_s(t) = s \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right), \quad s \in \{-1, 1\}, \quad -1 \leq t \leq 1.$$

Each chart has angular speed $2/(1+t^2) \leq 2$. Round t to the nearest j/B , $-B \leq j \leq B$, with a fixed rule at ties. The circular error is at most $1/B$. Thus any integer $B \geq \max\{1, K_{N,\lambda}/\delta\}$ yields a cover with at most $4B + 2$ centres. For rational inputs, an entirely rational choice is

$$(A.11) \quad B = \max \left\{ 1, \min \left\{ \left\lceil \frac{4N\lambda}{\delta} \right\rceil, \left\lceil \sqrt{\frac{16N\lambda^2}{\eta\delta^2}} \right\rceil \right\} \right\},$$

omitting the second entry when $\eta = 0$. Integer squaring and rational comparison compute the ceiling square root exactly. Both entries are ceilings of the corresponding real bounds, so their minimum is $\lceil 4K_{N,\lambda}/\delta \rceil$. For $z = (x, y)$ on the rational unit circle choose $s = 1$ when $x \geq 0$ and $s = -1$ otherwise, and set $t = sy/(1+sx)$. Its denominator is at least one, so it is rational and belongs to $[-1, 1]$.

The chart digit (s, j) determines a legal effect pair with eigenvalues $(1 \pm \lambda)/2$. No approximate positivity test, irrational normalization or search over real angles is needed. A fixed-length index over $4B + 2$ slots includes the chart sign. Its length gives (A.5). Redundant chart endpoints cause at most a constant overhead. At $\lambda = 0$ dispense with the chart and emit the single fair, input-erasing measurement. $\square$

Corollary A.5 (A continuous family of horizon exponents). *Fix $0 < \delta \leq 2^{-10}$. If $\lambda_N \rightarrow 1$ and $1 - \lambda_N^2 = N^{-\alpha+o(1)}$ for $\alpha \geq 0$, then*

$$\frac{\log_2 \mathcal{C}_{N,\lambda_N}(\delta)}{\log_2 N} \longrightarrow \frac{1 + \min\{\alpha, 1\}}{2}.$$

If $\lambda_N = N^{-\beta+o(1)} \rightarrow 0$, with $\beta \geq 0$, the limit is $\max\{1/2 - \beta, 0\}$. The projective and erased endpoints have coefficients one and zero, respectively.

Proof. Substitute the two regimes in (A.2) and use (A.4). The term $1 + K/\delta$ handles a vanishing or subpolynomial scale without taking the logarithm of zero. $\square$

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